

MASCO101

Tools for Success — Your Human Edge in an AI World

EU Business School · Masters Level 7 · Term 1 · 5 ECTS · AY 2026/27

Framework Sheet Library

Master Index · 16 Slibraries · 80 Reference Sheets

8

Course Units

foundations, self-assessment,
thinking, branding, communication,
projects, negotiation, AI

16

Slibraries

thematic clusters, two per unit,
colour-coded

80

Framework Sheets

nine sections each, print-ready A4

320+

Harvard Sources

peer-reviewed and practitioner

Design philosophy

Each sheet follows a consistent nine-section layout: professional definition, who developed it, why it was developed, a custom illustrative figure, how / when / by whom it is used, the innovation relative to the predecessor tool, related frameworks in this course, and Harvard-referenced sources. Every figure is drawn from the framework's own structure, so a model described as having six levels is drawn with six. The structure lets the library be read as a reference work — a mental Slibrary in the sense of Meier (2025) — scaling from single-framework lookup to reasoning across connected tools.

Dr. Hildegard Haas

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Unit 1

Introduction to Tools for Success & Digital/AI Literacy

Guided Discovery · 2 Slibraries · 10 framework sheets

CORE MESSAGE

Your value at work is no longer what you can execute — it is what you can judge, and judgment has to be built on purpose.

Learning focus

Foundations & digital fluency · ILO 1

ILO 1 — Systematically select key concepts and principles of operations and technology management that need to be implemented to support value creation.

SLIBRARY 1 · Learning & Goal Architecture

5 sheets · pp. 11–15

1	SMART Goal Frame	p. 11
2	Kolb Experiential Learning Cycle	p. 12
3	Bloom's Taxonomy (Revised)	p. 13
4	Dreyfus Skill Acquisition Model	p. 14
5	Deliberate Practice Loop	p. 15

SLIBRARY 2 · Digital & AI Literacy

5 sheets · pp. 16–20

1	DigComp 2.2 Competence Wheel	p. 16
2	CRAAP Source Test	p. 17
3	Technology Acceptance Model	p. 18
4	Automation Bias Curve	p. 19
5	Co-Intelligence Working Rules	p. 20

Unit 2

Know Yourself: Self-Assessment as a Strategic Tool

Self-Diagnostic Workshop · 2 Slibraries · 10 framework sheets

CORE MESSAGE

You cannot position what you have not diagnosed — self-knowledge is the one input no model can generate for you.

Learning focus Self-assessment & personal positioning · **ILO 1**

ILO 1 — Systematically select key concepts and principles of operations and technology management that need to be implemented to support value creation.

SLIBRARY 3 · Self-Diagnosis

5 sheets · pp. 21–25

1	Personal SWOT Matrix	p. 21
2	Johari Window	p. 22
3	Ikigai Purpose Venn	p. 23
4	Firm Foundation: Vision, Mission, Values	p. 24
5	Personal Energy Model	p. 25

SLIBRARY 4 · Style & Preference Profiling

5 sheets · pp. 26–30

1	MBTI Four Dichotomies	p. 26
2	DISC Behavioural Quadrant	p. 27
3	Big Five (OCEAN)	p. 28
4	Honey & Mumford Learning Styles	p. 29
5	Growth vs. Fixed Mindset Fork	p. 30

Unit 3

Think Better: Critical Thinking & Analytical Reasoning

Head-to-Head Analysis Lab · 2 Slibraries · 10 framework sheets

CORE MESSAGE

AI accelerates answers; only structured thinking decides whether the answer was the right one to look for.

Learning focus Critical thinking & structured problem-solving · ILO 1

ILO 1 — Systematically select key concepts and principles of operations and technology management that need to be implemented to support value creation.

SLIBRARY 5 · Structured Problem-Solving

5 sheets · pp. 31–35

1	SCQA Structure	p. 31
2	MECE Issue Tree	p. 32
3	First Principles Decomposition	p. 33
4	Double Diamond	p. 34
5	Cynefin Framework	p. 35

SLIBRARY 6 · Judgment & Bias

5 sheets · pp. 36–40

1	System 1 / System 2	p. 36
2	Confirmation Bias Loop	p. 37
3	Anchoring & Adjustment	p. 38
4	Ladder of Inference	p. 39
5	Dunning–Kruger Curve	p. 40

Unit 4

Present & Position Yourself: Personal Branding & Presentation Skills

Pitch Arena · 2 Slibraries · 10 framework sheets

CORE MESSAGE

A personal brand is not what you claim about yourself; it is the one sentence other people can repeat when you are not in the room.

Learning focus Personal branding & persuasive delivery · **ILO 5**

ILO 5 — Critically review the communication process, stakeholder communication and communication in diverse work groups.

SLIBRARY 7 · Personal Brand & Positioning

5 sheets · pp. 41–45

1	Brand Pyramid	p. 41
2	Twelve Brand Archetypes	p. 42
3	StoryBrand SB7 Framework	p. 43
4	Golden Circle	p. 44
5	Personal Value Proposition Canvas	p. 45

SLIBRARY 8 · Pitch & Delivery

5 sheets · pp. 46–50

1	Message Strategy Spectrum	p. 46
2	Problem → Insight → Recommendation	p. 47
3	Assertion–Evidence Slide Model	p. 48
4	Mehrabian 7–38–55	p. 49
5	Anxiety Reappraisal	p. 50

Unit 5

Communicate with Impact: Executive Presence & Business Communication

Message Clinic · 2 Slibraries · 10 framework sheets

CORE MESSAGE

Executives do not read to discover your conclusion — put it first, then make every sentence after it earn its place.

Learning focus

Business communication & executive presence · ILO 5

ILO 5 — Critically review the communication process, stakeholder communication and communication in diverse work groups.

SLIBRARY 9 · Argument Architecture		5 sheets · pp. 51–55
1	Pyramid Principle	p. 51
2	Distil → Decompose → Prioritise	p. 52
3	Toulmin Argument Model	p. 53
4	BLUF Executive Structure	p. 54
5	Storyline Canvas	p. 55

SLIBRARY 10 · Audience & Culture		5 sheets · pp. 56–60
1	Shannon–Weaver Model	p. 56
2	Hofstede Cultural Dimensions	p. 57
3	The Culture Map (Eight Scales)	p. 58
4	High- and Low-Context Communication	p. 59
5	Power–Interest Grid	p. 60

Unit 6

Deliver Results: Project Management Fundamentals

Project Simulation · 2 Slibraries · 10 framework sheets

CORE MESSAGE

A plan is not a schedule — it is an argument about scope, risk and trade-offs that has to survive contact with reality.

Learning focus Project planning, scheduling & risk · **ILO 2 & 3**
ILO 2 — Critically assess the roles and responsibilities of a project manager as part of an organisation's strategy.
ILO 3 — Investigate how to select, develop, plan, schedule and measure project outcomes and risks.

SLIBRARY 11 · Project Lifecycle & Planning

5 sheets · pp. 61–65

1	Project Lifecycle	p. 61
2	Iron Triangle	p. 62
3	Work Breakdown Structure	p. 63
4	Gantt Chart	p. 64
5	Critical Path Method	p. 65

SLIBRARY 12 · Agile, Risk & Control

5 sheets · pp. 66–70

1	Agile Manifesto Values	p. 66
2	Scrum Sprint Cycle	p. 67
3	Stacey Matrix	p. 68
4	Probability–Impact Risk Matrix	p. 69
5	Earned Value S-Curve	p. 70

Unit 7

Negotiate & Influence: Getting What You Need Without Formal Authority

Negotiation Simulation · 2 Slibraries · 10 framework sheets

CORE MESSAGE

Leverage is prepared, not improvised — know your walk-away before you walk in.

Learning focus Negotiation, influence & ethical boundaries · ILO 4

ILO 4 — Evaluate the keys to successful negotiation and differentiate negotiation styles, strategies, and tactics.

SLIBRARY 13 · Negotiation Architecture

5 sheets · pp. 71–75

1	BATNA	p. 71
2	Zone of Possible Agreement	p. 72
3	Principled Negotiation	p. 73
4	Dual Concerns Conflict Modes	p. 74
5	Distributive ↔ Integrative Spectrum	p. 75

SLIBRARY 14 · Influence Without Authority

5 sheets · pp. 76–80

1	Six Principles of Influence	p. 76
2	French & Raven Power Bases	p. 77
3	Influence Currencies	p. 78
4	Tactical Empathy	p. 79
5	Persuasion ↔ Manipulation Line	p. 80

Unit 8

Work Smarter with AI: Productivity, Tools and Professional Judgment

Mini-Hackathon · Capstone · 2 Slibraries · 10 framework sheets

CORE MESSAGE

Delegate the task, never the judgment — and be able to show exactly where you drew the line.

Learning focus

Productivity systems & AI workflow judgment · ILO 1 & 3

ILO 1 — Systematically select key concepts and principles of operations and technology management that need to be implemented to support value creation.

ILO 3 — Investigate how to select, develop, plan, schedule and measure project outcomes and risks.

SLIBRARY 15 · Prioritisation & Energy		5 sheets · pp. 81–85
1	Eisenhower Matrix	p. 81
2	Objectives & Key Results	p. 82
3	The Rule of 3	p. 83
4	Monday Vision, Daily Wins, Friday Reflection	p. 84
5	Pareto 80/20	p. 85

SLIBRARY 16 · AI Workflow & Judgment		5 sheets · pp. 86–90
1	Prompt Anatomy	p. 86
2	Delegate–Verify–Own Loop	p. 87
3	Automation–Augmentation Spectrum	p. 88
4	The Jagged Frontier	p. 89
5	AI Attribution Ladder	p. 90

How to Use This Library

Three ways in, one fixed page for every framework

For students

Each Slibrary is a thematic cluster of five frameworks, designed to be held in mind as a set rather than memorised individually — a Slibrary in the sense of Meier's Thumbnail Thinking (2025). Use a single sheet to look up one framework, a whole Slibrary for a thematic revision pass, and the related-frameworks pills at the foot of each sheet to move between connected ideas across units. The nine sections are deliberately uniform so that comparison between frameworks is possible.

For practising managers

The library works as a decision-support reference. Facing a negotiation, a project plan or a difficult message, open the relevant Slibrary rather than a single sheet: the frameworks in a cluster are chosen to be complementary, and the useful move is usually to apply two of them against each other rather than to follow one.

Finding a sheet by hand

Sheets run in Slibrary order, five to a Slibrary:
page = 10 + 5 × (Slibrary – 1) + sheet
Cover p. 1 · Master Index pp. 2–9 · this page p. 10 · sheets pp. 11–90

STARTS	UNIT	SESSION FORMAT
p. 11	Unit 1 — Introduction to Tools for Success & Digital/AI Literacy	Guided Discovery
p. 21	Unit 2 — Know Yourself: Self-Assessment as a Strategic Tool	Self-Diagnostic Workshop
p. 31	Unit 3 — Think Better: Critical Thinking & Analytical Reasoning	Head-to-Head Analysis Lab
p. 41	Unit 4 — Present & Position Yourself: Personal Branding & Presentation Skills	Pitch Arena
p. 51	Unit 5 — Communicate with Impact: Executive Presence & Business Communication	Message Clinic
p. 61	Unit 6 — Deliver Results: Project Management Fundamentals	Project Simulation
p. 71	Unit 7 — Negotiate & Influence: Getting What You Need Without Formal Authority	Negotiation Simulation
p. 81	Unit 8 — Work Smarter with AI: Productivity, Tools and Professional Judgment	Mini-Hackathon · Capstone

Citing a framework

Every sheet carries the original sources in Harvard style. Cite those, never this library: the sheets are a teaching synthesis, not the scholarship itself. Where a framework is contested, widely over-generalised, or supported by weaker evidence than its popularity suggests, the innovation section says so — carry that caveat with you when you use it.

A note on the figures

Each figure is generated from the structure of the framework it illustrates, so the number of levels, stages or quadrants drawn always matches the number described in the definition. The figures are schematic memory hooks, not data visualisations; read the definition and the sources before relying on the picture.

SMART Goal Frame

SLIBRARY 01
SHEET 1 / 5

Specific · Measurable · Achievable · Relevant · Time-bound — five tests, one commitment

PROFESSIONAL DEFINITION

The SMART Goal Frame is a five-criterion screen applied to any stated objective before it is accepted as a commitment. An objective qualifies only if it names a specific actor and action, specifies the evidence by which completion will be measured, is achievable with the resources actually available, is relevant to the wider strategy it serves, and carries a fixed date rather than an open horizon. Objectives failing any one criterion are rewritten, not merely noted (Doran, 1981).

WHO DEVELOPED IT

George T. Doran, a consultant and former director of corporate planning at Washington Water Power Company, set out the acronym in a 1981 article in *Management Review*. Doran presented it as a practical writing discipline for managers, not a motivational theory. The criteria were later widely re-lettered — 'achievable' and 'attainable', 'relevant' and 'realistic' — a drift Doran himself did not authorise (Doran, 1981).

WHY IT WAS DEVELOPED

Doran observed that most corporate objectives were unactionable because they named a direction rather than a destination: 'improve customer service' cannot be started, finished, or verified. The frame forces the ambiguity out at the point of writing, when it is cheap to fix, rather than at review, when it is not. It complements rather than replaces goal-setting theory, which shows that specific and difficult goals outperform vague ones (Locke and Latham, 2002).

FIGURE 1 · THE FIVE SMART CRITERIA

Specific	who does what
Measurable	evidence of completion
Achievable	fits actual capacity
Relevant	serves the strategy
Time-bound	carries a date

An objective must pass all five tests, not most of them

Figure 1. The five criteria applied in sequence. Failure on any single band returns the objective for rewriting; a goal that satisfies four of five is not a SMART goal (Doran, 1981).

HOW IT IS USED

Draft the objective in one sentence, then test it against each band in turn, rewriting until all five pass. The measurable band is the usual failure point.

WHEN IT IS USED

At the start of any piece of work with a deliverable, and whenever a manager or client states an expectation in directional rather than concrete terms.

BY WHOM IT IS USED

Individuals setting personal development goals, project managers writing objectives, and anyone converting a brief received verbally into a written commitment.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Management by Objectives (Drucker, 1954) established that objectives should be agreed rather than imposed, but offered no test of whether a given objective was well formed. SMART supplies exactly that test in a form that can be applied in under a minute and taught in one session. Its weakness is the mirror of its strength: the frame rewards what is easy to measure, which can quietly displace goals that matter but resist quantification (Ordóñez et al., 2009).

RELATED FRAMEWORKS IN THIS COURSE

The Rule of 3 · [Slibrary 15](#)

Objectives & Key Results · [Slibrary 15](#)

Deliberate Practice Loop · [Slibrary 1](#)

Monday Vision, Daily Wins, Friday Reflection · [Slibrary 15](#)

SOURCES (HARVARD REFERENCING STYLE)

Doran, G.T. (1981) 'There's a S.M.A.R.T. way to write management's goals and objectives', *Management Review*, 70(11), pp. 35–36.

Drucker, P.F. (1954) *The Practice of Management*. New York: Harper & Row.

Locke, E.A. and Latham, G.P. (2002) 'Building a practically useful theory of goal setting and task motivation', *American Psychologist*, 57(9), pp. 705–717.

Ordóñez, L.D., Schweitzer, M.E., Galinsky, A.D. and Bazerman, M.H. (2009) 'Goals gone wild', *Academy of Management Perspectives*, 23(1), pp. 6–16.

Kolb Experiential Learning Cycle

SLIBRARY 01
SHEET 2 / 5

Experience · Reflect · Conceptualise · Test — four stages, one continuous loop

PROFESSIONAL DEFINITION

Kolb's Experiential Learning Cycle models learning as a four-stage loop in which a concrete experience is followed by reflective observation, then by abstract conceptualisation that turns the reflection into a transferable rule, and finally by active experimentation that tests the rule in a new situation, generating the next concrete experience. Learning is held to occur through the transformation of experience, not through its accumulation, so a stage skipped is learning lost (Kolb, 1984).

WHO DEVELOPED IT

David A. Kolb, then at Case Western Reserve University, published the cycle in *Experiential Learning* (1984), synthesising Dewey's pragmatism, Lewin's action research and Piaget's developmental psychology. Peter Honey and Alan Mumford later mapped four learner preferences onto the same four stages, producing the learning-styles instrument widely used in management development (Honey and Mumford, 1986).

WHY IT WAS DEVELOPED

Kolb was addressing the observation that experience alone is a poor teacher: practitioners can repeat a task for years and improve very little. The cycle identifies why. Without the reflective and conceptualising stages, an experience produces a memory rather than a rule, and nothing transfers to the next situation. The model gives educators a structure for designing activity that closes the loop rather than stopping at the doing.

FIGURE 2 · THE FOUR-STAGE LEARNING CYCLE

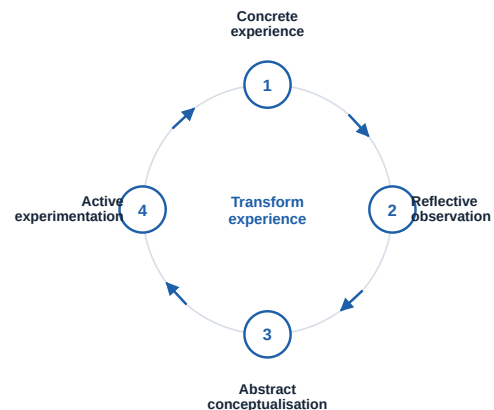


Figure 2. The four stages run continuously; entry is possible at any point, but a full circuit is required before an experience becomes transferable knowledge (Kolb, 1984).

HOW IT IS USED

After any significant task, work the cycle explicitly: what happened, what did I notice, what rule explains it, and what will I deliberately do differently next time.

WHEN IT IS USED

After project milestones, presentations, negotiations and failures — anywhere a repeatable situation is likely to recur.

BY WHOM IT IS USED

Students building reflective practice, educators designing activity-based sessions, and teams running retrospectives and after-action reviews.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Behaviourist training treated learning as stimulus and reinforcement, with the learner passive. Kolb reframed the learner as the agent who transforms experience, which made reflection a designable stage rather than an optional habit. The empirical support for the derived learning-styles instruments is weak and the 'match teaching to style' claim is not supported (Coffield et al., 2004); the cycle itself remains valuable as a design structure.

RELATED FRAMEWORKS IN THIS COURSE

Deliberate Practice Loop · [Slibrary 1](#)Honey & Mumford Learning Styles · [Slibrary 4](#)Ladder of Inference · [Slibrary 6](#)Monday Vision, Daily Wins, Friday Reflection · [Slibrary 15](#)

SOURCES (HARVARD REFERENCING STYLE)

Kolb, D.A. (1984) *Experiential Learning: Experience as the Source of Learning and Development*. Englewood Cliffs, NJ: Prentice Hall.

Honey, P. and Mumford, A. (1986) *The Manual of Learning Styles*. Maidenhead: Peter Honey Publications.

Coffield, F., Moseley, D., Hall, E. and Ecclestone, K. (2004) *Learning Styles and Pedagogy in Post-16 Learning*. London: Learning and Skills Research Centre.

Dewey, J. (1938) *Experience and Education*. New York: Macmillan.

Bloom's Taxonomy (Revised)

SLIBRARY 01
SHEET 3 / 5

Remember · Understand · Apply · Analyse · Evaluate · Create — six levels of cognitive demand

PROFESSIONAL DEFINITION

The revised Bloom's Taxonomy classifies cognitive tasks into six ascending levels of demand: remembering retrieves facts, understanding explains them, applying uses them in a familiar situation, analysing breaks a whole into parts and finds relationships, evaluating judges against criteria, and creating produces something original. The 2001 revision restated the categories as verbs and swapped the top two levels, placing creation above evaluation (Anderson and Krathwohl, 2001).

WHO DEVELOPED IT

The original taxonomy was produced in 1956 by a committee of American college examiners chaired by Benjamin Bloom at the University of Chicago. Lorin Anderson, a former student of Bloom's, led the 2001 revision with David Krathwohl, a co-author of the original, adding a second knowledge dimension running from factual to metacognitive (Bloom et al., 1956; Anderson and Krathwohl, 2001).

WHY IT WAS DEVELOPED

The 1956 committee wanted a shared vocabulary so that examiners at different institutions could agree on what a question actually demanded. The problem it solved is still live: an assessment brief saying 'discuss' may expect anything from summary to original synthesis. At FHEQ Level 7 the expectation sits at the top three levels, and work pitched at remembering and understanding will not meet the criteria however well written.

FIGURE 3 · THE SIX LEVELS OF COGNITIVE DEMAND

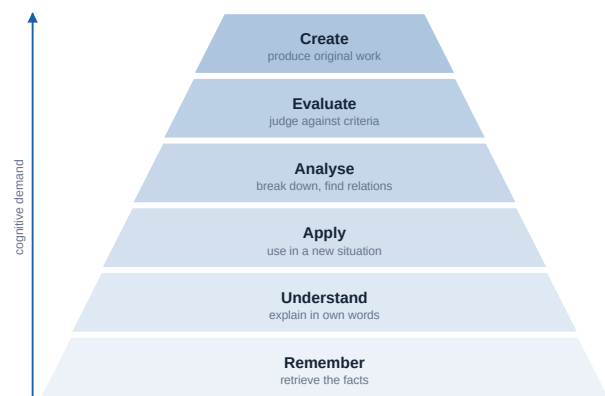


Figure 3. Six levels from retrieval at the base to original production at the apex. Masters level assessment is written for the top three (Anderson and Krathwohl, 2001).

HOW IT IS USED

Read the verb in the assessment brief, locate it on the six levels, and pitch the response there. 'Critically evaluate' is not 'describe' with more words.

WHEN IT IS USED

Before starting any assignment, when writing learning outcomes, and when a piece of work has been marked lower than expected despite being accurate.

BY WHOM IT IS USED

Students calibrating assignment responses, educators writing outcomes and rubrics, and anyone designing training that must move beyond recall.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Before 1956 examiners had no shared language for difficulty, so 'hard' meant more content rather than deeper thinking. The taxonomy separated the two. The 2001 revision matters because it converted nouns into verbs, making the levels usable for writing outcomes, and because promoting creation above evaluation reflected evidence that original synthesis is the more demanding act. The strict hierarchy is contested; the levels overlap in practice.

RELATED FRAMEWORKS IN THIS COURSE

Dreyfus Skill Acquisition Model · [Slibrary 1](#)

MECE Issue Tree · [Slibrary 5](#)

System 1 / System 2 · [Slibrary 6](#)

AI Attribution Ladder · [Slibrary 16](#)

SOURCES (HARVARD REFERENCING STYLE)

Anderson, L.W. and Krathwohl, D.R. (eds.) (2001) *A Taxonomy for Learning, Teaching, and Assessing: A Revision of Bloom's Taxonomy of Educational Objectives*. New York: Longman.

Bloom, B.S., Engelhart, M.D., Furst, E.J., Hill, W.H. and Krathwohl, D.R. (1956) *Taxonomy of Educational Objectives: The Classification of Educational Goals, Handbook I*. New York: David McKay.

Krathwohl, D.R. (2002) 'A revision of Bloom's taxonomy: an overview', *Theory Into Practice*, 41(4), pp. 212–218.

Dreyfus Skill Acquisition Model

SLIBRARY 01
SHEET 4 / 5

Novice · Advanced beginner · Competent · Proficient · Expert — five stages from rules to judgment

PROFESSIONAL DEFINITION

The Dreyfus model describes skill acquisition as a five-stage progression in which reliance on explicit rules is progressively replaced by situational discrimination. Novices apply context-free rules; advanced beginners recognise recurring situational elements; competent performers select a plan and accept responsibility for the outcome; proficient performers read situations holistically while still deliberating on action; and experts perceive situation and appropriate action as one, deliberating only when something is unusual (Dreyfus and Dreyfus, 1980).

WHO DEVELOPED IT

Stuart Dreyfus, an operations researcher, and Hubert Dreyfus, a philosopher of phenomenology, both at the University of California, Berkeley, developed the model in a 1980 report for the US Air Force Office of Scientific Research. Their wider project was a critique of the claim that human expertise could be captured in the rule sets of symbolic artificial intelligence (Dreyfus, 1972; Dreyfus and Dreyfus, 1986).

WHY IT WAS DEVELOPED

The brothers were arguing against the assumption, then dominant in AI research, that expert performance consists of applying more and better rules. Their observation from studying pilots, chess players and drivers was the reverse: rules are scaffolding that experts discard. The model matters for AI-era work because it locates precisely what a rule-following system can and cannot supply — and situational judgment sits at the stages it cannot reach.

FIGURE 4 · THE FIVE STAGES OF SKILL ACQUISITION

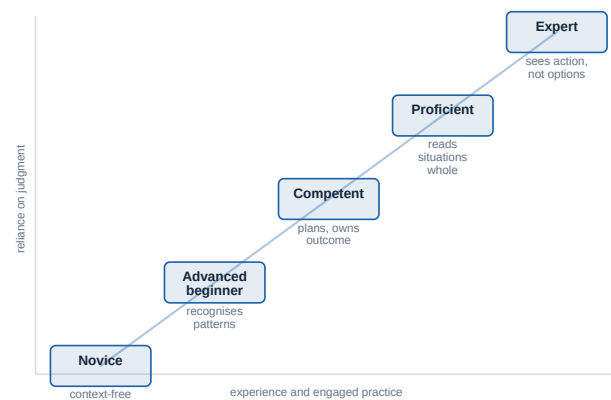


Figure 4. Rule-following falls as situational judgment rises across the five stages. The shift to owning the outcome occurs at stage three (Dreyfus and Dreyfus, 1980).

HOW IT IS USED

Locate your current stage for a named skill, then set development at the next stage rather than at expert. Stage-appropriate practice differs: novices need rules, competents need responsibility.

WHEN IT IS USED

When planning personal development, when designing training, and when deciding which tasks can safely be delegated to a rule-following system.

BY WHOM IT IS USED

Individuals mapping capability honestly, managers assigning stretch work, and curriculum designers sequencing a programme.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Earlier training models treated skill as a single continuum of accumulated knowledge, implying that more instruction always helps. The Dreyfus model shows that instruction which helps a novice actively hinders an expert, because explicit rule-following interrupts fluent performance. That insight reframed both professional education and, later, the debate on which parts of expert work can be automated (Dreyfus and Dreyfus, 1986).

RELATED FRAMEWORKS IN THIS COURSE

[Bloom's Taxonomy \(Revised\) · Slibrary 1](#)
[Deliberate Practice Loop · Slibrary 1](#)
[Dunning–Kruger Curve · Slibrary 6](#)
[Automation–Augmentation Spectrum · Slibrary 16](#)

SOURCES (HARVARD REFERENCING STYLE)

Dreyfus, S.E. and Dreyfus, H.L. (1980) *A Five-Stage Model of the Mental Activities Involved in Directed Skill Acquisition*. Berkeley, CA: University of California, Operations Research Center.

Dreyfus, H.L. and Dreyfus, S.E. (1986) *Mind over Machine: The Power of Human Intuition and Expertise in the Era of the Computer*. New York: Free Press.

Dreyfus, H.L. (1972) *What Computers Can't Do: A Critique of Artificial Reason*. New York: Harper & Row.

Benner, P. (1982) 'From novice to expert', *American Journal of Nursing*, 82(3), pp. 402–407.

Deliberate Practice Loop

SLIBRARY 01
SHEET 5 / 5

Stretch task · Immediate feedback · Targeted repetition — why hours alone do not compound

PROFESSIONAL DEFINITION

Deliberate practice is a structured form of training distinguished from ordinary experience by four conditions: the task sits just beyond current ability, a specific weakness is targeted rather than the whole performance, feedback is immediate and informative, and the learner repeats with correction until the weakness resolves. Ericsson's studies of musicians, athletes and chess players found accumulated deliberate practice, not undifferentiated experience, predicted attained performance (Ericsson, Krampe and Tesch-Römer, 1993).

WHO DEVELOPED IT

K. Anders Ericsson, a Swedish psychologist at Florida State University, established the construct with Ralf Krampe and Clemens Tesch-Römer in a 1993 *Psychological Review* paper studying violinists at the Berlin Music Academy. Ericsson spent the rest of his career objecting to the popular '10,000-hour rule' derived from his work, which he regarded as a misreading (Ericsson and Pool, 2016; Gladwell, 2008).

WHY IT WAS DEVELOPED

The research question was why some performers improve for decades while others plateau after a few years despite continued activity. The answer located in the structure of the practice, not its volume: comfortable repetition automates a skill and then stops improving it. The loop matters at Masters level because professional work supplies experience abundantly and feedback rarely, which is precisely the configuration that produces plateaus.

FIGURE 5 · THE DELIBERATE PRACTICE LOOP

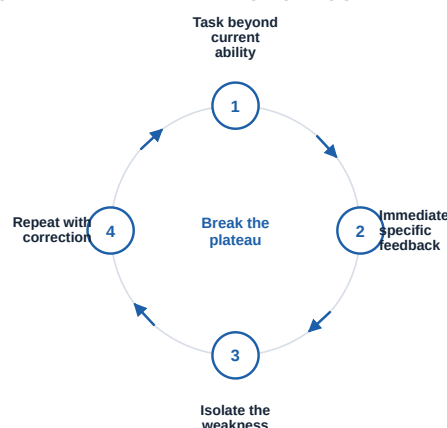


Figure 5. Four conditions in a repeating loop. Remove feedback and the loop degrades into ordinary experience, which produces plateaus rather than gains (Ericsson et al., 1993).

HOW IT IS USED

Name one specific weakness, design a task that isolates it, arrange feedback within minutes rather than weeks, and repeat until the weakness resolves. Then choose the next weakness.

WHEN IT IS USED

Whenever performance has plateaued despite continued activity, and when preparing for a high-stakes performance such as a pitch, viva or negotiation.

BY WHOM IT IS USED

Individuals building a professional skill, coaches and mentors designing practice, and anyone who has done something many times without getting measurably better at it.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The prevailing view attributed elite performance to innate talent, which made training design largely irrelevant. Ericsson's contribution was to show that the structure of practice accounts for far more variance than had been assumed, making expertise a design problem. Later meta-analysis found deliberate practice explains a substantial but smaller share of variance than the original claim, and less in less-structured domains (Macnamara et al., 2014).

RELATED FRAMEWORKS IN THIS COURSE

Kolb Experiential Learning Cycle · [Slibrary 1](#)

Dreyfus Skill Acquisition Model · [Slibrary 1](#)

Growth vs. Fixed Mindset Fork · [Slibrary 4](#)

SMART Goal Frame · [Slibrary 1](#)

SOURCES (HARVARD REFERENCING STYLE)

Ericsson, K.A., Krampe, R.T. and Tesch-Römer, C. (1993) 'The role of deliberate practice in the acquisition of expert performance', *Psychological Review*, 100(3), pp. 363–406.

Ericsson, K.A. and Pool, R. (2016) *Peak: Secrets from the New Science of Expertise*. Boston, MA: Houghton Mifflin Harcourt.

Macnamara, B.N., Hambrick, D.Z. and Oswald, F.L. (2014) 'Deliberate practice and performance', *Psychological Science*, 25(8), pp. 1608–1618.

Gladwell, M. (2008) *Outliers: The Story of Success*. New York: Little, Brown.

DigComp 2.2 Competence Wheel

SLIBRARY 02
SHEET 1 / 5

Five competence areas, twenty-one competences, eight proficiency levels

PROFESSIONAL DEFINITION

DigComp is the European Commission's reference framework for citizens' digital competence. It organises capability into five areas — information and data literacy, communication and collaboration, digital content creation, safety, and problem solving — subdivided into twenty-one competences, each described at eight proficiency levels from foundation to highly specialised. The 2.2 update added several hundred examples covering interaction with AI systems (Vuorikari, Kluzer and Punie, 2022).

WHO DEVELOPED IT

DigComp is maintained by the European Commission's Joint Research Centre. Version 1.0 appeared in 2013 under Anusca Ferrari; version 2.2 was published in 2022 by Riina Vuorikari, Stefano Kluzer and Yves Punie. It underpins Europass, national digital-skills strategies and the Digital Education Action Plan, giving it regulatory as well as pedagogical weight (European Commission, 2021).

WHY IT WAS DEVELOPED

Employers and educators lacked a shared vocabulary for digital capability, so job adverts asked for 'IT skills' and CVs answered with software names. DigComp separates competence from tools, which matters because tools turn over far faster than the underlying capability. The 2.2 update was driven specifically by generative AI: the Commission judged that critical evaluation of AI-produced content had become a citizen-level competence.

FIGURE 6 · THE FIVE COMPETENCE AREAS



Figure 6. The five areas of DigComp 2.2. Each subdivides into competences described at eight proficiency levels, from foundation to highly specialised (Vuorikari et al., 2022).

HOW IT IS USED

Self-assess against each of the five areas, identify the weakest, and set development against the proficiency descriptors rather than against a software product.

WHEN IT IS USED

At the start of a programme or role, when writing a CV or development plan, and when an organisation is auditing capability rather than tool licences.

BY WHOM IT IS USED

Students auditing their own capability, employers writing role specifications, and educators aligning curricula with European standards.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Earlier digital-skills frameworks were built around named applications, so they aged with the software and encouraged training in menus rather than judgment. DigComp is tool-agnostic and proficiency-graded, so the same framework describes a first-year student and a specialist. Its limitation is breadth: as a citizen-level framework it does not reach the depth a specialist role requires.

RELATED FRAMEWORKS IN THIS COURSE

CRAAP Source Test · [Slibrary 2](#)

Co-Intelligence Working Rules · [Slibrary 2](#)

Personal SWOT Matrix · [Slibrary 3](#)

AI Attribution Ladder · [Slibrary 16](#)

SOURCES (HARVARD REFERENCING STYLE)

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CRAAP Source Test

SLIBRARY 02
SHEET 2 / 5

Currency · Relevance · Authority · Accuracy · Purpose — five questions before you cite

PROFESSIONAL DEFINITION

The CRAAP test is a five-criterion screen applied to any source before it is admitted into an argument. Currency asks when the information was produced and whether that still holds; relevance asks whether it addresses the actual question; authority asks who produced it and on what standing; accuracy asks what evidence supports it and whether it is verifiable; and purpose asks why it was produced and who benefits if the reader believes it (Blakeslee, 2004).

WHO DEVELOPED IT

Sarah Blakeslee and colleagues at the Meriam Library, California State University, Chico, devised the test in 2004 as a teaching device for undergraduate information literacy. It has since been adopted very widely in higher education, and equally widely criticised as a checklist that can be completed without genuine evaluation (Caulfield, 2019).

WHY IT WAS DEVELOPED

Blakeslee's students could find sources easily and evaluate them badly, treating retrievability as a proxy for reliability. The test makes the evaluation explicit and memorable. Generative AI sharpens the problem rather than changing it: model output has no author, no publication date and no evidence trail, so it fails authority and accuracy by construction, whatever its fluency suggests.

FIGURE 7 · THE FIVE-CRITERION SOURCE SCREEN

Currency	when was it produced?
Relevance	does it answer this question?
Authority	who says it, on what standing?
Accuracy	what evidence, verifiable?
Purpose	who benefits if I believe it?

Five questions, asked before the source enters your notes

Figure 7. Sources pass through five successive questions. Generative output fails authority and accuracy by construction, since it carries neither author nor evidence (Blakeslee, 2004).

HOW IT IS USED

Apply the five questions to each source before it enters your notes, not after it has entered your draft. Record the answer to 'purpose' — it is the one most often skipped.

WHEN IT IS USED

During any literature search, when a striking statistic is encountered, and whenever a claim arrives without a traceable origin.

BY WHOM IT IS USED

Students building a reference list, researchers screening literature, and professionals verifying a claim before repeating it to a client.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Earlier information-literacy teaching relied on source hierarchies — peer-reviewed above trade press above web — which broke down as publishing formats blurred. CRAAP evaluates the individual item rather than its container. Caulfield's SIFT method argues the checklist encourages surface compliance and that lateral reading, investigating the source elsewhere, is more effective; the two are best used together (Caulfield, 2019).

RELATED FRAMEWORKS IN THIS COURSE

DigComp 2.2 Competence Wheel · [Slibrary 2](#)Automation Bias Curve · [Slibrary 2](#)Confirmation Bias Loop · [Slibrary 6](#)Toulmin Argument Model · [Slibrary 9](#)

SOURCES (HARVARD REFERENCING STYLE)

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Technology Acceptance Model

SLIBRARY 02
SHEET 3 / 5

Perceived usefulness × perceived ease of use → intention → actual use

PROFESSIONAL DEFINITION

The Technology Acceptance Model predicts whether an information technology will actually be used from two beliefs held by the prospective user: perceived usefulness, the degree to which the technology is expected to improve performance, and perceived ease of use, the degree to which it is expected to be effortless. These beliefs shape attitude and behavioural intention, which in turn predicts actual system use. Perceived usefulness is consistently the stronger predictor (Davis, 1989).

WHO DEVELOPED IT

Fred D. Davis developed the model in his 1985 MIT doctoral thesis, publishing the core paper in *MIS Quarterly* in 1989. It adapts Fishbein and Ajzen's theory of reasoned action to the technology setting. Venkatesh and Davis later extended it as TAM2, and Venkatesh and colleagues folded it into the broader UTAUT model (Venkatesh et al., 2003).

WHY IT WAS DEVELOPED

Organisations were buying systems that employees then did not use, and the failure was being diagnosed as resistance or poor training. Davis showed the determinants were beliefs about the technology, formed early and often accurately. The model matters for AI adoption because it predicts the observed pattern: tools perceived as useful get adopted despite awkward interfaces, while easy tools with unclear benefit are abandoned after a fortnight.

FIGURE 8 · FROM BELIEF TO ACTUAL USE



Perceived ease of use feeds usefulness as well as attitude

Figure 8. Two beliefs drive attitude, intention and finally use. Perceived usefulness has the stronger path; ease of use acts partly through it (Davis, 1989).

HOW IT IS USED

Before rolling out a tool, test both beliefs with intended users. If usefulness is unclear, training will not rescue adoption — the benefit case has to be made first.

WHEN IT IS USED

Before introducing any new tool to a team, and when diagnosing why an available system is going unused.

BY WHOM IT IS USED

Managers introducing systems, educators mandating platforms, and individuals deciding whether a new tool warrants the switching cost.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Diffusion of innovations (Rogers, 1962) explained adoption at population level through adopter categories, but could not predict whether a given individual would use a given system. TAM is individual-level, measurable before deployment, and specific enough to act on. Its limitation is that it models voluntary use and explains little in mandatory settings, which UTAUT later addressed by adding social influence and facilitating conditions.

RELATED FRAMEWORKS IN THIS COURSE

Co-Intelligence Working Rules · [Slibrary 2](#)

Automation Bias Curve · [Slibrary 2](#)

Growth vs. Fixed Mindset Fork · [Slibrary 4](#)

Prompt Anatomy · [Slibrary 16](#)

SOURCES (HARVARD REFERENCING STYLE)

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Automation Bias Curve

SLIBRARY 02
SHEET 4 / 5

Use · Misuse · Disuse · Abuse — how trust in a reliable system stops verification

PROFESSIONAL DEFINITION

Automation bias is the tendency of operators to over-rely on an automated aid, producing two characteristic failures: errors of commission, where an incorrect automated recommendation is followed despite contradicting evidence, and errors of omission, where a problem the automation did not flag goes undetected. Parasuraman and Riley frame the wider phenomenon as four states — appropriate use, misuse through over-trust, disuse through under-trust, and abuse through poor design (Parasuraman and Riley, 1997).

WHO DEVELOPED IT

Raja Parasuraman, a human-factors psychologist at George Mason University, and Victor Riley published the framing in *Human Factors* in 1997, drawing on aviation and process-control research. Linda Skitka and colleagues demonstrated both error types experimentally in flight simulation, showing that verification decays even when operators are told the aid is imperfect (Skitka, Mosier and Burdick, 1999).

WHY IT WAS DEVELOPED

Aviation and medicine had accumulated incidents where highly trained operators followed automation into avoidable accidents. The research question was why competence did not protect them. The answer is that reliability itself is the hazard: a system correct ninety-five per cent of the time trains the operator to stop checking, and the remaining five per cent then arrives unexamined. Generative AI reproduces this configuration precisely.

FIGURE 9 · HOW RELIANCE OUTGROWS VERIFICATION

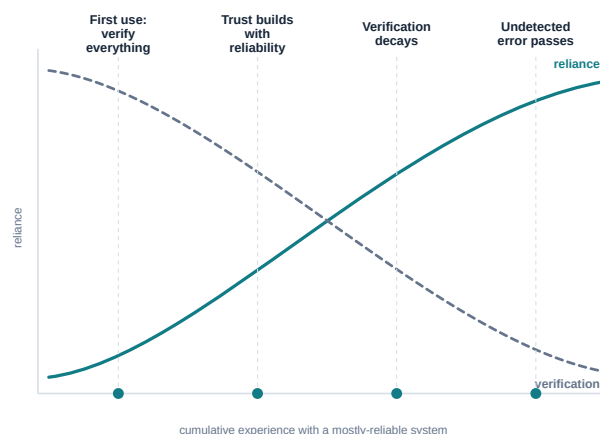


Figure 9. Reliance rises with experienced reliability while verification effort falls. The gap between them is where undetected errors pass (Parasuraman and Riley, 1997).

HOW IT IS USED

Set a verification rule in advance and tie it to consequence, not to confidence: decide which outputs must be checked against a source regardless of how right the system has been lately.

WHEN IT IS USED

Whenever a tool has been reliable for a sustained period, and before any output reaches a client, an examiner or a publication.

BY WHOM IT IS USED

Anyone working with AI assistance, professionals in regulated settings, and managers designing review processes around automated tools.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Earlier human-factors work treated automation failure as a technical reliability problem to be engineered out. Parasuraman and Riley reframed it as a calibration problem in the human operator, which cannot be engineered out and must be designed around. That shift produced the human-in-the-loop design patterns now standard in high-stakes AI deployment (Amershi et al., 2019).

RELATED FRAMEWORKS IN THIS COURSE

Co-Intelligence Working Rules · [Slibrary 2](#)Delegate–Verify–Own Loop · [Slibrary 16](#)Dunning–Kruger Curve · [Slibrary 6](#)[CRAAP Source Test](#) · [Slibrary 2](#)

SOURCES (HARVARD REFERENCING STYLE)

Parasuraman, R. and Riley, V. (1997) 'Humans and automation: use, misuse, disuse, abuse', *Human Factors*, 39(2), pp. 230–253.

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Co-Intelligence Working Rules

SLIBRARY 02
SHEET 5 / 5

Invite it to the table · Be the human in the loop · Tell it who to be · Assume it improves

PROFESSIONAL DEFINITION

Mollick's four rules are operating principles for working with general-purpose AI. First, invite AI to every task, since capability cannot be predicted in advance and must be discovered empirically. Second, remain the human in the loop, retaining judgment and accountability for what the system produces. Third, give the model a specific persona and context, because unspecified output defaults to generic. Fourth, treat the current model as the weakest you will ever use, and design accordingly (Mollick, 2024).

WHO DEVELOPED IT

Ethan Mollick, associate professor of management at the Wharton School, University of Pennsylvania, set out the rules in *Co-Intelligence* (2024). They draw on his field experiments with organisations, including the Boston Consulting Group study of consultant performance with and without GPT-4 that produced the jagged-frontier finding (Dell'Acqua et al., 2023).

WHY IT WAS DEVELOPED

Organisations were oscillating between banning AI outright and adopting it uncritically, because neither position required understanding where the tools actually work. Mollick's rules replace policy with practice: capability is jagged and unmapped, so the only reliable method is systematic trial with a human retaining judgment. The fourth rule addresses planning — designing workflows around today's limitations tends to age badly.

FIGURE 10 · THE FOUR WORKING RULES

- 1 Always invite AI to the table**
capability is discovered, not predicted
- 2 Be the human in the loop**
judgment and accountability stay with you
- 3 Treat it like a person**
but tell it precisely which person
- 4 Assume this is the worst AI you will ever use**
design for the next model, not this one

Four operating rules, applied together rather than selectively

Figure 10. Four rules for practical use. Rule two is the constraint on rule one: broad experimentation is safe only while judgment is retained (Mollick, 2024).

HOW IT IS USED

Add AI to a task you already know how to do, so you can evaluate the output. Specify the role and context. Verify before use. Re-test tasks that previously failed.

WHEN IT IS USED

When deciding whether a task is worth delegating, when onboarding a team to new tools, and periodically re-testing tasks that a model could not previously handle.

BY WHOM IT IS USED

Professionals building an AI workflow, managers setting team norms, and students establishing practice that will survive academic-integrity scrutiny.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Prior guidance was either prohibition, which forfeits the capability, or vendor enthusiasm, which ignores the failure modes. Mollick's contribution is a middle position grounded in field experiment rather than assertion: capability is uneven, the boundary is invisible from outside, so empirical trial under human judgment is the only reliable method. The rules are heuristics, not validated instruments, and should be read as such.

RELATED FRAMEWORKS IN THIS COURSE

The Jagged Frontier · [Slibrary 16](#)Delegate–Verify–Own Loop · [Slibrary 16](#)Automation Bias Curve · [Slibrary 2](#)Prompt Anatomy · [Slibrary 16](#)

SOURCES (HARVARD REFERENCING STYLE)

Mollick, E. (2024) *Co-Intelligence: Living and Working with AI*. New York: Portfolio/Penguin.

Dell'Acqua, F. et al. (2023) 'Navigating the jagged technological frontier: field experimental evidence of the effects of AI on knowledge worker productivity and quality', *Harvard Business School Working Paper 24-013*.

Noy, S. and Zhang, W. (2023) 'Experimental evidence on the productive effects of generative artificial intelligence', *Science*, 381(6654), pp. 187–192.

Personal SWOT Matrix

SLIBRARY 03
SHEET 1 / 5

Strengths · Weaknesses · Opportunities · Threats — two internal quadrants, two external

PROFESSIONAL DEFINITION

The Personal SWOT Matrix applies the strategic-planning grid to an individual rather than a firm. Strengths and weaknesses are internal and largely within the person's control — capability, credentials, temperament, network. Opportunities and threats are external and are not — market movements, technological substitution, organisational change. The discipline of the tool is the internal/external boundary: an item placed in the wrong half produces a plan that cannot be acted on (Weihrich, 1982).

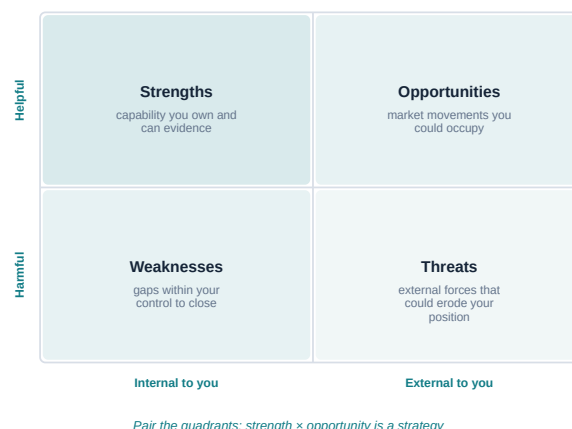
WHO DEVELOPED IT

The SWOT grid emerged from Harvard Business School policy teaching in the 1960s, commonly associated with Kenneth Andrews and Albert Humphrey's Stanford research programme. Heinz Weihrich formalised the paired analysis as the TOWS Matrix in 1982, adding the crucial step of combining quadrants into strategies rather than merely listing entries (Weihrich, 1982; Andrews, 1971).

WHY IT WAS DEVELOPED

Individuals asked to plan a career typically produce either an inventory of skills with no market reference, or a list of desired jobs with no capability reference. SWOT forces both onto one page and, in Weihrich's extension, forces them into contact: a strength matched to an opportunity is a strategy, while a weakness exposed to a threat is a risk requiring mitigation now rather than acknowledgement later.

FIGURE 11 · THE FOUR QUADRANTS OF PERSONAL SWOT



Pair the quadrants: strength × opportunity is a strategy

Figure 11. Four quadrants divided by two axes — origin and effect. Weihrich's contribution was to pair them rather than list them, converting description into strategy (Weihrich, 1982).

HOW IT IS USED

Populate all four quadrants with evidence rather than adjectives, then pair them: strength × opportunity, weakness × threat. Each pairing produces one action.

WHEN IT IS USED

Before a career decision, at the start of a programme, and whenever an external change — a restructure, a new technology — alters the market for your capability.

BY WHOM IT IS USED

Individuals planning career moves, mentors structuring a development conversation, and teams assessing their collective position before a strategy cycle.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Unstructured self-assessment mixes what a person is with what the world is doing, so the resulting plan addresses neither. SWOT separates the two and the TOWS extension recombines them deliberately. The tool's well-documented weakness is that it invites unweighted, unevidenced lists; it produces value only when each entry carries evidence and the quadrants are paired (Hill and Westbrook, 1997).

RELATED FRAMEWORKS IN THIS COURSE

[Johari Window · Slibrary 3](#)
[Personal Value Proposition Canvas · Slibrary 7](#)
[DigComp 2.2 Competence Wheel · Slibrary 2](#)
[Power–Interest Grid · Slibrary 10](#)

SOURCES (HARVARD REFERENCING STYLE)

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Johari Window

SLIBRARY 03
SHEET 2 / 5

Open · Blind · Hidden · Unknown — four panes formed by what you know and what others know

PROFESSIONAL DEFINITION

The Johari Window maps self-knowledge against the knowledge others hold about a person, producing four panes. The open area is known to self and others; the blind area is visible to others but not to the self; the hidden area is known to the self and withheld; and the unknown area is available to neither. The two productive moves are disclosure, which enlarges the open area at the expense of the hidden, and feedback, which enlarges it at the expense of the blind (Luft and Ingham, 1955).

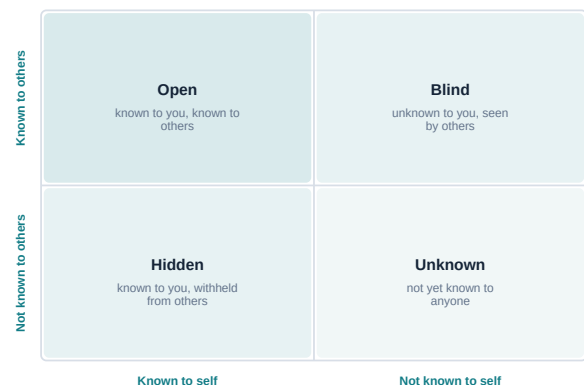
WHO DEVELOPED IT

Joseph Luft and Harrington Ingham devised the model in 1955 at a group-development laboratory at the University of California, Los Angeles; the name is a contraction of their forenames. It became a standard instrument in the T-group and sensitivity-training movement and remains embedded in management and clinical education (Luft, 1969).

WHY IT WAS DEVELOPED

Luft and Ingham were working on why interpersonal feedback in groups so often failed to change behaviour. Their answer was structural rather than motivational: people cannot act on information they do not have, and the blind pane is by definition invisible from inside. The window makes the asymmetry visible and, in doing so, identifies feedback as the only mechanism that can reach the blind area at all.

FIGURE 12 · THE FOUR PANES OF THE WINDOW



Disclosure widens the open pane; feedback shrinks the blind

Figure 12. Four panes formed by two knowledge axes. Only feedback from others can reduce the blind pane, which is why it cannot be closed by reflection alone (Luft and Ingham, 1955).

HOW IT IS USED

Name one behaviour you suspect sits in your blind pane, then ask two people who see you work for a specific example rather than a general impression.

WHEN IT IS USED

At the start of a team, after a performance review, and when the same criticism has arrived from more than one independent source.

BY WHOM IT IS USED

Individuals seeking calibrated self-knowledge, facilitators building psychological safety in new teams, and managers structuring developmental feedback.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Earlier trait models described a person as a fixed set of attributes, implicitly assuming the person was the authority on their own characteristics. The Johari Window makes self-knowledge relational and therefore incomplete by construction. Its limitation is that it is a heuristic rather than a measured instrument: the panes have no scale, and their relative sizes are illustrative rather than diagnostic.

RELATED FRAMEWORKS IN THIS COURSE

Personal SWOT Matrix · [Slibrary 3](#)

Ladder of Inference · [Slibrary 6](#)

Dunning–Kruger Curve · [Slibrary 6](#)

Growth vs. Fixed Mindset Fork · [Slibrary 4](#)

SOURCES (HARVARD REFERENCING STYLE)

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Ikigai Purpose Venn

SLIBRARY 03
SHEET 3 / 5

Love · Skill · Need · Payment — four circles, one overlap worth working toward

PROFESSIONAL DEFINITION

The Ikigai Purpose Venn locates sustainable working purpose at the intersection of four conditions: what you love, what you are good at, what the world needs, and what you can be paid for. Pairwise overlaps yield passion, mission, vocation and profession; the centre, where all four hold, is presented as ikigai. The diagram is a Western synthesis and should be distinguished from the Japanese concept it borrows its name from (García and Miralles, 2016; Kamiya, 1966).

WHO DEVELOPED IT

The four-circle diagram derives from a purpose model published by Andrés Zuzunaga in 2011, relabelled 'ikigai' by Marc Winn in a 2014 blog post, and popularised worldwide by Héctor García and Francesc Miralles in *Ikigai* (2016). The underlying Japanese concept was analysed far earlier by the psychiatrist Mieko Kamiya, whose account carries no requirement of income or vocation at all (Kamiya, 1966).

WHY IT WAS DEVELOPED

Career advice tends to optimise a single variable — follow your passion, or follow the money — and each produces a predictable failure: unpaid enthusiasm or well-paid disengagement. The diagram's value is that it makes the missing circle diagnosable. Someone describing burnout in a well-paid, competent role is usually reporting an absence in 'what the world needs' or 'what you love', and can name which.

FIGURE 13 · THE FOUR CONDITIONS AND THEIR OVERLAP

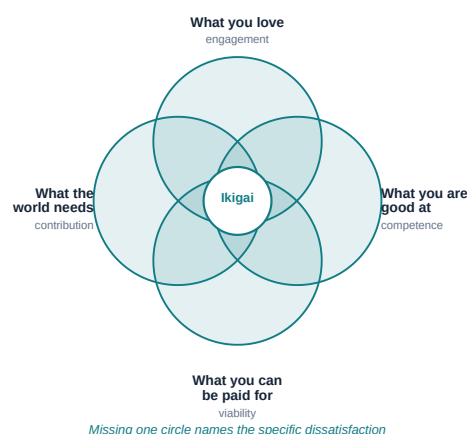


Figure 13. Four conditions overlapping at a single centre. Diagnostic value lies in the missing circle, not the centre: name which one is absent (García and Miralles, 2016).

HOW IT IS USED

Fill each circle with concrete evidence rather than aspiration, then identify which single circle is thinnest. That circle is the development question for the year ahead.

WHEN IT IS USED

When choosing a specialisation, when a role is objectively good but subjectively wrong, and when weighing a move that trades income against meaning.

BY WHOM IT IS USED

Students choosing a career direction, professionals considering a change, and coaches structuring a purpose conversation.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Single-factor career advice cannot explain why competent, well-paid people leave. The four-circle model makes the trade-off structure explicit and gives dissatisfaction a vocabulary. Two caveats belong with it: the diagram is not the Japanese concept, which does not require paid work; and it has no empirical validation, so it functions as a conversation structure rather than as an assessment (Kamiya, 1966; Kumano, 2018).

RELATED FRAMEWORKS IN THIS COURSE

Firm Foundation: Vision, Mission, Values · [Slibrary 3](#)Personal SWOT Matrix · [Slibrary 3](#)Golden Circle · [Slibrary 7](#)Personal Value Proposition Canvas · [Slibrary 7](#)

SOURCES (HARVARD REFERENCING STYLE)

García, H. and Miralles, F. (2016) *Ikigai: The Japanese Secret to a Long and Happy Life*. London: Hutchinson.

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Firm Foundation: Vision, Mission, Values

SLIBRARY 03
SHEET 4 / 5

Where you are going · what you do · what you will not trade — three pillars under every plan

PROFESSIONAL DEFINITION

Meier's Firm Foundation sets three durable reference points beneath short-cycle planning. Vision states the future condition being worked toward, in the present tense and specific enough to be recognised if reached. Mission states the work undertaken now to move toward it. Values state the constraints that hold when the two conflict with an opportunity. Together they supply the criteria by which weekly priorities are chosen rather than merely listed (Meier, 2010).

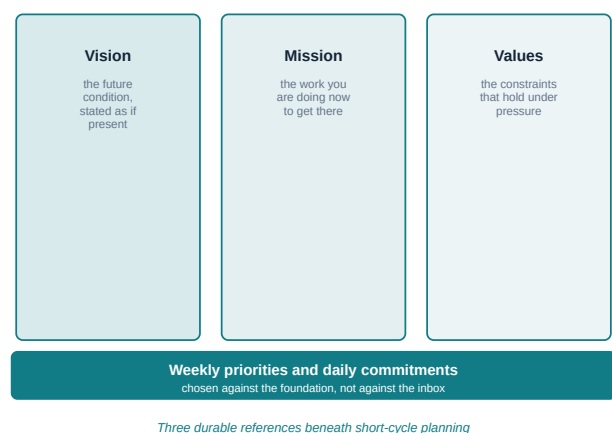
WHO DEVELOPED IT

J.D. Meier, a longstanding programme manager at Microsoft, set out the Firm Foundation as the grounding layer of *Getting Results the Agile Way* (2010) and returned to it as a thinking structure in *Thumbnail Thinking* (2025). The three-part formulation itself descends from strategic-planning practice, notably Collins and Porras's work on core ideology and envisioned future (Collins and Porras, 1996).

WHY IT WAS DEVELOPED

Meier's observation from managing multi-year programmes was that productivity systems fail not at execution but at selection: a person with a good weekly method and no stable criteria simply executes the wrong things efficiently. The foundation supplies the criteria. Values are included explicitly because the hard cases are not choices between good and bad options but between two attractive ones.

FIGURE 14 · THE THREE PILLARS AND WHAT THEY CARRY



Three durable references beneath short-cycle planning

Figure 14. Three pillars carrying the short-cycle layer above them. Priorities selected without them are efficient execution of unexamined work (Meier, 2010).

HOW IT IS USED

Write each in one sentence, then test the coming week's commitments against them. Anything that serves none of the three is a candidate for removal or renegotiation.

WHEN IT IS USED

At the start of a programme or year, when a tempting opportunity appears off-track, and whenever weekly planning has become reactive.

BY WHOM IT IS USED

Individuals building a personal operating system, managers setting team direction, and anyone whose calendar no longer reflects their stated intentions.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Conventional productivity systems optimise throughput and are silent on selection, which makes them efficient at whatever arrives. Meier's contribution is to put the selection criteria one layer below the method, where they are stable across weeks. The concepts derive from corporate strategy rather than from psychological research, and their evidence base is practitioner experience rather than trial.

RELATED FRAMEWORKS IN THIS COURSE

Ikigai Purpose Venn · [Slibrary 3](#)

Monday Vision, Daily Wins, Friday Reflection · [Slibrary 15](#)

The Rule of 3 · [Slibrary 15](#)

Golden Circle · [Slibrary 7](#)

SOURCES (HARVARD REFERENCING STYLE)

Meier, J.D. (2010) *Getting Results the Agile Way: A Personal Results System for Work and Life*. Bellevue, WA: Innovation Playhouse.

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Personal Energy Model

SLIBRARY 03
SHEET 5 / 5

Physical · Emotional · Mental · Spiritual — capacity is renewed, not extended

PROFESSIONAL DEFINITION

The Personal Energy Model treats performance as a function of managed energy rather than allocated time. Four capacities are distinguished: physical energy, which sets the quantity available; emotional energy, which sets its quality; mental energy, which sets its focus; and spiritual energy, meaning alignment with purpose, which sets its force. Each is expended and must be deliberately renewed through oscillation between exertion and recovery (Loehr and Schwartz, 2003; Meier, 2010).

WHO DEVELOPED IT

Jim Loehr and Tony Schwartz developed the four-capacity model at LGE Performance Systems, generalising from athletic-performance work to executives in *The Power of Full Engagement* (2003). J.D. Meier incorporated energy management as a first-class dimension of personal results alongside time and technique, arguing that hours are a poor proxy for capacity (Meier, 2010; Schwartz and McCarthy, 2007).

WHY IT WAS DEVELOPED

Time-management systems assume that hours are interchangeable, which anyone who has attempted analytical work at the end of a long day knows to be false. The model was developed because executives were already at their time limit and could improve only by changing the quality of the hours they had. Recovery is treated as productive rather than residual — the central reversal of the model.

FIGURE 15 · THE FOUR ENERGY CAPACITIES

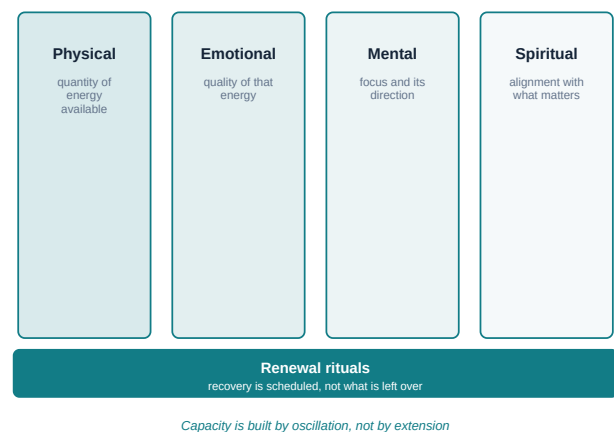


Figure 15. Four capacities resting on deliberate renewal. Each is expended by use and restored only by scheduled recovery (Loehr and Schwartz, 2003).

HOW IT IS USED

Identify which capacity is depleted rather than assuming the answer is time, then attach one specific renewal ritual to it and schedule your most demanding work at your peak physical hours.

WHEN IT IS USED

When output falls despite unchanged hours, when preparing for a sustained high-demand period, and when a to-do system is being followed but is not producing results.

BY WHOM IT IS USED

Individuals designing a working week, managers setting realistic team load, and anyone whose difficulty is depletion rather than disorganisation.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Classical time management treats the person as a container of interchangeable hours and responds to overload by seeking more of them. The energy model reframes capacity as renewable and makes recovery a designed input, which changes the intervention from scheduling to ritual design. The supporting evidence is largely practitioner case data rather than controlled trial, and the spiritual capacity in particular resists measurement.

RELATED FRAMEWORKS IN THIS COURSE

Eisenhower Matrix · [Slibrary 15](#)The Rule of 3 · [Slibrary 15](#)Firm Foundation: Vision, Mission, Values · [Slibrary 3](#)[Deliberate Practice Loop · Slibrary 1](#)

SOURCES (HARVARD REFERENCING STYLE)

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MBTI Four Dichotomies

SLIBRARY 04
SHEET 1 / 5

E–I · S–N · T–F · J–P — four preference pairs, sixteen reported types

PROFESSIONAL DEFINITION

The Myers–Briggs Type Indicator sorts self-reported preference on four dichotomies: where attention is directed (extraversion–introversion), how information is taken in (sensing–intuition), how decisions are reached (thinking–feeling), and how the outer world is approached (judging–perceiving). Each respondent is assigned one pole per dichotomy, producing sixteen four-letter types. The instrument reports preference, not competence (Myers and Myers, 1980).

WHO DEVELOPED IT

Katharine Cook Briggs and her daughter Isabel Briggs Myers, neither professionally trained in psychology, constructed the indicator from the 1940s onward as an operationalisation of Carl Jung's *Psychological Types* (1921). It is published commercially and is among the most widely administered instruments in organisational settings, with well-documented psychometric objections in the peer-reviewed literature (Pittenger, 2005).

WHY IT WAS DEVELOPED

Briggs and Myers wanted to help people entering wartime employment find roles suited to their dispositions, and to make Jung's typology usable outside clinical practice. The framework's durable contribution is a shared, non-pejorative vocabulary for difference: it lets a team say that a colleague needs processing time before speaking without either party treating this as a deficiency.

FIGURE 16 · THE FOUR PREFERENCE DICHOTOMIES



Figure 16. Four bipolar scales. The instrument forces a pole on each, although the underlying scores are continuous and cluster near the midpoint (Pittenger, 2005).

HOW IT IS USED

Use the vocabulary to describe working preferences within a team, and treat any reported type as a hypothesis to check against behaviour rather than as a finding.

WHEN IT IS USED

During team formation and communication workshops, where a shared language for difference is useful — never for selection, promotion or role assignment.

BY WHOM IT IS USED

Facilitators running team-development sessions, and individuals reflecting on working preferences with the instrument's limitations understood.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Jung's typology was a theoretical scheme with no measurement procedure; Briggs and Myers made it administrable, which is why it spread. That is also where the difficulty lies: the underlying dimensions are continuous, so dichotomising them at the midpoint creates unstable categories, and test–retest studies report substantial numbers of respondents changing type within weeks. Predictive validity for job performance is weak (Pittenger, 2005; Boyle, 1995).

RELATED FRAMEWORKS IN THIS COURSE

DISC Behavioural Quadrant · [Slibrary 4](#)Big Five (OCEAN) · [Slibrary 4](#)Honey & Mumford Learning Styles · [Slibrary 4](#)[Johari Window](#) · [Slibrary 3](#)

SOURCES (HARVARD REFERENCING STYLE)

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DISC Behavioural Quadrant

SLIBRARY 04
SHEET 2 / 5

Dominance · Influence · Steadiness · Conscientiousness — observable behaviour, not personality

PROFESSIONAL DEFINITION

DISC classifies observable workplace behaviour on two axes: whether a person orients primarily to tasks or to people, and whether they act at a fast or a measured pace. The four resulting quadrants are dominance, direct and results-driven; influence, expressive and persuasive; steadiness, cooperative and consistent; and conscientiousness, precise and standards-driven. DISC describes behavioural style in context, not underlying personality (Marston, 1928).

WHO DEVELOPED IT

William Moulton Marston, a Harvard-trained psychologist, described the four factors in *Emotions of Normal People* (1928) without producing an instrument. Walter Clarke built the first assessment from Marston's categories in 1956, and the profile is now published by several competing commercial providers whose versions differ in wording and scoring (Marston, 1928).

WHY IT WAS DEVELOPED

Marston was interested in the behaviour of ordinary people rather than clinical populations, which was unusual for the period. The practical problem the profile addresses is friction that is stylistic rather than substantive: a fast, task-focused colleague and a measured, people-focused one frequently agree on the objective while each experiencing the other as the obstacle. Naming the axes defuses that misattribution.

FIGURE 17 · THE FOUR BEHAVIOURAL QUADRANTS

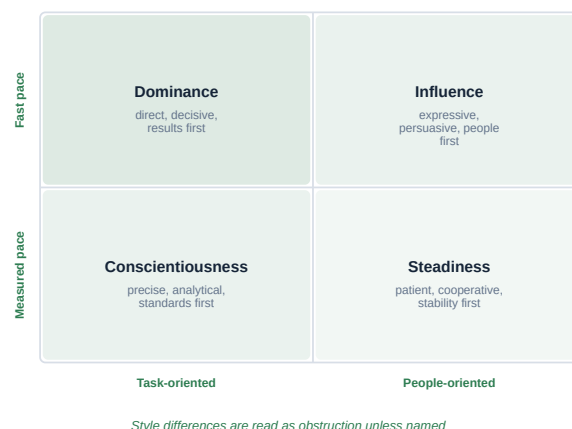


Figure 17. Four quadrants on orientation and pace. Style is situational: the same person may show a different quadrant under pressure than in routine work (Marston, 1928).

HOW IT IS USED

Identify your dominant quadrant, then flex deliberately toward the other person's: give a conscientiousness colleague data and time, give a dominance colleague the conclusion first.

WHEN IT IS USED

In team formation, before a difficult conversation with a colleague whose style differs, and when recurring friction is about how rather than what.

BY WHOM IT IS USED

Managers building mixed teams, salespeople adapting approach to a buyer, and individuals preparing for a stakeholder conversation.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Trait personality models describe stable dispositions but say little about what to do differently on Monday. DISC restricts itself to observable behaviour in a work context and yields an immediate action — flex your style toward the other quadrant. The trade-off is rigour: commercial DISC instruments are not built on a common validated model, peer-reviewed evidence is thin, and the four categories compress far more variation than they reveal.

RELATED FRAMEWORKS IN THIS COURSE

[MBTI Four Dichotomies · Slibrary 4](#)
[Big Five \(OCEAN\) · Slibrary 4](#)
[Dual Concerns Conflict Modes · Slibrary 13](#)
[The Culture Map \(Eight Scales\) · Slibrary 10](#)

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Big Five (OCEAN)

SLIBRARY 04
SHEET 3 / 5

Openness · Conscientiousness · Extraversion · Agreeableness · Neuroticism — five continuous traits

PROFESSIONAL DEFINITION

The Five Factor Model describes personality as five continuous dimensions rather than discrete types: openness to experience, conscientiousness, extraversion, agreeableness, and neuroticism or emotional stability. Each is a spectrum on which an individual is located relative to a population, not a category to which they belong. The factors emerged empirically from analyses of trait vocabulary and have been replicated across languages and cultures (Costa and McCrae, 1992).

WHO DEVELOPED IT

The structure was converged on by several independent research programmes from the 1960s onward — Tupes and Christal, Norman, and Goldberg — and operationalised as the NEO-PI-R by Paul Costa and Robert McCrae at the US National Institute on Aging. Unlike most instruments in commercial use, it originated in academic psychology rather than consulting practice (Goldberg, 1993).

WHY IT WAS DEVELOPED

Trait research before the 1980s was fragmented across incompatible instruments, so findings could not be compared. The Big Five was developed to establish how many dimensions were actually needed to describe personality, using factor analysis of natural-language trait terms. The answer converged on five, and its practical importance is that conscientiousness predicts job performance across nearly every occupation studied (Barrick and Mount, 1991).

FIGURE 18 · THE FIVE CONTINUOUS TRAIT DIMENSIONS

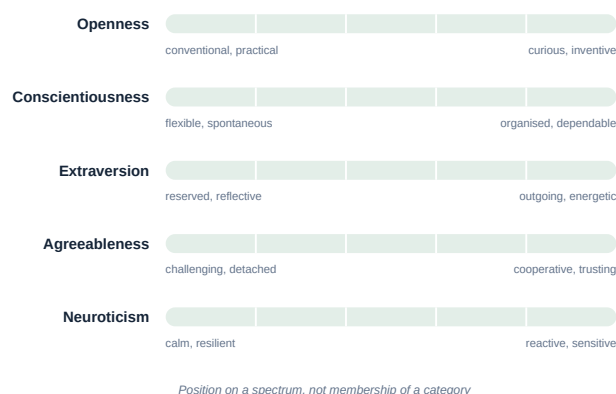


Figure 18. Five dimensions, each continuous. An individual is described by a position on all five at once, which is why type labels lose information (Costa and McCrae, 1992).

HOW IT IS USED

Read your profile as five positions rather than a label, and use conscientiousness and neuroticism scores to design working conditions rather than to explain outcomes after the fact.

WHEN IT IS USED

When evidence-based self-assessment is required, in structured selection processes where validity matters legally, and when comparing a commercial profile against a validated baseline.

BY WHOM IT IS USED

Researchers and practitioners requiring psychometric validity, occupational psychologists in selection, and individuals wanting a defensible profile rather than a memorable one.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Type-based instruments assign a person to one of a small number of boxes, discarding the distance between two people placed in the same box. The Big Five retains that information by keeping every dimension continuous, which is why it survives replication where type models do not. Its cost is memorability: five numeric positions are harder to use in a workshop than a four-letter code, which partly explains why weaker instruments remain more popular.

RELATED FRAMEWORKS IN THIS COURSE

MBTI Four Dichotomies · [Slibrary 4](#)

DISC Behavioural Quadrant · [Slibrary 4](#)

Growth vs. Fixed Mindset Fork · [Slibrary 4](#)

Personal SWOT Matrix · [Slibrary 3](#)

SOURCES (HARVARD REFERENCING STYLE)

Costa, P.T. and McCrae, R.R. (1992) *Revised NEO Personality Inventory (NEO-PI-R) Professional Manual*. Odessa, FL: Psychological Assessment Resources.

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Honey & Mumford Learning Styles

SLIBRARY 04
SHEET 4 / 5

Activist · Reflector · Theorist · Pragmatist — four preferences on the Kolb cycle

PROFESSIONAL DEFINITION

Honey and Mumford's model identifies four learning preferences corresponding to the four stages of Kolb's cycle. Activists engage first and think afterwards, learning from immersion; reflectors gather observations before committing; theorists require a coherent model before accepting a practice; and pragmatists want an application they can test at work. The preference indicates a favoured entry point to the cycle, not a restriction on the stages a learner can use (Honey and Mumford, 1986).

WHO DEVELOPED IT

Peter Honey, an occupational psychologist, and Alan Mumford, a management-development specialist, published the Learning Styles Questionnaire in 1982 and the accompanying manual in 1986. They built directly on Kolb's cycle but replaced his instrument with items describing workplace behaviour rather than abstract self-description, which improved face validity for managers (Kolb, 1984).

WHY IT WAS DEVELOPED

Honey and Mumford were designing management-development programmes and observed that identically delivered training produced very different results across participants. Their diagnosis was that most programmes were built around a single stage of the cycle — usually theory or activity — so learners whose preference lay elsewhere disengaged. The remedy was to design activity covering all four stages rather than to match delivery to individuals.

FIGURE 19 · THE FOUR PREFERENCES ON THE CYCLE

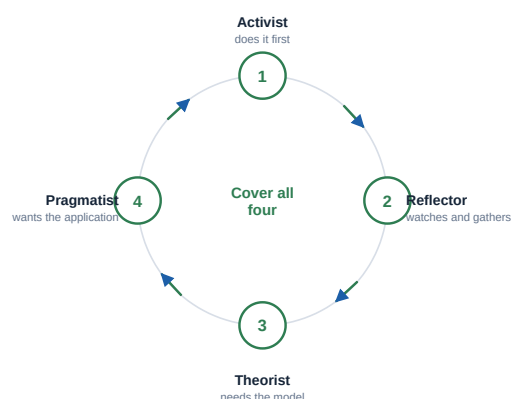


Figure 19. Four preferences mapped to the four stages of the Kolb cycle. Well-designed learning covers every stage rather than matching delivery to a preference (Honey and Mumford, 1986).

HOW IT IS USED

Identify your weakest preference and deliberately add that stage to how you learn — a pragmatist should force a theory pass; a theorist should force a trial.

WHEN IT IS USED

When designing a session or programme for a mixed group, and when a subject you understand conceptually keeps failing in application.

BY WHOM IT IS USED

Educators and trainers designing balanced activity, team leaders planning knowledge transfer, and individuals diagnosing why learning is not transferring.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Kolb's own instrument asked learners to rank abstract descriptors, which managers found difficult to answer honestly; Honey and Mumford substituted behavioural statements about work, raising completion quality. The important caveat applies to the whole learning-styles field: systematic reviews find no support for improving outcomes by matching teaching to a diagnosed style, so the model earns its place as a design checklist, not a personalisation rule (Coffield et al., 2004; Pashler et al., 2008).

RELATED FRAMEWORKS IN THIS COURSE

Kolb Experiential Learning Cycle · [Slibrary 1](#)Deliberate Practice Loop · [Slibrary 1](#)MBTI Four Dichotomies · [Slibrary 4](#)Bloom's Taxonomy (Revised) · [Slibrary 1](#)

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Growth vs. Fixed Mindset Fork

SLIBRARY 04
SHEET 5 / 5

Ability as trainable or as fixed — one belief, two divergent responses to difficulty

PROFESSIONAL DEFINITION

Dweck's model distinguishes two implicit theories of ability. Under a fixed mindset, ability is treated as a stable quantity a person possesses, so difficulty is evidence of a limit and effort signals its absence. Under a growth mindset, ability is treated as developable through effort and strategy, so difficulty is information about what to practise next. The belief is domain-specific and situational rather than a global personality trait (Dweck, 2006).

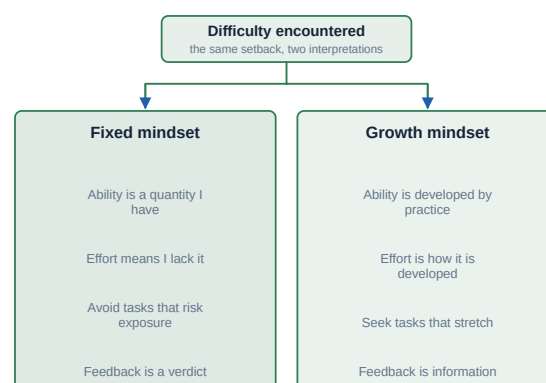
WHO DEVELOPED IT

Carol S. Dweck, professor of psychology at Stanford University, developed the account across three decades of research on children's responses to failure, beginning with work on learned helplessness with Carol Diener, and set it out for a general readership in *Mindset* (2006) (Dweck and Leggett, 1988).

WHY IT WAS DEVELOPED

Dweck's early experiments found that children of equal ability responded to the same failure in opposite ways — some disengaging, others intensifying effort — and that the difference tracked what they believed ability to be. The construct matters at Masters level because the point at which a fixed mindset produces withdrawal is exactly the point at which the most learning is available, and AI tools make avoidance of difficulty unusually easy.

FIGURE 20 · ONE BELIEF, TWO RESPONSES TO DIFFICULTY



The fork is situational: the same person differs by domain

Figure 20. A single interpretation of difficulty branches into opposite behaviour. The divergence appears at setback, not at success (Dweck, 2006).

HOW IT IS USED

Catch the interpretation at the moment of difficulty and restate it as a strategy question: not 'am I good at this' but 'what specifically am I not yet doing'.

WHEN IT IS USED

After critical feedback, when a task has been avoided repeatedly, and when using AI in a way that removes the difficulty rather than the drudgery.

BY WHOM IT IS USED

Students facing demanding assessment, managers giving developmental feedback, and educators framing failure as diagnostic rather than final.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Motivation was previously explained largely through ability and incentive; Dweck introduced an intervening belief that predicted response to failure independently of both, and one that is modifiable. The claim requires care: large-scale replications and meta-analyses find average effects on achievement that are small and concentrated among lower-attaining students, so the model is best used descriptively rather than as a guaranteed intervention (Sisk et al., 2018).

RELATED FRAMEWORKS IN THIS COURSE

Deliberate Practice Loop · **Slibrary 1**

Dunning–Kruger Curve · **Slibrary 6**

Kolb Experiential Learning Cycle · **Slibrary 1**

Anxiety Reappraisal · **Slibrary 8**

SOURCES (HARVARD REFERENCING STYLE)

Dweck, C.S. (2006) *Mindset: The New Psychology of Success*. New York: Random House.

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SCQA Structure

SLIBRARY 05
SHEET 1 / 5

Situation · Complication · Question · Answer — four moves that make a problem worth solving

PROFESSIONAL DEFINITION

SCQA is a framing sequence placed before any analysis. The situation states what the audience already accepts as true; the complication states the change or tension that makes the situation no longer tenable; the question is the single query the complication forces; and the answer is the governing thought the rest of the document supports. The sequence is diagnostic: if no complication can be stated, there is no problem to solve (Minto, 1987).

WHO DEVELOPED IT

Barbara Minto developed SCQA at McKinsey & Company in the 1960s and 1970s as the introduction structure of the Pyramid Principle, having been the firm's first female consultant and later its writing tutor in Europe. The pattern draws on classical narrative structure, which is why it reads as a story rather than as a report opening (Minto, 1987).

WHY IT WAS DEVELOPED

Minto observed that consultants opened documents with background, on the assumption that context precedes argument, and that readers consequently could not tell why they were reading. The complication supplies that reason. The structure also acts as a filter on the analyst's own work: an analysis that cannot be traced back to a stated complication is answering a question nobody asked.

FIGURE 21 · THE FOUR FRAMING MOVES



No stateable complication means no problem worth a document

Figure 21. Four moves in fixed order. The complication is the load-bearing element: it converts background into a reason to keep reading (Minto, 1987).

HOW IT IS USED

Write the four sentences before you open a spreadsheet. If the complication is vague, the analysis will be too — rewrite it until it names a specific change.

WHEN IT IS USED

Before beginning any report, proposal or recommendation, and when a piece of work has grown large without a clear purpose.

BY WHOM IT IS USED

Consultants and analysts framing engagements, students structuring assignments, and anyone briefing a colleague on a problem they have not yet met.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Conventional report openings summarise background, which forces the reader to infer the purpose from the content. SCQA supplies purpose first and makes it testable. It also functions upstream of writing as a scoping device, which distinguishes it from presentation formulas: the structure fails visibly when the underlying problem is ill-defined, and that failure is the useful signal.

RELATED FRAMEWORKS IN THIS COURSE

Pyramid Principle · [Slibrary 9](#)

MECE Issue Tree · [Slibrary 5](#)

Problem → Insight → Recommendation · [Slibrary 8](#)

Storyline Canvas · [Slibrary 9](#)

SOURCES (HARVARD REFERENCING STYLE)

Minto, B. (1987) *The Pyramid Principle: Logic in Writing and Thinking*. London: Minto International.

Rasiel, E.M. (1999) *The McKinsey Way*. New York: McGraw-Hill.

Davis, S. (2015) 'Structured communication in professional services', *Journal of Business and Technical Communication*, 29(3), pp. 279–303.

MECE Issue Tree

SLIBRARY 05
SHEET 2 / 5

Mutually exclusive, collectively exhaustive — decomposition you can trust to be complete

PROFESSIONAL DEFINITION

A MECE issue tree decomposes a question into sub-questions that do not overlap and together cover the whole problem. Mutual exclusivity prevents the same driver being analysed twice under different names; collective exhaustiveness prevents a driver being missed entirely. The tree is complete when each branch can be tested independently and the branches together account for the parent question without residue (Rasiel, 1999; Minto, 1987).

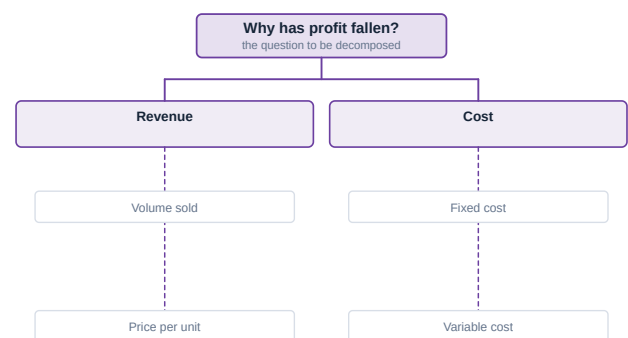
WHO DEVELOPED IT

The principle was codified at McKinsey & Company, articulated by Barbara Minto as the logical requirement underlying pyramid structure and popularised as an explicit acronym by Ethan Rasiel in *The McKinsey Way* (1999). Its roots lie in the partition rules of formal logic and set theory rather than in management research (Minto, 1987).

WHY IT WAS DEVELOPED

Analysts under time pressure investigate the drivers that occur to them first, which produces coverage that is unknowably incomplete: nothing in the output reveals what was never considered. MECE makes completeness a property of the structure rather than of the analyst's memory. It is particularly relevant when AI generates plausible-looking analysis, since fluent output offers no indication of what it omitted.

FIGURE 22 · A DECOMPOSITION THAT LEAVES NO RESIDUE



Branches must not overlap, and must leave nothing out

Figure 22. Each level partitions the level above it. Two branches that overlap double-count a driver; two that under-cover hide one entirely (Rasiel, 1999).

HOW IT IS USED

Decompose the question one level at a time and test each level twice: could two branches capture the same thing, and could something real fall outside all of them?

WHEN IT IS USED

At the start of any diagnostic problem, when a team is duplicating work, and when a recommendation feels incomplete but the gap cannot be named.

BY WHOM IT IS USED

Consultants structuring engagements, analysts scoping investigations, and anyone auditing AI-generated analysis for silent omissions.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Brainstormed lists of causes offer no test of coverage, so an analyst cannot distinguish a thorough list from a lucky one. MECE supplies that test at every level. The cost is that strict exclusivity is difficult in genuinely interdependent systems — feedback loops resist clean partition — so the discipline works best on decomposable problems and should not be forced onto complex ones (Snowden and Boone, 2007).

RELATED FRAMEWORKS IN THIS COURSE

SCQA Structure · [Slibrary 5](#)First Principles Decomposition · [Slibrary 5](#)Pyramid Principle · [Slibrary 9](#)Cynefin Framework · [Slibrary 5](#)

SOURCES (HARVARD REFERENCING STYLE)

Rasiel, E.M. (1999) *The McKinsey Way*. New York: McGraw-Hill.

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First Principles Decomposition

SLIBRARY 05
SHEET 3 / 5

Strip the assumption · reach the fundamentals · rebuild from there

PROFESSIONAL DEFINITION

First-principles decomposition solves a problem by reducing it to propositions that are known to be true independently of convention, then reasoning upward from those alone. The procedure has three moves: identify the assumptions carried by the current framing, break the problem down until only physically or logically necessary constraints remain, and construct a solution from that base rather than from an existing example. It is the deliberate opposite of reasoning by analogy (Aristotle, c. 350 BCE; Wertheimer, 1945).

WHO DEVELOPED IT

The concept is Aristotelian in origin — the *archai* from which demonstration proceeds — and enters modern problem-solving through Descartes's method of analysis and Max Wertheimer's Gestalt work on productive thinking, which distinguished genuine restructuring of a problem from reproductive application of a remembered method (Wertheimer, 1945).

WHY IT WAS DEVELOPED

Wertheimer's experimental finding was that learners who had memorised a procedure could not solve a structurally identical problem in altered form, while those who had grasped the underlying structure could. Reasoning by analogy is fast and usually adequate, but it inherits every constraint of the analogy, including those that were never necessary — which is why it reliably fails at exactly the moments when conditions have changed.

FIGURE 23 · THREE MOVES BACK TO THE FUNDAMENTALS



Analogy inherits constraints that were never necessary

Figure 23. Three moves: strip, reduce, reconstruct. The middle move is the hard one, because conventions are experienced as constraints (Wertheimer, 1945).

HOW IT IS USED

Write the problem as currently framed, list every assumption it carries, and strike each one that is convention rather than necessity. Rebuild from what survives.

WHEN IT IS USED

When an established approach has stopped working, when cost or capability is asserted as fixed, and when every available option is a variation on the same idea.

BY WHOM IT IS USED

Engineers and product designers challenging cost structures, strategists confronting industry orthodoxy, and students asked to produce original rather than derivative analysis.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Analogical reasoning — benchmark the competitor, adapt best practice — is efficient but cannot generate a solution outside the space of existing examples. First-principles reasoning can, at a much higher cost in time and cognitive effort. The honest limitation is that it is wasteful when applied indiscriminately: most problems are well served by analogy, and the skill lies in recognising the minority that are not.

RELATED FRAMEWORKS IN THIS COURSE

MECE Issue Tree · [Slibrary 5](#)Double Diamond · [Slibrary 5](#)Cynefin Framework · [Slibrary 5](#)Ladder of Inference · [Slibrary 6](#)

SOURCES (HARVARD REFERENCING STYLE)

Wertheimer, M. (1945) *Productive Thinking*. New York: Harper & Brothers.

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Double Diamond

SLIBRARY 05
SHEET 4 / 5

Discover · Define · Develop · Deliver — two divergences, two convergences

PROFESSIONAL DEFINITION

The Double Diamond models design as four phases alternating between divergent and convergent thinking. Discover explores the problem space broadly; define narrows it to a single framed problem statement; develop generates multiple candidate solutions; and deliver converges on one and tests it. The central constriction is the point at which the problem, not the solution, is fixed — and it is the step most often skipped (Design Council, 2005).

WHO DEVELOPED IT

The UK Design Council published the model in 2005 after studying design processes in eleven global companies, including Microsoft, Starbucks and LEGO. It was revised in 2019 as the Framework for Innovation, adding engagement principles and explicit leadership and connection methods around the original two diamonds (Design Council, 2019).

WHY IT WAS DEVELOPED

The Council found that organisations treated design as the styling stage at the end of development, arriving with the problem already assumed and only the execution open. The model makes problem definition a distinct, funded phase with its own divergence. Its persistent message is that effort spent widening the problem space before narrowing it is not delay but the substance of the work.

FIGURE 24 · TWO DIAMONDS, FOUR PHASES

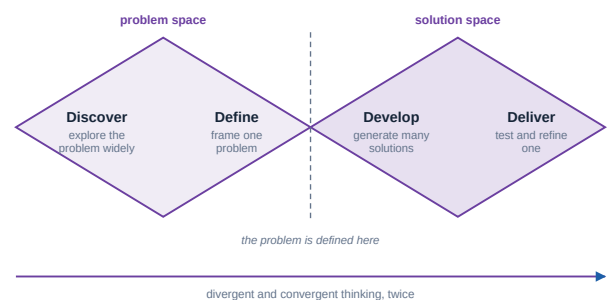


Figure 24. Divergence then convergence, twice. The waist between the diamonds is where the problem is fixed, before any solution is developed (Design Council, 2005).

HOW IT IS USED

Name which of the four phases you are in and refuse to mix modes: do not evaluate during a divergent phase, and do not generate new options during a convergent one.

WHEN IT IS USED

At the start of any design, innovation or improvement project, and when a team is arguing about solutions before agreeing what the problem is.

BY WHOM IT IS USED

Design and innovation teams, project managers structuring discovery, and facilitators protecting divergent phases from premature judgment.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Linear stage-gate development treats requirements as an input to be received rather than a question to be investigated, so an incorrectly framed problem is discovered only at launch. The Double Diamond makes reframing a designed phase. Its own limitation is that the diagram implies a single pass, whereas real projects loop repeatedly — a point the 2019 revision acknowledged explicitly.

RELATED FRAMEWORKS IN THIS COURSE

First Principles Decomposition · [Slibrary 5](#)Cynefin Framework · [Slibrary 5](#)Agile Manifesto Values · [Slibrary 12](#)[Project Lifecycle](#) · [Slibrary 11](#)

SOURCES (HARVARD REFERENCING STYLE)

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Cynefin Framework

SLIBRARY 05
SHEET 5 / 5

Clear · Complicated · Complex · Chaotic — match the method to the kind of problem

PROFESSIONAL DEFINITION

Cynefin is a sense-making framework that sorts situations by the nature of the relation between cause and effect. In clear contexts the relation is self-evident and best practice applies; in complicated contexts it is knowable through analysis or expertise; in complex contexts it is apparent only in retrospect, so safe-to-fail probes are required; and in chaotic contexts no relation is discernible and action must precede analysis. A fifth domain, confusion, is not knowing which of the four applies (Snowden and Boone, 2007).

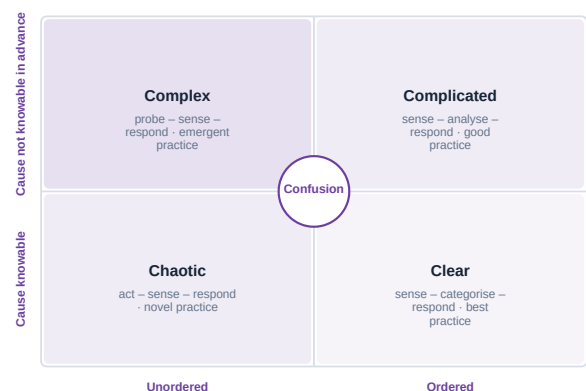
WHO DEVELOPED IT

Dave Snowden developed the framework at IBM's Institute for Knowledge Management from 1999 onward, drawing on complexity science and anthropology; the Welsh word cynefin denotes the multiple belongings that shape how a person perceives a place. It reached general management readership through his 2007 *Harvard Business Review* article with Mary Boone (Kurtz and Snowden, 2003).

WHY IT WAS DEVELOPED

Snowden's argument was that management failure is frequently a category error rather than an execution error: best practice imported into a complex domain produces confident action on a misread situation, and rigorous analysis applied to a chaotic one wastes the window for response. The framework's demand is that the domain be diagnosed before a method is chosen.

FIGURE 25 · FOUR DOMAINS AND THE METHOD EACH DEMANDS



The dangerous move is treating a complex problem as complicated

Figure 25. Four domains with a central state of not knowing which applies. Each domain licenses a different sequence of action and analysis (Snowden and Boone, 2007).

HOW IT IS USED

Diagnose the domain before selecting a method. If cause and effect are only clear afterwards, run small reversible probes instead of a plan.

WHEN IT IS USED

At the start of an initiative, when a proven method is failing unexpectedly, and during a crisis when the instinct is to analyse rather than act.

BY WHOM IT IS USED

Leaders choosing a management approach, project managers deciding between plan-driven and adaptive delivery, and teams diagnosing why best practice is not working.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Contingency theories of management said that the right approach depends on the situation but offered little help in classifying situations. Cynefin supplies the classification and attaches a distinct action sequence to each class. It is a sense-making device rather than a categorisation model — domains are assigned by judgment, and Snowden explicitly warns against treating it as a matrix to be filled in.

RELATED FRAMEWORKS IN THIS COURSE

Double Diamond · [Slibrary 5](#)Stacey Matrix · [Slibrary 12](#)MECE Issue Tree · [Slibrary 5](#)Agile Manifesto Values · [Slibrary 12](#)

SOURCES (HARVARD REFERENCING STYLE)

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System 1 / System 2

SLIBRARY 06
SHEET 1 / 5

Fast, automatic, always on · slow, effortful, easily displaced

PROFESSIONAL DEFINITION

Kahneman's two-system account distinguishes System 1, which produces impressions, intuitions and judgments automatically, continuously and without felt effort, from System 2, which allocates attention to effortful mental activities including complex computation and deliberate self-monitoring. System 2 generally endorses System 1's proposals with little modification, which is efficient in most conditions and is the origin of systematic bias in the rest (Kahneman, 2011).

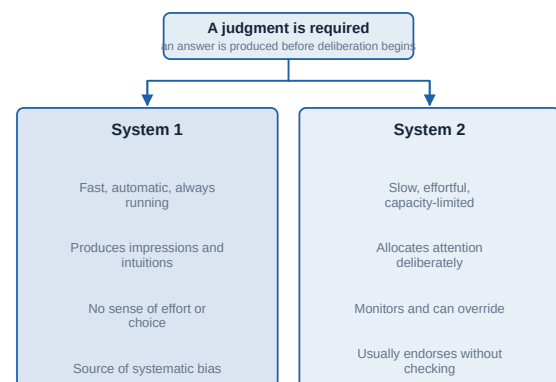
WHO DEVELOPED IT

Daniel Kahneman, an Israeli-American psychologist at Princeton University and Nobel laureate in economics (2002), set out the account in *Thinking, Fast and Slow* (2011), synthesising four decades of work with Amos Tversky on heuristics and judgment under uncertainty. The labelling of the two systems is borrowed from Stanovich and West (Stanovich and West, 2000).

WHY IT WAS DEVELOPED

Kahneman and Tversky's experimental programme established that errors of judgment are not random noise but predictable departures from normative reasoning, arising from heuristics that work well in most circumstances. The two-system framing explains why expertise and intelligence do not protect against these errors: the fast system produces an answer before the slow one is engaged, and the slow system is lazy about checking.

FIGURE 26 · TWO SYSTEMS, ONE ENDORSEMENT PROBLEM



The default is endorsement, not scrutiny

Figure 26. Two modes and the relationship between them. System 2's habitual endorsement of System 1 output is where predictable error enters (Kahneman, 2011).

HOW IT IS USED

Build the check into the process rather than relying on noticing: pre-commit to a decision checklist for consequential judgments, since the feeling of certainty is a System 1 output.

WHEN IT IS USED

On high-stakes decisions, under time pressure or fatigue, and whenever an answer arrives immediately and feels obviously right.

BY WHOM IT IS USED

Anyone making consequential judgments, managers designing decision processes, and professionals working with fluent AI output that invites immediate acceptance.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Classical economics modelled the decision-maker as consistently rational, so observed departures were treated as noise. The heuristics-and-biases programme demonstrated the departures were systematic and predictable, founding behavioural economics. Kahneman himself cautioned that the two systems are expository fictions rather than brain structures, and parts of the surrounding priming literature have failed replication (Kahneman, 2011).

RELATED FRAMEWORKS IN THIS COURSE

Confirmation Bias Loop · [Slibrary 6](#)Anchoring & Adjustment · [Slibrary 6](#)Automation Bias Curve · [Slibrary 2](#)Bloom's Taxonomy (Revised) · [Slibrary 1](#)

SOURCES (HARVARD REFERENCING STYLE)

Kahneman, D. (2011) *Thinking, Fast and Slow*. New York: Farrar, Straus and Giroux.

Tversky, A. and Kahneman, D. (1974) 'Judgment under uncertainty: heuristics and biases', *Science*, 185(4157), pp. 1124–1131.

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Confirmation Bias Loop

SLIBRARY 06
SHEET 2 / 5

Hypothesis · selective search · biased interpretation · reinforced belief

PROFESSIONAL DEFINITION

Confirmation bias is the tendency to seek, interpret, favour and recall information in a way that supports an existing belief. It operates as a closed loop: an initial hypothesis shapes what evidence is searched for, the evidence found is interpreted in the hypothesis's favour, disconfirming data is scrutinised more harshly than confirming data, and the strengthened belief then narrows the next search. The process is largely unintentional (Nickerson, 1998).

WHO DEVELOPED IT

Peter Wason demonstrated the effect experimentally in the 1960s with his rule-discovery and selection tasks, in which participants overwhelmingly sought confirming instances rather than the falsifying tests logic requires. Raymond S. Nickerson's 1998 review in *Review of General Psychology* consolidated the evidence across domains and remains the standard reference (Wason, 1960; Nickerson, 1998).

WHY IT WAS DEVELOPED

Wason was testing whether ordinary reasoning follows the falsificationist logic Popper prescribed for science. It does not: people test hypotheses by looking for cases that fit. The loop matters more, not less, with modern information tools, because search engines, recommendation systems and conversational AI all return material matched to the framing of the query — so a leading question retrieves a confirming answer.

FIGURE 27 · THE SELF-SEALING FOUR-STAGE LOOP

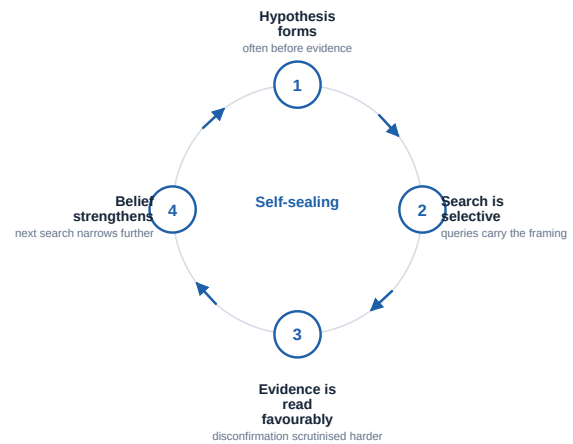


Figure 27. Four stages forming a closed loop. Each pass narrows the evidence admitted, so the belief grows more confident on less information (Nickerson, 1998).

HOW IT IS USED

Break the loop at the search stage: write the query that would produce disconfirming evidence and run it, and ask what specific observation would change your mind.

WHEN IT IS USED

When forming a view on contested evidence, before committing to a recommendation, and whenever research has produced only supportive findings.

BY WHOM IT IS USED

Researchers and analysts, managers evaluating a favoured proposal, and anyone prompting an AI system, since the framing of the prompt determines the evidence returned.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Earlier accounts explained biased conclusions through motivation — people believe what they want to believe. Wason and Nickerson showed the distortion enters much earlier and without motivation, at the point where evidence is selected, which is why it resists exhortations to be objective. That relocation makes the countermeasure procedural: change what is searched for, not how hard you try to be fair.

RELATED FRAMEWORKS IN THIS COURSE

System 1 / System 2 · [Slibrary 6](#)

CRAAP Source Test · [Slibrary 2](#)

Ladder of Inference · [Slibrary 6](#)

Anchoring & Adjustment · [Slibrary 6](#)

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Anchoring & Adjustment

SLIBRARY 06
SHEET 3 / 5

An arbitrary starting number moves the final estimate — and the adjustment is never enough

PROFESSIONAL DEFINITION

Anchoring is the disproportionate influence of an initially considered value on a subsequent numerical judgment. Estimators begin from the anchor and adjust toward what they believe correct, but adjustment terminates prematurely, leaving the final estimate biased toward the starting point. The effect holds when the anchor is transparently irrelevant, when it is self-generated, and when participants are warned about it and paid for accuracy (Tversky and Kahneman, 1974).

WHO DEVELOPED IT

Amos Tversky and Daniel Kahneman identified the heuristic in their 1974 *Science* paper, using a wheel of fortune whose visibly random outcome shifted participants' estimates of an unrelated quantity. Later work by Northcraft and Neale demonstrated the effect in professional estate agents valuing real property, and by Englich and colleagues in judicial sentencing (Northcraft and Neale, 1987; Englich et al., 2006).

WHY IT WAS DEVELOPED

Tversky and Kahneman were mapping the shortcuts by which people make estimates under uncertainty. Anchoring proved the most robust and the least correctable: expertise reduces it only modestly, and awareness barely at all. For negotiation and pricing this is a practical matter rather than a curiosity — the first number stated shapes the settlement range, so who opens, and with what, is a decision.

FIGURE 28 · WHY THE ADJUSTMENT STOPS TOO SOON

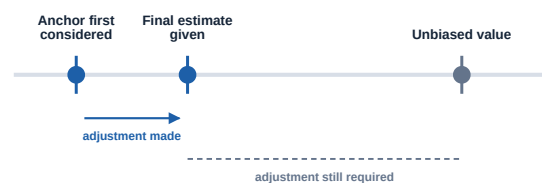


Figure 28. Estimation begins at the anchor and moves toward the correct value, but stops early. The residual gap is the measured bias (Tversky and Kahneman, 1974).

HOW IT IS USED

Prepare your own number before exposure to theirs, and treat any figure you encounter first as contamination: re-derive the estimate from independent inputs rather than adjusting.

WHEN IT IS USED

Before any negotiation, budget or valuation discussion, and whenever a figure has been mentioned in passing before the analysis is complete.

BY WHOM IT IS USED

Negotiators deciding whether to open, managers setting budgets, and anyone accepting a starting figure from a colleague, a client or a model's first draft.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Prior models of estimation assumed judgment converges on a value from available evidence, treating irrelevant information as ignorable. Anchoring showed that a demonstrably irrelevant number alters the outcome, which forced a procedural rather than motivational remedy: generate your estimate before exposure. Effect sizes vary by domain, and some incentive-based studies find weaker effects than the classic demonstrations.

RELATED FRAMEWORKS IN THIS COURSE

BATNA · [Slibrary 13](#)Zone of Possible Agreement · [Slibrary 13](#)System 1 / System 2 · [Slibrary 6](#)Confirmation Bias Loop · [Slibrary 6](#)

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Ladder of Inference

SLIBRARY 06
SHEET 4 / 5

Data · selection · meaning · assumption · conclusion · belief · action — seven rungs,
climbed in a second

PROFESSIONAL DEFINITION

The Ladder of Inference describes the near-instantaneous progression from observable data to action. From the available data a subset is selected; meaning is added from cultural and personal context; assumptions are formed on that meaning; conclusions are drawn; beliefs adopted; and action taken. Beliefs then determine what data is selected on the next pass, closing a reflexive loop. Disagreements are frequently disputes about different rungs (Argyris, 1990; Senge, 1990).

WHO DEVELOPED IT

Chris Argyris, professor at Harvard Business School and a founder of organisational learning research, developed the ladder within his work on theories of action and defensive routines. Peter Senge popularised it as a team-learning discipline in *The Fifth Discipline* (1990), which is the form in which most practitioners encounter it (Argyris and Schön, 1974).

WHY IT WAS DEVELOPED

Argyris's field research found that skilled professionals were remarkably poor at examining the reasoning behind their own conclusions, defending positions whose derivation they could not reconstruct. The ladder makes the invisible steps nameable, which is what allows two people who disagree to locate the rung at which they diverged instead of arguing at the top.

FIGURE 29 · SEVEN RUNGS FROM OBSERVATION TO ACTION



Beliefs then govern which data is selected next time

Figure 29. Seven rungs traversed almost instantly. Beliefs at the top determine selection at the bottom, making the ladder a loop rather than a line (Argyris, 1990).

HOW IT IS USED

When you disagree strongly, walk down: what did I actually observe, what did I select, and what meaning did I add that the other person may not share?

WHEN IT IS USED

In conflict where both parties believe the facts are obvious, in feedback conversations, and when a strong reaction has formed very quickly.

BY WHOM IT IS USED

Teams working through disagreement, managers examining their own judgments of people, and facilitators surfacing the reasoning behind stated positions.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Communication training had focused on how positions are expressed. Argyris relocated the problem to how they are formed, arguing that most organisational conflict concerns unexamined inference rather than incompatible interests. The framework has little quantitative validation and functions as a reflective discipline; its value lies in giving the steps names that can be used out loud in a meeting.

RELATED FRAMEWORKS IN THIS COURSE

Johari Window · [Slibrary 3](#)

Confirmation Bias Loop · [Slibrary 6](#)

Kolb Experiential Learning Cycle · [Slibrary 1](#)

Tactical Empathy · [Slibrary 14](#)

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Dunning–Kruger Curve

SLIBRARY 06
SHEET 5 / 5

Least competent, least able to know it — why self-assessment fails where it matters most

PROFESSIONAL DEFINITION

The Dunning–Kruger effect is the finding that people with low competence in a domain systematically overestimate their ability, because the knowledge required to perform well is largely the same knowledge required to recognise poor performance. Highly competent people tend to underestimate their relative standing, assuming that tasks they find manageable are manageable for others. Both errors are failures of metacognition (Kruger and Dunning, 1999).

WHO DEVELOPED IT

Justin Kruger and David Dunning, then at Cornell University, published the studies in 1999 in the *Journal of Personality and Social Psychology*, testing participants on logical reasoning, grammar and humour. The popular staircase graph with a 'peak of Mount Stupid' does not appear in their paper and is a later illustrative invention (Kruger and Dunning, 1999).

WHY IT WAS DEVELOPED

Dunning and Kruger were investigating why incompetent performance so often coexists with confidence. Their explanation was structural rather than characterological: the deficit that causes poor performance also removes the means of detecting it. The practical implication is that self-assessment is least reliable precisely where the stakes of misjudgment are highest, so external calibration is required rather than recommended.

FIGURE 30 · CONFIDENCE AGAINST ACTUAL COMPETENCE

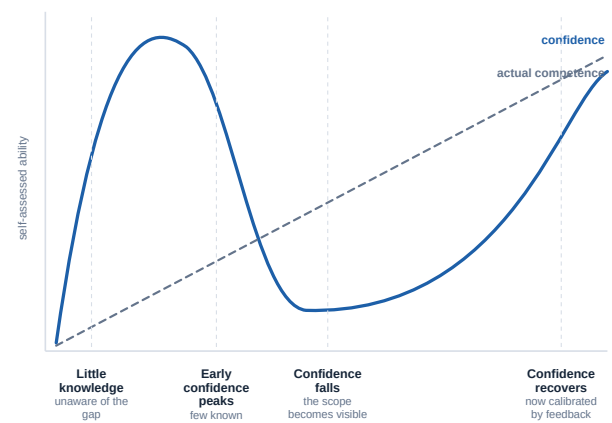


Figure 30. Self-assessed ability against measured competence. The two converge only where external feedback is available, which is the model's practical claim (Kruger and Dunning, 1999).

HOW IT IS USED

Treat your own confidence as uninformative in unfamiliar domains and seek external calibration: a measured test, a benchmark, or an expert's judgment of your work.

WHEN IT IS USED

When entering a new field, when appraising your own work with no comparison, and when adopting a tool whose failure modes you have not yet seen.

BY WHOM IT IS USED

Students assessing their readiness, managers evaluating capability claims, and anyone adopting AI tools in a domain where they cannot yet check the output.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Poor self-assessment had been attributed to ego protection. Dunning and Kruger identified a cognitive rather than motivational cause and thereby predicted the specific asymmetry between novices and experts. The effect is genuine but frequently overstated: some of the pattern in the original graphs is reproduced by regression to the mean and statistical artefact, and the popular curve exaggerates the shape considerably (Nuhfer et al., 2017).

RELATED FRAMEWORKS IN THIS COURSE

[Dreyfus Skill Acquisition Model · Slibrary 1](#)
[Johari Window · Slibrary 3](#)
[Automation Bias Curve · Slibrary 2](#)
[Growth vs. Fixed Mindset Fork · Slibrary 4](#)

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Brand Pyramid

SLIBRARY 07
SHEET 1 / 5

Attributes · Functional benefits · Emotional rewards · Values · Essence — five ascending levels

PROFESSIONAL DEFINITION

The Brand Pyramid arranges what a brand offers into five ascending levels of abstraction. Attributes are the verifiable features; functional benefits are what those features do for the user; emotional rewards are how the user feels as a result; values express what the brand stands for; and the essence is a single distilled idea. Each level must be supported by the one beneath it, so an essence with no attribute trail is a claim rather than a position (Aaker, 1996; Keller, 2001).

WHO DEVELOPED IT

David A. Aaker, professor emeritus at the Haas School of Business, University of California, Berkeley, established the modern architecture of brand identity and equity across *Managing Brand Equity* (1991) and *Building Strong Brands* (1996). Kevin Lane Keller's customer-based brand equity pyramid and Millward Brown's BrandDynamics gave the hierarchy its familiar pyramid form (Keller, 2001).

WHY IT WAS DEVELOPED

Aaker's concern was that brands were being described either as lists of features, which competitors could copy, or as abstract adjectives, which meant nothing to a buyer. The pyramid connects the two ends: it requires that every emotional claim be traceable downward to something concrete. Applied to a person, it is the difference between 'I am strategic' and a demonstrable record that makes the word credible.

FIGURE 31 · FIVE ASCENDING LEVELS OF ABSTRACTION

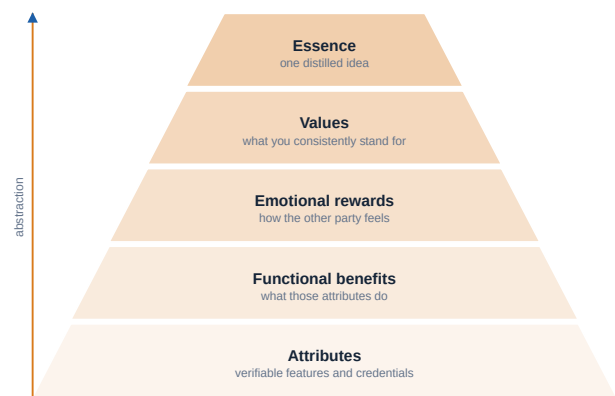


Figure 31. Five levels rising from the verifiable to the distilled. Any level unsupported from below is an assertion rather than a position (Aaker, 1996).

HOW IT IS USED

Build upward, not downward: list attributes you can evidence, derive the benefit each delivers, and only then attempt the one-line essence.

WHEN IT IS USED

When constructing a personal brand statement, when rewriting a CV summary, and whenever a positioning line sounds impressive but cannot be substantiated.

BY WHOM IT IS USED

Individuals building a professional profile, brand managers auditing consistency, and anyone whose self-description has drifted away from what they can prove.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Feature-based positioning is easily matched and gives a buyer no reason to prefer one provider. Purely emotional positioning is unmatched but also unsupported, and collapses on contact with evidence. The pyramid's contribution is the requirement of vertical support throughout. Its limitation is that the upper levels resist measurement, so the model is far better at diagnosing an unsupported claim than at validating a strong one.

RELATED FRAMEWORKS IN THIS COURSE

Golden Circle · [Slibrary 7](#)

Personal Value Proposition Canvas · [Slibrary 7](#)

Twelve Brand Archetypes · [Slibrary 7](#)

Personal SWOT Matrix · [Slibrary 3](#)

SOURCES (HARVARD REFERENCING STYLE)

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Twelve Brand Archetypes

SLIBRARY 07
SHEET 2 / 5

Twelve recurring characters — a vocabulary for consistent personality

PROFESSIONAL DEFINITION

The archetype system assigns a brand one of twelve recurring character patterns — Innocent, Explorer, Sage, Hero, Outlaw, Magician, Everyman, Lover, Jester, Caregiver, Creator and Ruler — each defined by a core motivation and a characteristic voice. The archetype is a consistency device: it constrains tone, imagery and behaviour across channels so that decisions made by different people at different times still cohere (Mark and Pearson, 2001).

WHO DEVELOPED IT

Margaret Mark, a brand strategist, and Carol S. Pearson, a scholar of archetypal psychology, set out the system in *The Hero and the Outlaw* (2001). It adapts Carl Jung's theory of archetypes as universal patterns in the collective unconscious, by way of Pearson's earlier work applying those patterns to personal development (Jung, 1959; Pearson, 1991).

WHY IT WAS DEVELOPED

Mark and Pearson were addressing the practical problem that brand guidelines specify what a brand looks like but not how it behaves, so consistency collapses whenever a new situation arises. An archetype supplies a decision rule that generalises: a Sage brand explains, a Jester brand disarms, and each responds differently to the same complaint without needing a rule written in advance.

FIGURE 32 · THE TWELVE ARCHETYPES AND THEIR MOTIVATIONS

Innocent safety, simplicity	Explorer freedom, discovery	Sage understanding, truth	Hero mastery, courage
Outlaw disruption, revolution	Magician transformation	Everyman belonging, realism	Lover intimacy, beauty
Jester enjoyment, release	Caregiver service, protection	Creator innovation, expression	Ruler control, order

Twelve patterns; a brand or a person occupies one, not several

Figure 32. Twelve archetypes, each with a distinct core motivation. Occupying more than one dissolves the consistency the system exists to provide (Mark and Pearson, 2001).

HOW IT IS USED

Choose the single archetype that fits how you actually behave rather than how you would like to be seen, then test recent communications for consistency with it.

WHEN IT IS USED

When defining a personal or organisational voice, when messaging feels inconsistent across channels, and when a rebrand needs a defensible tone decision.

BY WHOM IT IS USED

Brand and communications managers, founders shaping company voice, and individuals building a recognisable professional presence.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Positioning statements describe what a brand claims but give no guidance on how to behave in an unanticipated situation. Archetypes supply that generative rule, which is why they survive contact with new channels. The framework rests on Jungian theory that is not empirically testable in the usual sense, and the twelve categories are a convention rather than a finding — useful as a discipline, not as evidence.

RELATED FRAMEWORKS IN THIS COURSE

Brand Pyramid · **Slibrary 7**

Golden Circle · **Slibrary 7**

StoryBrand SB7 Framework · **Slibrary 7**

The Culture Map (Eight Scales) · **Slibrary 10**

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StoryBrand SB7 Framework

SLIBRARY 07
SHEET 3 / 5

The customer is the hero; you are the guide — seven elements of a clarifying story

PROFESSIONAL DEFINITION

SB7 structures a message as a story with seven elements: a character with a want; a problem, at external, internal and philosophical levels; a guide who shows empathy and authority; a plan the character can follow; a call to action; the failure that is avoided; and the success that results. The framework's governing rule is positional — the customer is the hero of the story and the brand is the guide, never the other way round (Miller, 2017).

WHO DEVELOPED IT

Donald Miller, an American author and founder of the StoryBrand consultancy, published the framework in *Building a StoryBrand* (2017). It compresses classical narrative theory, notably Joseph Campbell's monomyth as adapted for screenwriting, into a marketing structure usable without literary training (Campbell, 1949).

WHY IT WAS DEVELOPED

Miller's diagnosis of failing marketing was not that it lacked information but that it supplied too much of the wrong kind: organisations describe themselves, which requires the audience to work out its own relevance, and audiences do not. Casting the customer as hero reverses the burden. The internal-problem layer matters most in practice, since purchase is usually driven by how a problem feels rather than by what it costs.

FIGURE 33 · SEVEN ELEMENTS OF THE SB7 STORY

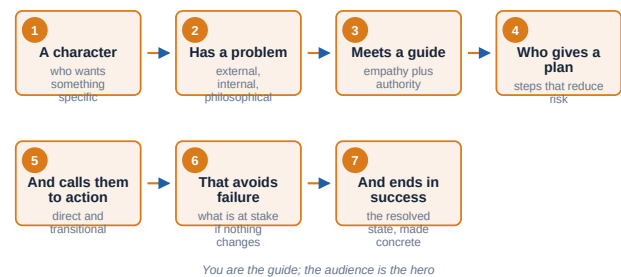


Figure 33. Seven elements in narrative order. Reversing the hero and guide roles is the failure the framework is designed to prevent (Miller, 2017).

HOW IT IS USED

Write one sentence for each element about your audience, not about yourself, then check that every element names them rather than you.

WHEN IT IS USED

When writing a pitch, a personal profile, a landing page or a cover letter, and whenever feedback says a message is unclear rather than unconvincing.

BY WHOM IT IS USED

Founders and marketers writing customer-facing copy, job applicants framing a profile, and anyone whose message describes their own capability at length.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Feature-and-benefit copywriting keeps the organisation at the centre and leaves the audience to infer relevance. SB7 inverts the roles and makes the inference the writer's job. It is a practitioner formula rather than a researched model, and its rigidity is a real cost: applied uniformly it produces recognisably formulaic copy, which erodes the distinctiveness it was meant to create.

RELATED FRAMEWORKS IN THIS COURSE

Golden Circle · [Slibrary 7](#)

Twelve Brand Archetypes · [Slibrary 7](#)

Storyline Canvas · [Slibrary 9](#)

Problem → Insight → Recommendation · [Slibrary 8](#)

SOURCES (HARVARD REFERENCING STYLE)

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Campbell, J. (1949) *The Hero with a Thousand Faces*. New York: Pantheon Books.

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Golden Circle

SLIBRARY 07
SHEET 4 / 5

Why · How · What — communicate from the inside out

PROFESSIONAL DEFINITION

The Golden Circle is a three-ring model of communication order. The outer ring, what, states the product or activity; the middle ring, how, states the differentiating process; and the inner ring, why, states the purpose or belief that motivates the activity. Sinek's claim is that most organisations communicate from the outside in, and that those which inspire consistently begin at the centre and work outward (Sinek, 2009).

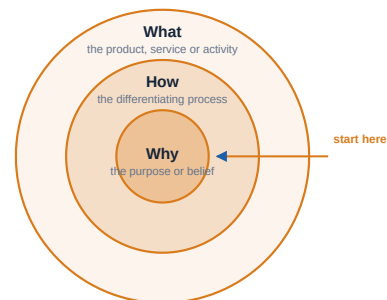
WHO DEVELOPED IT

Simon Sinek, a British-American author and speaker, presented the model in a 2009 TEDx talk and in *Start with Why* (2009). He supports it with an appeal to the limbic-system account of decision-making — a popular simplification of neuroanatomy that is not supported in the terms he states it (Sinek, 2009; Poldrack, 2011).

WHY IT WAS DEVELOPED

Sinek's observation was that organisations with comparable products and resources achieve very different levels of loyalty, and that the difference tracks whether they can articulate a purpose beyond the transaction. The model's practical use in this course is as an ordering rule for introductions and pitches: it forces the belief to the front, where an audience decides whether to keep listening.

FIGURE 34 · THREE RINGS, WORKED FROM THE INSIDE OUT



Most communication runs outside in; the model reverses it

Figure 34. Three concentric rings. The claim is about sequence rather than content: begin at the centre and work outward (Sinek, 2009).

HOW IT IS USED

Draft your introduction three times — once beginning with what, once with how, once with why — and compare which version an unfamiliar listener stays with.

WHEN IT IS USED

When opening a pitch, an interview answer or a networking introduction, and when a proposal has plenty of detail but no evident motivation.

BY WHOM IT IS USED

Founders framing a venture, professionals introducing themselves, and leaders explaining a change to people who did not request it.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Conventional communication training optimised the content of a message; Sinek's contribution is a claim about its order, which is easier to apply and easier to test. Two cautions belong with it: the neuroscientific justification is not supportable as stated, and the supporting evidence is selected case narrative rather than systematic comparison. Use it as a rhetorical structure, not as a finding about the brain.

RELATED FRAMEWORKS IN THIS COURSE

Brand Pyramid · [Slibrary 7](#)StoryBrand SB7 Framework · [Slibrary 7](#)Firm Foundation: Vision, Mission, Values · [Slibrary 3](#)Ikigai Purpose Venn · [Slibrary 3](#)

SOURCES (HARVARD REFERENCING STYLE)

Sinek, S. (2009) *Start with Why: How Great Leaders Inspire Everyone to Take Action*. New York: Portfolio.

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Personal Value Proposition Canvas

SLIBRARY 07
SHEET 5 / 5

Their jobs, pains and gains against your services, relievers and creators — six blocks, one fit

PROFESSIONAL DEFINITION

The Value Proposition Canvas pairs two profiles. The audience side records the jobs they are trying to get done, the pains that obstruct them, and the gains they seek. The offer side records the products or services provided, the pain relievers, and the gain creators. Fit is achieved when specific elements on the offer side address specific, high-priority items on the audience side — not when both lists are merely long (Osterwalder et al., 2014).

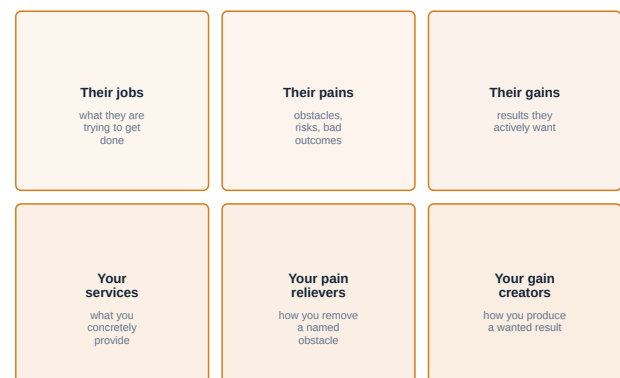
WHO DEVELOPED IT

Alexander Osterwalder, Yves Pigneur, Gregory Bernarda and Alan Smith published the canvas in *Value Proposition Design* (2014) as a zoom-in on two blocks of the earlier Business Model Canvas. The jobs-to-be-done concept on the audience side derives from Clayton Christensen's work on why customers hire products (Christensen et al., 2016).

WHY IT WAS DEVELOPED

Osterwalder's observation was that teams could describe their offer in detail and their customer only in demographics, so claimed value propositions were untestable. The canvas forces both sides into the same, comparable structure. Applied to a career, it converts a skills inventory into a claim about a specific employer's specific difficulty, which is what makes an application answerable.

FIGURE 35 · SIX BLOCKS, TWO PROFILES, ONE FIT



Fit is item-to-item, not list-to-list

Figure 35. Two three-part profiles set against one another. Fit means a named reliever answers a named pain, ranked by the audience's priority (Osterwalder et al., 2014).

HOW IT IS USED

Complete the audience side first and rank it before writing anything about yourself, then draw a line from each of your items to the specific pain or gain it addresses.

WHEN IT IS USED

Before an application or pitch, when tailoring a CV to a role, and when an offer is being described in terms of what it contains rather than what it resolves.

BY WHOM IT IS USED

Job applicants and consultants scoping proposals, product teams testing propositions, and anyone whose pitch lists capability without naming a problem.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Earlier positioning statements asserted value in a single sentence with no visible derivation, so they could be neither tested nor improved. The canvas makes the derivation explicit and therefore falsifiable: an unmatched item on either side is visible immediately. Its weakness is that the audience side is usually filled in from assumption; the canvas is a structure for research, not a substitute for it.

RELATED FRAMEWORKS IN THIS COURSE

Brand Pyramid · [Slibrary 7](#)

Personal SWOT Matrix · [Slibrary 3](#)

Problem → Insight → Recommendation · [Slibrary 8](#)

Influence Currencies · [Slibrary 14](#)

SOURCES (HARVARD REFERENCING STYLE)

Osterwalder, A., Pigneur, Y., Bernarda, G. and Smith, A. (2014) *Value Proposition Design*. Hoboken, NJ: John Wiley & Sons.

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Message Strategy Spectrum

SLIBRARY 08
SHEET 1 / 5

Convince ← → Clarify — every message is a blend, and the blend is a decision

PROFESSIONAL DEFINITION

Bostelaar's spectrum places any message between two poles. At the convince end the message is motivational: it establishes why the topic matters and builds trust in the message itself, typically through inspiring stories and introductory-level content. At the clarify end it is informational: it unpacks what the thing is, through explanation and detail for a more advanced audience. Messages sit at neither extreme but as a deliberate blend of the two (Bostelaar, 2024).

WHO DEVELOPED IT

Kurt Bostelaar, a learning-experience designer, sets out the spectrum in his playbook *Clarify Your Core Message*, alongside the distil–decompose–prioritise method. The underlying distinction between motivational and informational communication has a long history in instructional design, notably in Keller's ARCS model of motivation (Keller, 1987).

WHY IT WAS DEVELOPED

The playbook addresses a failure familiar to anyone who has taught: a technically excellent explanation delivered to an audience that has not yet accepted the topic matters, or an inspiring case made to an audience that already agreed and now needs the detail. Both are well-made messages aimed at the wrong point of the spectrum, and the diagnosis is invisible without the axis.

FIGURE 36 · FIVE READINGS THAT LOCATE THE BLEND

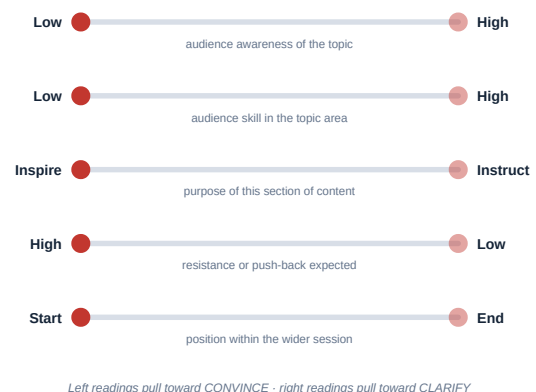


Figure 36. Five assessments, averaged left to right, locate the message on the spectrum. Low awareness and high resistance pull toward convince (Bostelaar, 2024).

HOW IT IS USED

Rate the five dimensions before drafting, average them, and sketch the intended blend as a flow — why, then what, then why again — with a rough percentage for each.

WHEN IT IS USED

Before designing any teaching session, pitch or briefing, and when a well-prepared message has not landed with a particular audience.

BY WHOM IT IS USED

Educators and trainers sequencing content, managers announcing change to a sceptical team, and anyone presenting the same material to audiences of differing readiness.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Presentation advice generally treats clarity as the sole objective, which is correct only for audiences who already accept the premise. Bostelaar makes the convince–clarify balance an explicit design variable set by audience diagnosis rather than by preference. It is a practitioner instrument without formal validation; its value is that it turns an intuition into a decision that can be recorded and reviewed.

RELATED FRAMEWORKS IN THIS COURSE

Distil → Decompose → Prioritise · **Slibrary 9**

Storyline Canvas · **Slibrary 9**

Pyramid Principle · **Slibrary 9**

High- and Low-Context Communication · **Slibrary 10**

SOURCES (HARVARD REFERENCING STYLE)

Bostelaar, K. (2024) *Clarify Your Core Message: A Playbook*. Leverlo.

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Keller, J.M. (1987) 'Development and use of the ARCS model of instructional design', *Journal of Instructional Development*, 10(3), pp. 2–10.

Problem → Insight → Recommendation

SLIBRARY 08
SHEET 2 / 5

Three moves that turn analysis into a decision someone can take

PROFESSIONAL DEFINITION

The Problem–Insight–Recommendation structure organises a persuasive briefing into three moves. The problem states what is wrong and what it costs, in the audience's own terms. The insight states the non-obvious finding that reframes the problem — the element the audience did not already know. The recommendation states the specific action requested, with its owner, cost and first step. A briefing without an insight is a status report (Minto, 1987; Gallo, 2014).

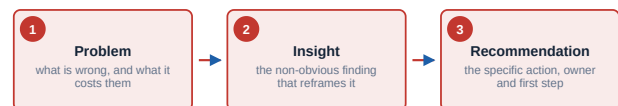
WHO DEVELOPED IT

The structure is standard practice in management consulting, descending from Barbara Minto's pyramid work at McKinsey. Its articulation for presenters — in particular the emphasis on the single unexpected element an audience will remember — is associated with Carmine Gallo's analysis of TED presentations and communication in *Talk Like TED* (2014).

WHY IT WAS DEVELOPED

Two failure modes recur in professional presentations. The first is analysis presented in the order it was performed, which asks the audience to reconstruct the argument. The second is a complete, accurate account containing nothing the audience did not already believe. The insight move exists to prevent the second: it forces the presenter to name what actually changes as a result of the work.

FIGURE 37 · THREE MOVES FROM ANALYSIS TO DECISION



No insight means you have written a status report

Figure 37. Three moves in fixed order. The middle move is the test: if the audience already knew it, the briefing has no reason to exist (Gallo, 2014).

HOW IT IS USED

Write the insight sentence first and check it against the audience's prior belief. If they would have said it themselves, the analysis is not finished.

WHEN IT IS USED

Before any recommendation to a decision-maker, in interview case questions, and when compressing extensive analysis into a short slot.

BY WHOM IT IS USED

Consultants and analysts briefing clients, managers proposing investment, and students presenting case analysis under time constraint.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Chronological reporting mirrors the analyst's process and places the conclusion last, where a senior audience may never reach it. This structure front-loads relevance and isolates the contribution as a named element. It is practitioner convention rather than a validated model, and it compresses badly when a decision genuinely rests on several independent findings.

RELATED FRAMEWORKS IN THIS COURSE

Pyramid Principle · [Slibrary 9](#)

SCQA Structure · [Slibrary 5](#)

BLUF Executive Structure · [Slibrary 9](#)

Assertion–Evidence Slide Model · [Slibrary 8](#)

SOURCES (HARVARD REFERENCING STYLE)

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Gallo, C. (2014) *Talk Like TED: The 9 Public Speaking Secrets of the World's Top Minds*. New York: St Martin's Press.

Duarte, N. (2010) *Resonate: Present Visual Stories that Transform Audiences*. Hoboken, NJ: John Wiley & Sons.

Assertion–Evidence Slide Model

SLIBRARY 08
SHEET 3 / 5

A sentence headline carrying the claim, supported by a visual — not a bulleted topic

PROFESSIONAL DEFINITION

The assertion–evidence model replaces the conventional slide of a topic phrase above bullet points with two components: a complete sentence headline stating the claim the slide makes, and visual evidence supporting it. Words that are not part of the headline are spoken rather than displayed. The design follows from evidence that a slide read and heard simultaneously competes for the same verbal channel and reduces comprehension (Alley and Neeley, 2005).

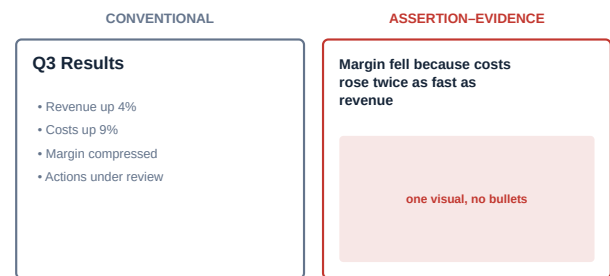
WHO DEVELOPED IT

Michael Alley, professor of engineering communication at Pennsylvania State University, developed and tested the model with colleagues from the mid-1990s. Its theoretical basis is Richard Mayer's cognitive theory of multimedia learning, in particular the redundancy and modality effects, which have been demonstrated repeatedly under controlled conditions (Mayer, 2009; Alley et al., 2006).

WHY IT WAS DEVELOPED

Alley's work began from a practical failure in engineering education: audiences retained little from technically sound presentations. The mechanism identified is redundancy — an audience reading a text bullet while hearing a spoken version processes both through the verbal channel and does neither well. Comparative studies of the assertion–evidence format report measurably better retention than the topic-and-bullet default.

FIGURE 38 · THE DEFAULT SLIDE AND THE ALTERNATIVE



The headline carries the claim; the body carries the evidence

Figure 38. A topic phrase leaves the claim to be inferred; a sentence headline states it, leaving the body free to carry evidence rather than text (Alley and Neeley, 2005).

HOW IT IS USED

Write the headline as a full sentence stating what the slide proves, then ask what single visual would demonstrate it. Everything else is said, not shown.

WHEN IT IS USED

When preparing any technical or analytical presentation, and when slides are being used as speaker notes rather than as audience support.

BY WHOM IT IS USED

Presenters of technical and analytical material, educators designing lecture slides, and anyone whose deck is being read aloud.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The default topic-and-bullet layout is inherited from software templates rather than from research on comprehension, and it obliges the audience to infer the claim. Assertion–evidence states the claim and supplies evidence, with support from controlled studies rather than preference. The cost is preparation time and the loss of the deck as a standalone document — a real trade-off where slides are circulated instead of presented.

RELATED FRAMEWORKS IN THIS COURSE

Problem → Insight → Recommendation · **Slibrary 8**

Pyramid Principle · **Slibrary 9**

BLUF Executive Structure · **Slibrary 9**

Mehrabian 7–38–55 · **Slibrary 8**

SOURCES (HARVARD REFERENCING STYLE)

Alley, M. and Neeley, K.A. (2005) 'Rethinking the design of presentation slides', *Technical Communication*, 52(4), pp. 417–426.

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Mayer, R.E. (2009) *Multimedia Learning*. 2nd edn. Cambridge: Cambridge University Press.

Mehrabian 7–38–55

SLIBRARY 08
SHEET 4 / 5

Words · tone · face — the most misquoted numbers in communication training

PROFESSIONAL DEFINITION

Mehrabian's studies found that when a speaker's words, vocal tone and facial expression convey inconsistent messages about feelings and attitudes, listeners weight the channels approximately 7 per cent verbal, 38 per cent vocal and 55 per cent facial in judging the speaker's attitude. The result applies to inconsistent affective signals only. It does not state that 93 per cent of all communication is non-verbal, a claim Mehrabian has repeatedly disowned (Mehrabian and Ferris, 1967).

WHO DEVELOPED IT

Albert Mehrabian, professor emeritus of psychology at the University of California, Los Angeles, conducted the underlying experiments in 1967 with Morton Wiener and Susan Ferris using single spoken words and photographs. He has published explicit corrections of the popular generalisation, which persists in communication training regardless (Mehrabian, 1971).

WHY IT WAS DEVELOPED

The research question was narrow and legitimate: when channels disagree, which one does a listener believe? The answer — the non-verbal channels — matters for presentation because a speaker whose delivery contradicts their content will be read through the delivery. The wider myth is instructive in its own right as a case of a specific finding stripped of its conditions and repeated until it became common knowledge.

FIGURE 39 · THE THREE CHANNELS AND THEIR WEIGHTS



These weights apply only when the three channels contradict each other about feeling or attitude — not to communication in general

Figure 39. The three weights as measured. The scope condition is the whole finding: inconsistent signals about attitude, not communication at large (Mehrabian and Ferris, 1967).

HOW IT IS USED

Use it as a consistency check rather than a ratio: rehearse until tone and expression match the claim, and treat any mismatch as the message the audience will actually receive.

WHEN IT IS USED

When rehearsing a high-stakes delivery, when feedback reports that a message did not feel sincere, and whenever the 93 per cent claim is cited in your presence.

BY WHOM IT IS USED

Presenters and negotiators rehearsing delivery, trainers correcting a widespread misattribution, and anyone delivering difficult news.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Before Mehrabian, the relative credibility of communication channels was asserted rather than measured; his contribution was to measure it under controlled conditions. The framework now serves a double purpose in this course: a genuine finding about consistency of delivery, and a documented example of how a narrow experimental result acquires a false general claim once its scope conditions are dropped.

RELATED FRAMEWORKS IN THIS COURSE

Assertion–Evidence Slide Model · **Slibrary 8**Anxiety Reappraisal · **Slibrary 8**CRAAP Source Test · **Slibrary 2**Tactical Empathy · **Slibrary 14**

SOURCES (HARVARD REFERENCING STYLE)

Mehrabian, A. and Ferris, S.R. (1967) 'Inference of attitudes from nonverbal communication in two channels', *Journal of Consulting Psychology*, 31(3), pp. 248–252.

Mehrabian, A. and Wiener, M. (1967) 'Decoding of inconsistent communications', *Journal of Personality and Social Psychology*, 6(1), pp. 109–114.

Mehrabian, A. (1971) *Silent Messages*. Belmont, CA: Wadsworth.

Lapakko, D. (1997) 'Three cheers for language: a closer examination of a widely cited study of nonverbal communication', *Communication Education*, 46(1), pp. 63–67.

Anxiety Reappraisal

SLIBRARY 08
SHEET 5 / 5

Say 'I am excited' — relabelling arousal outperforms trying to calm down

PROFESSIONAL DEFINITION

Anxiety reappraisal is the deliberate reinterpretation of pre-performance arousal as excitement rather than anxiety. Both states share high physiological activation and differ in valence, so reframing requires only a change of interpretation, whereas attempting to calm down requires a change of arousal. Across experiments in singing, public speaking and mathematics, participants instructed to say 'I am excited' outperformed those told to calm down or say nothing (Brooks, 2014).

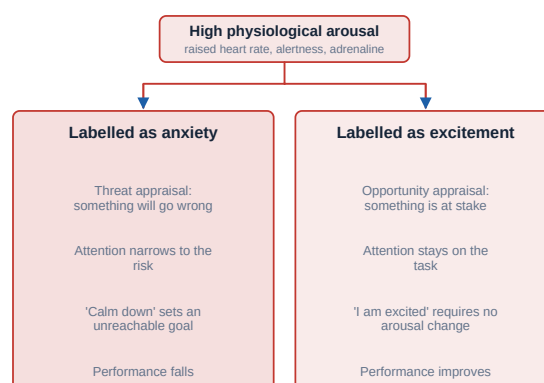
WHO DEVELOPED IT

Alison Wood Brooks, professor at Harvard Business School, established the effect in a 2014 paper in the *Journal of Experimental Psychology: General*. It builds on Schachter and Singer's two-factor theory of emotion, in which an emotional state results from physiological arousal plus a cognitive label, and on Jamieson's work on stress appraisal (Schachter and Singer, 1962; Jamieson et al., 2012).

WHY IT WAS DEVELOPED

The standard advice before a performance is to relax, which sets an objective the performer cannot meet: arousal is already high and attempting to suppress it supplies further evidence that something is wrong. Brooks reasoned that shifting the label rather than the arousal would be both easier and more effective, and that reframing toward an opportunity mindset would improve performance rather than merely feelings.

FIGURE 40 · ONE AROUSAL STATE, TWO AVAILABLE LABELS



The arousal is identical; only the appraisal differs

Figure 40. A single physiological state with two possible labels. Reappraisal changes the label, which is achievable, rather than the arousal, which is not (Brooks, 2014).

HOW IT IS USED

Say 'I am excited' aloud before the performance and name one specific opportunity in the situation. Do not attempt to reduce the physical arousal.

WHEN IT IS USED

Immediately before a pitch, presentation, viva, interview or negotiation, and whenever advice to relax has failed.

BY WHOM IT IS USED

Presenters and candidates preparing to perform, educators coaching students through assessment nerves, and anyone whose preparation is undermined by anticipatory anxiety.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Prior interventions targeted arousal downward through breathing and relaxation, which asks for a physiological change under time pressure. Reappraisal targets the appraisal instead, which is under direct voluntary control and can be applied in seconds. The effects reported are modest in size and drawn largely from short laboratory tasks, so it is best treated as a low-cost adjunct to preparation rather than a substitute for it.

RELATED FRAMEWORKS IN THIS COURSE

Growth vs. Fixed Mindset Fork · **Slibrary 4**

Mehrabian 7–38–55 · **Slibrary 8**

Deliberate Practice Loop · **Slibrary 1**

Tactical Empathy · **Slibrary 14**

SOURCES (HARVARD REFERENCING STYLE)

- Brooks, A.W. (2014) 'Get excited: reappraising pre-performance anxiety as excitement', *Journal of Experimental Psychology: General*, 143(3), pp. 1144–1158.
- Schachter, S. and Singer, J.E. (1962) 'Cognitive, social, and physiological determinants of emotional state', *Psychological Review*, 69(5), pp. 379–399.
- Jamieson, J.P., Nock, M.K. and Mendes, W.B. (2012) 'Mind over matter: reappraising arousal improves cardiovascular and cognitive responses to stress', *Journal of Experimental Psychology: General*, 141(3), pp. 417–422.

Pyramid Principle

SLIBRARY 09
SHEET 1 / 5

Answer first, then grouped support, then data — an argument shaped for a busy reader

PROFESSIONAL DEFINITION

The Pyramid Principle organises writing so that the governing thought appears first and everything below it supports it. Ideas at any level summarise the ideas grouped beneath them; ideas in each group are of the same kind; and they are ordered logically — by time, structure, or degree. The reader can therefore stop at any level and still hold a complete, correct argument (Minto, 1987).

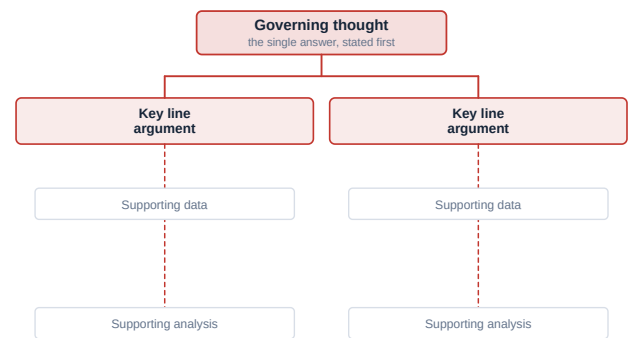
WHO DEVELOPED IT

Barbara Minto joined McKinsey & Company in 1963 as its first female consultant and was later appointed to teach writing across the firm's European offices. She published the method independently in 1987 and it has since become the default structure of professional consulting communication and much executive writing beyond it (Minto, 1987).

WHY IT WAS DEVELOPED

Minto's observation was that writers organise documents by the order in which they did the work, while readers need the order in which they will make the decision. Those two orders are almost never the same. The pyramid resolves the conflict by putting the conclusion at the top, which also imposes discipline on the writer: an argument that cannot be summarised in one sentence has not been finished.

FIGURE 41 · ONE GOVERNING THOUGHT, SUPPORTED DOWNWARD



Each level summarises the level beneath it

Figure 41. Ideas summarise what sits below them and group with what sits beside them. A reader who stops early still holds a complete argument (Minto, 1987).

HOW IT IS USED

Write the single governing sentence first, then list the three or four arguments that would make a sceptical reader accept it, then gather evidence under each.

WHEN IT IS USED

Before writing any report, memo or proposal, and when a draft is accurate but readers keep asking what the point is.

BY WHOM IT IS USED

Consultants and analysts, managers writing to senior stakeholders, and students structuring written assessment at Masters level.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Academic and technical writing conventionally builds toward a conclusion, which suits a reader obliged to read the whole document and fails with one who is not. Minto inverted the order and specified the logical rules that make the inversion sound. The structure is demanding to produce and is poorly suited to genuinely exploratory writing, where the conclusion is not yet known.

RELATED FRAMEWORKS IN THIS COURSE

SCQA Structure · [Slibrary 5](#)BLUF Executive Structure · [Slibrary 9](#)MECE Issue Tree · [Slibrary 5](#)Problem → Insight → Recommendation · [Slibrary 8](#)

SOURCES (HARVARD REFERENCING STYLE)

Minto, B. (1987) *The Pyramid Principle: Logic in Writing and Thinking*. London: Minto International.

Minto, B. (2009) *The Pyramid Principle: Logic in Writing and Thinking*. 3rd edn. Harlow: Pearson Education.

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Distil → Decompose → Prioritise

SLIBRARY 09
SHEET 2 / 5

Write long, break it apart, force a ranking — three passes to one clear sentence

PROFESSIONAL DEFINITION

Bostelaar's method reduces a complex message to a single accessible line in three passes. Distil: write the message at length, then compress it repeatedly — a paragraph, then three sentences, then one. Decompose: break the message into its component concepts as isolated items, so that unnecessary ones become visible. Prioritise: force-rank those concepts from most to least important, which reveals the primary emphasis rather than leaving it assumed (Bostelaar, 2024).

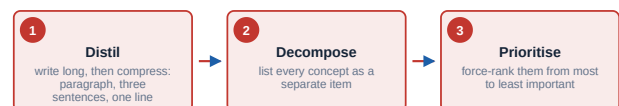
WHO DEVELOPED IT

Kurt Bostelaar, a learning-experience designer, sets out the method in his playbook *Clarify Your Core Message* as the operational core of message clarification, alongside the message strategy spectrum. It develops a long practitioner tradition of compression as a test of understanding, of which Feynman's explanation technique is the best-known instance.

WHY IT WAS DEVELOPED

The method addresses a specific failure: writers who cannot say which of their points is the most important, because they have never been made to choose. Distillation alone tends to produce a shorter version of the same confusion. Decomposition externalises the concepts so they can be compared, and forced ranking makes the choice unavoidable — which is why the third pass, not the first, produces the clarity.

FIGURE 42 · THREE PASSES OVER ONE MESSAGE



The forced ranking is where the emphasis is actually decided

Figure 42. Three passes in fixed order. Distillation shortens; decomposition separates; only forced ranking establishes the primary emphasis (Bostelaar, 2024).

HOW IT IS USED

Write the long version without editing, list the concepts it contains, rank them, then rewrite the one-liner around the top-ranked concept only.

WHEN IT IS USED

Before a lecture, pitch, abstract or executive summary, and whenever a message contains several good points and no evident main one.

BY WHOM IT IS USED

Educators preparing sessions, professionals writing summaries, and anyone whose message is correct but leaves the audience unsure what to take away.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Conventional editing advice — cut words, remove jargon — improves readability without resolving emphasis, so a shortened message can remain unclear about its own priority. This method makes prioritisation a separate, explicit pass with a visible output. It is a practitioner instrument rather than a validated one; its value is procedural, in that it prevents the ranking step from being skipped.

RELATED FRAMEWORKS IN THIS COURSE

Message Strategy Spectrum · **Slibrary 8**

Storyline Canvas · **Slibrary 9**

Pyramid Principle · **Slibrary 9**

The Rule of 3 · **Slibrary 15**

SOURCES (HARVARD REFERENCING STYLE)

Bostelaar, K. (2024) *Clarify Your Core Message: A Playbook*. Leverlo.

Heath, C. and Heath, D. (2007) *Made to Stick: Why Some Ideas Survive and Others Die*. New York: Random House.

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Toulmin Argument Model

SLIBRARY 09
SHEET 3 / 5

Claim · grounds · warrant · backing · qualifier · rebuttal — the anatomy of a real argument

PROFESSIONAL DEFINITION

Toulmin's model sets out six functional components of a practical argument. The claim is the position asserted; the grounds are the facts offered in support; the warrant is the principle licensing the move from grounds to claim; the backing supports the warrant itself; the qualifier states the strength claimed, such as 'probably'; and the rebuttal states the conditions under which the claim would not hold. The warrant is usually left implicit, which is where arguments fail (Toulmin, 1958).

WHO DEVELOPED IT

Stephen E. Toulmin, a British philosopher trained at Cambridge under Wittgenstein and later professor at the University of Southern California, published the model in *The Uses of Argument* (1958). The book was received poorly by formal logicians and became foundational in rhetoric, communication studies and legal reasoning instead (Toulmin, Rieke and Janik, 1984).

WHY IT WAS DEVELOPED

Toulmin's argument was that formal deductive logic, modelled on mathematics, describes almost nothing of how people reason in law, medicine, management or ethics, where conclusions are probable and defeasible rather than certain. He proposed a structure drawn from jurisprudence instead. Its practical payoff is diagnostic: most disagreements turn out to concern an unstated warrant rather than the facts.

FIGURE 43 · THE SIX COMPONENTS OF AN ARGUMENT



The warrant is usually unstated — and usually where the dispute lies

Figure 43. Six components, of which three are habitually left implicit. Naming the warrant locates most disagreements precisely (Toulmin, 1958).

HOW IT IS USED

Write out all six for your own argument, especially the warrant and the rebuttal. If you cannot state a rebuttal condition, the claim is being overstated.

WHEN IT IS USED

When preparing a contested recommendation, when writing at Masters level where qualification is expected, and when a disagreement is circling without progress.

BY WHOM IT IS USED

Students constructing academic arguments, professionals defending recommendations, and anyone analysing why two parties reading the same evidence disagree.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Syllogistic logic evaluates arguments as valid or invalid, a binary that discards the qualified, defeasible reasoning used in every applied field. Toulmin added the qualifier and the rebuttal, making degrees of strength and stated exceptions part of the argument's structure rather than signs of weakness. The model describes and diagnoses arguments; it does not adjudicate between competing ones.

RELATED FRAMEWORKS IN THIS COURSE

Pyramid Principle · **Slibrary 9**

CRAAP Source Test · **Slibrary 2**

Ladder of Inference · **Slibrary 6**

Confirmation Bias Loop · **Slibrary 6**

SOURCES (HARVARD REFERENCING STYLE)

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BLUF Executive Structure

SLIBRARY 09
SHEET 4 / 5

Bottom line up front — the conclusion in the first sentence, support after it

PROFESSIONAL DEFINITION

BLUF places the conclusion, decision or request in the opening sentence, followed by supporting detail in decreasing order of importance and background last. The structure assumes a reader who may stop at any point, so the most consequential information is never left to the end. Doumont formalises the underlying principle as optimising the message for the audience's attention rather than for the writer's completeness (Doumont, 2009).

WHO DEVELOPED IT

The convention originates in US military staff writing, where doctrine requires the recommendation before the analysis. Jean-luc Doumont, an engineer and communication specialist trained at Louvain and Stanford, gives it a systematic treatment in *Trees, Maps, and Theorems* (2009), deriving it from the constraints of attention and memory rather than from convention (Doumont, 2009).

WHY IT WAS DEVELOPED

The problem addressed is asymmetric attention: senior readers allocate a fraction of the time the writer spent, and read declining amounts as a document proceeds. A conclusion at the end is therefore frequently never reached. BLUF also has a drafting benefit — a writer unable to state the bottom line in one sentence has usually not finished the thinking.

FIGURE 44 · INFORMATION IN DESCENDING ORDER OF CONSEQUENCE

Bottom line	the decision, conclusion or request
Key support	the two or three reasons it holds
Detail	evidence, method, figures
Background	context, for those who need it

The reader who stops after line one still has what they need

Figure 44. Four blocks in descending consequence. The order is the opposite of the order in which the analysis was performed (Doumont, 2009).

HOW IT IS USED

Write the document, then move the last paragraph to the front and delete whatever now repeats it. Test it by reading only the first sentence.

WHEN IT IS USED

In emails to senior stakeholders, in executive summaries, and any time a reader has less time than the document requires.

BY WHOM IT IS USED

Anyone writing upward in an organisation, project managers reporting status, and professionals whose messages routinely go unanswered.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Narrative and academic conventions build toward a conclusion, which rewards the complete reader and penalises the partial one. BLUF optimises for partial reading, which is the realistic case in professional settings. The trade-off is genuine: it suits decisions and reduces persuasive build-up, so it is poorly suited to writing whose purpose is to bring a resistant reader gradually to an unwelcome conclusion.

RELATED FRAMEWORKS IN THIS COURSE

[Pyramid Principle · Slibrary 9](#)[Problem → Insight → Recommendation · Slibrary 8](#)[Assertion–Evidence Slide Model · Slibrary 8](#)[The Rule of 3 · Slibrary 15](#)

SOURCES (HARVARD REFERENCING STYLE)

Doumont, J.-L. (2009) *Trees, Maps, and Theorems: Effective Communication for Rational Minds*. Kraainem: Principiae.

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Storyline Canvas

SLIBRARY 09

SHEET 5 / 5

Plot, character and story mapped across beginning, middle and end — nine cells

PROFESSIONAL DEFINITION

Bostelaar's Storyline Canvas maps a teaching story on a three-by-three grid. The plot row runs beginning, middle, end — setup, conflict, conclusion. The character row runs misbelief, insight, understanding: the protagonist first lives out the opposite of the lesson, then begins to grasp it, then fully embodies it. The story row is the narrated combination of the two. The message is the life lesson the character learns, never stated directly (Bostelaar, 2024).

WHO DEVELOPED IT

Kurt Bostelaar sets out the canvas in his playbook *Craft Stories to Teach and Explain*, following the companion playbook on clarifying the core message. The plot row uses the standard dramatic structure — inciting incident, point of no return, crisis, climax — while the character row applies the transformation arc familiar from screenwriting craft (Bostelaar, 2024).

WHY IT WAS DEVELOPED

Bostelaar's sequencing argument is that a story should be built only after the message has been distilled, because a story constructed first will communicate whatever it happens to contain. The canvas prevents the common failure of an anecdote that is engaging but teaches nothing: if the character does not begin with a misbelief and end with understanding, no lesson has been transmitted.

FIGURE 45 · NINE CELLS: PLOT, CHARACTER, STORY

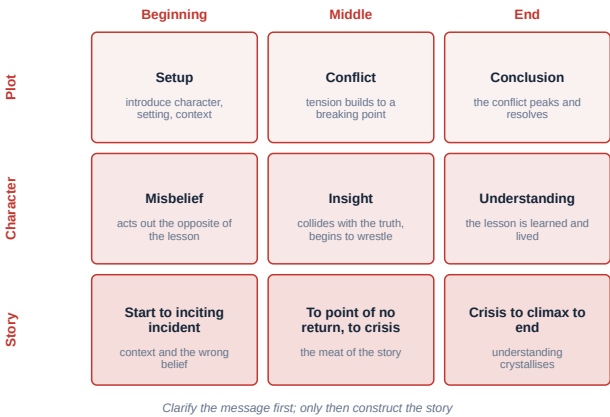


Figure 45. Three rows across three acts. The character row carries the message; the plot row carries the attention (Bostelaar, 2024).

HOW IT IS USED

Complete the character row first — misbelief, insight, understanding — then design plot events that force each transition, and write the story row last.

WHEN IT IS USED

When preparing a teaching case, an opening anecdote, or any explanation that must be remembered rather than merely understood.

BY WHOM IT IS USED

Educators building narrative teaching material, leaders communicating change, and presenters opening with a story that has to earn its place.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Storytelling advice usually supplies a plot template and leaves the lesson to emerge, which produces memorable stories with no reliable teaching function. The canvas separates the character transformation from the plot and requires the message to be fixed before either is designed. It is a design instrument grounded in narrative craft rather than in experimental research on retention.

RELATED FRAMEWORKS IN THIS COURSE

Distil → Decompose → Prioritise · **Slibrary 9**

Message Strategy Spectrum · **Slibrary 8**

StoryBrand SB7 Framework · **Slibrary 7**

Kolb Experiential Learning Cycle · **Slibrary 1**

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Shannon–Weaver Model

SLIBRARY 10
SHEET 1 / 5

Source · transmitter · channel · receiver · destination — and noise at every step

PROFESSIONAL DEFINITION

The Shannon–Weaver model represents communication as a linear transmission: an information source selects a message, a transmitter encodes it as a signal, the signal travels through a channel where noise may distort it, a receiver decodes the signal, and it reaches its destination. Noise is any interference between transmission and reception. Weaver's commentary extended the engineering account to human communication, adding semantic and effectiveness problems (Shannon and Weaver, 1949).

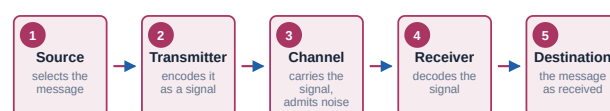
WHO DEVELOPED IT

Claude Shannon, a mathematician at Bell Telephone Laboratories, published the mathematical theory of communication in 1948 to solve a signal-transmission problem. Warren Weaver wrote the interpretive essay published with it in 1949, which extended the model beyond engineering — a move Shannon himself did not make and did not encourage (Shannon, 1948).

WHY IT WAS DEVELOPED

Shannon's problem was the reliable reproduction of a signal at a distance, irrespective of meaning. Weaver's contribution was to observe that the same structure exposes where human communication fails: encoding into words the receiver decodes differently, channel noise, and receivers whose decoding is shaped by their own context. Its enduring teaching value is that it makes noise a named, addressable component.

FIGURE 46 · FIVE STAGES AND THE NOISE BETWEEN THEM



Noise enters at the channel; decoding differs from encoding

Figure 46. Five stages in a linear chain. The model's limitation is its own shape: real communication is not one-directional (Shannon and Weaver, 1949).

HOW IT IS USED

Locate the failure precisely: was the message poorly encoded, the channel noisy, or the decoding shaped by a context you did not account for?

WHEN IT IS USED

When diagnosing a communication breakdown, when choosing a channel for a difficult message, and when the same message lands differently with different audiences.

BY WHOM IT IS USED

Communications professionals diagnosing failures, managers selecting channels, and students learning the vocabulary of communication theory.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Before Shannon there was no formal account of information transmission at all; his paper founded information theory and made noise a quantifiable term. The model's weakness is structural: it is linear and one-way, with no feedback and no shared field of experience, which is why transactional models such as Schramm's and Berlo's superseded it for human communication (Schramm, 1954).

RELATED FRAMEWORKS IN THIS COURSE

High- and Low-Context Communication · **Slibrary 10**

The Culture Map (Eight Scales) · **Slibrary 10**

Mehrabian 7–38–55 · **Slibrary 8**

Ladder of Inference · **Slibrary 6**

SOURCES (HARVARD REFERENCING STYLE)

Shannon, C.E. and Weaver, W. (1949) *The Mathematical Theory of Communication*. Urbana, IL: University of Illinois Press.

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Hofstede Cultural Dimensions

SLIBRARY 10
SHEET 2 / 5

Six dimensions on which national cultures differ — averages, never individuals

PROFESSIONAL DEFINITION

Hofstede's framework describes national cultures on six dimensions: power distance, the acceptance of unequal power; individualism versus collectivism; motivation toward achievement versus caring for others, originally termed masculinity–femininity; uncertainty avoidance, the tolerance of ambiguity; long-term versus short-term orientation; and indulgence versus restraint. Scores are country averages and carry no implication about any individual (Hofstede, 2001).

WHO DEVELOPED IT

Geert Hofstede, a Dutch social psychologist, derived the first four dimensions from an employee-attitude survey of more than one hundred thousand IBM staff across some seventy countries between 1967 and 1973. Long-term orientation was added with Michael Harris Bond from the Chinese Value Survey, and indulgence with Michael Minkov from World Values Survey data (Hofstede, Hofstede and Minkov, 2010).

WHY IT WAS DEVELOPED

Hofstede's data showed that IBM employees doing identical work under identical policies answered systematically differently by country, which established that national culture shaped workplace behaviour independently of the organisation. The dimensions gave international management its first quantified comparative vocabulary, replacing anecdote with something that could be tested.

FIGURE 47 · SIX DIMENSIONS AS CONTINUOUS SCALES



Figure 47. Six continuous dimensions on which national averages differ. Within-country variation exceeds between-country variation on every one (Hofstede, 2001).

HOW IT IS USED

Use it to anticipate where friction is likely — for instance around directness, deadlines or decision rights — then verify against the individual rather than assuming.

WHEN IT IS USED

Before working with an unfamiliar national context, when a cross-border team is misunderstanding intent, and when designing policy for multiple countries.

BY WHOM IT IS USED

International managers, students in multinational cohorts, and negotiators preparing for counterparts from unfamiliar business cultures.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Cross-cultural management before Hofstede was descriptive and anecdotal, with no way to compare countries systematically. He supplied quantified, comparable dimensions from a controlled sample. The framework is heavily criticised on solid grounds: a single company's employees, data now decades old, nations equated with cultures, and — most importantly for practice — variation within a country far exceeding the difference between countries (McSweeney, 2002).

RELATED FRAMEWORKS IN THIS COURSE

The Culture Map (Eight Scales) · **Slibrary 10**

High- and Low-Context Communication · **Slibrary 10**

Power–Interest Grid · **Slibrary 10**

Principled Negotiation · **Slibrary 13**

SOURCES (HARVARD REFERENCING STYLE)

Hofstede, G. (2001) *Culture's Consequences: Comparing Values, Behaviors, Institutions and Organizations Across Nations*. 2nd edn. Thousand Oaks, CA: Sage.

Hofstede, G., Hofstede, G.J. and Minkov, M. (2010) *Cultures and Organizations: Software of the Mind*. 3rd edn. New York: McGraw-Hill.

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The Culture Map (Eight Scales)

SLIBRARY 10
SHEET 3 / 5

Communicating · evaluating · persuading · leading · deciding · trusting · disagreeing · scheduling

PROFESSIONAL DEFINITION

Meyer's framework positions cultures on eight behavioural scales relevant to working together: communicating, from low-context to high-context; evaluating, from direct to indirect negative feedback; persuading, from principles-first to applications-first; leading, from egalitarian to hierarchical; deciding, from consensual to top-down; trusting, from task-based to relationship-based; disagreeing, from confrontational to confrontation-avoidant; and scheduling, from linear to flexible time. Position is relative, not absolute (Meyer, 2014).

WHO DEVELOPED IT

Erin Meyer, professor of management practice at INSEAD, developed the map from work with executive teams and published it in *The Culture Map* (2014). It builds on Edward T. Hall's context and time concepts and on Hofstede's dimensional tradition, reworked around observable workplace behaviours rather than values (Hall, 1976).

WHY IT WAS DEVELOPED

Meyer's argument is that value-level frameworks tell a manager that two cultures differ without telling them what to do on Monday morning. The eight scales are chosen because each generates a concrete, observable practice — how feedback is phrased, whether the decision after the meeting is final. The relative-position rule is central: a culture is direct or indirect only compared with another.

FIGURE 48 · EIGHT SCALES OF WORKING BEHAVIOUR

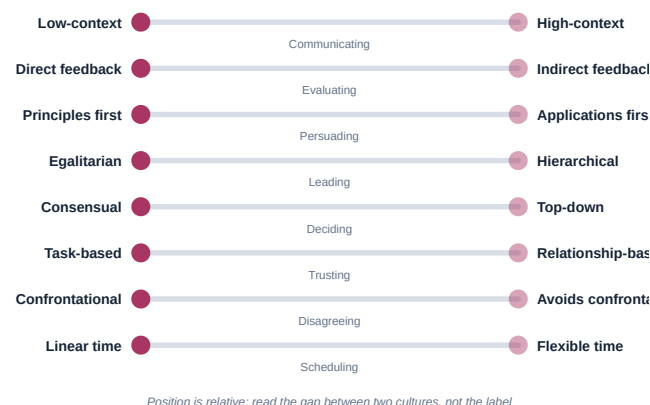


Figure 48. Eight scales, each an observable working practice. What matters is the distance between two cultures on a scale, not either one's absolute position (Meyer, 2014).

HOW IT IS USED

Plot both cultures on the scales that bear on the task, then adjust on the one or two with the widest gap rather than attempting to adapt on all eight.

WHEN IT IS USED

Before leading or joining a cross-cultural team, before giving feedback across cultures, and when a working relationship is deteriorating without a substantive disagreement.

BY WHOM IT IS USED

Managers of international teams, students working in multinational groups, and professionals relocating or negotiating across borders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Hofstede measured values; Meyer measures behaviours, which is what a manager can actually adjust, and adds the relative-position rule that prevents a culture being labelled 'direct' in the abstract. The evidence base is practitioner observation with executive cohorts rather than large-sample survey data, so it is stronger as a preparation and diagnosis tool than as a source of country scores.

RELATED FRAMEWORKS IN THIS COURSE

Hofstede Cultural Dimensions · **Slibrary 10**

High- and Low-Context Communication · **Slibrary 10**

Dual Concerns Conflict Modes · **Slibrary 13**

DISC Behavioural Quadrant · **Slibrary 4**

SOURCES (HARVARD REFERENCING STYLE)

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High- and Low-Context Communication

SLIBRARY 10
SHEET 4 / 5

How much meaning sits in the words, and how much in everything around them

PROFESSIONAL DEFINITION

Hall's distinction concerns where meaning is located. In low-context communication most of the meaning is carried explicitly in the words, so messages are literal, detailed and designed to be understood without shared background. In high-context communication much of the meaning resides in the setting, the relationship, the status of the speakers and what is deliberately left unsaid, so the same words carry more, and less, than they appear to (Hall, 1976).

WHO DEVELOPED IT

Edward T. Hall, an American anthropologist who worked for the US Foreign Service training personnel for overseas postings, introduced the distinction in *Beyond Culture* (1976), having earlier developed proxemics and the monochronic–polychronic account of time. His work founded the field of intercultural communication (Hall, 1959; Hall, 1966).

WHY IT WAS DEVELOPED

Hall was addressing repeated diplomatic and commercial failures in which both parties behaved correctly by their own standards. His finding was that the disagreement was often about how much should be said: low-context communicators experience high-context partners as evasive, while high-context communicators experience low-context partners as blunt and mistrustful. Each reads the other's competence as a character flaw.

FIGURE 49 · WHERE THE MEANING IS CARRIED

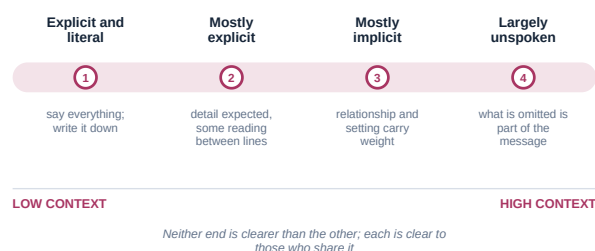


Figure 49. A continuum from meaning in the words to meaning in the surroundings. Misreadings run in both directions and are usually attributed to character (Hall, 1976).

HOW IT IS USED

Adjust your own position rather than expecting adjustment: put decisions in writing with high-context partners, and read what is not said with low-context ones.

WHEN IT IS USED

In cross-cultural negotiation, when written agreements are interpreted differently, and when a counterpart seems either evasive or abrupt.

BY WHOM IT IS USED

Negotiators and managers working internationally, students in mixed cohorts, and anyone whose written communication is being read differently from how it was meant.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Communication theory before Hall treated meaning as residing in the message, making misunderstanding a matter of noise or poor encoding. Hall relocated part of the meaning outside the message entirely, which explains failures where the message was transmitted perfectly. The distinction is a continuum applied to whole cultures and has limited quantitative support, so it is best used to generate hypotheses about a counterpart, not conclusions.

RELATED FRAMEWORKS IN THIS COURSE

The Culture Map (Eight Scales) · **Slibrary 10**

Hofstede Cultural Dimensions · **Slibrary 10**

Shannon–Weaver Model · **Slibrary 10**

Tactical Empathy · **Slibrary 14**

SOURCES (HARVARD REFERENCING STYLE)

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Power–Interest Grid

SLIBRARY 10
SHEET 5 / 5

Who can affect the outcome, and who cares — four engagement strategies

PROFESSIONAL DEFINITION

The Power–Interest Grid classifies stakeholders on two axes: their power to affect the outcome, and their interest in it. High power with high interest requires close management and genuine participation; high power with low interest requires keeping satisfied, since dissatisfaction converts latent power into active opposition; low power with high interest requires keeping informed; and low power with low interest requires monitoring only (Mendelow, 1981).

WHO DEVELOPED IT

Aubrey L. Mendelow presented the grid at the 1981 International Conference on Information Systems in Cambridge, Massachusetts, within work on stakeholder mapping for strategic planning. It is now standard in project management and corporate strategy texts and sits within the broader stakeholder tradition associated with R. Edward Freeman (Freeman, 1984).

WHY IT WAS DEVELOPED

Mendelow's concern was that organisations consulted stakeholders either uniformly, which is expensive and dilutes attention, or by seniority, which misses influential parties without formal authority. The grid allocates engagement effort by consequence. Its most useful cell is high power with low interest: these stakeholders are ignored until something raises their interest, at which point their power is already fully formed.

FIGURE 50 · FOUR ENGAGEMENT STRATEGIES BY POWER AND INTEREST



Interest can rise overnight; power rarely falls as quickly

Figure 50. Four quadrants setting engagement effort against consequence. The upper-left cell is the one most often mishandled (Mendelow, 1981).

HOW IT IS USED

Map every stakeholder before the project begins, then re-map at each phase change: positions move, particularly on the interest axis.

WHEN IT IS USED

At project initiation, before a change programme, and whenever an unanticipated objection has come from someone who was not being consulted.

BY WHOM IT IS USED

Project managers planning engagement, change leaders sequencing communication, and anyone seeking approval from a group with differing stakes.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Stakeholder theory established that organisations answer to more parties than shareholders but gave no method for allocating limited attention among them. The grid supplies one, using the two variables that determine consequence. Its limitations are that both axes are assessed by judgment rather than measurement, that it is silent on stakeholder attitude — supportive or hostile — and that it produces a static picture of a shifting field.

RELATED FRAMEWORKS IN THIS COURSE

Hofstede Cultural Dimensions · **Slibrary 10**

Influence Currencies · **Slibrary 14**

French & Raven Power Bases · **Slibrary 14**

Project Lifecycle · **Slibrary 11**

SOURCES (HARVARD REFERENCING STYLE)

Mendelow, A.L. (1981) 'Environmental scanning: the impact of the stakeholder concept', *Proceedings of the Second International Conference on Information Systems*, Cambridge, MA, pp. 407–417.

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Project Lifecycle

SLIBRARY 11
SHEET 1 / 5

Initiating · planning · executing · monitoring and controlling · closing

PROFESSIONAL DEFINITION

The project lifecycle organises project work into five process groups. Initiating authorises the project and identifies stakeholders; planning establishes scope, schedule, cost and risk baselines; executing performs the work and manages the team; monitoring and controlling runs throughout, comparing performance against the baselines and managing change; and closing formally completes the work and captures lessons. The groups overlap in time rather than running in strict sequence (Project Management Institute, 2021).

WHO DEVELOPED IT

The five process groups were codified by the Project Management Institute, founded in 1969, and published in successive editions of the *PMBOK Guide* from 1996. The seventh edition (2021) reframed the standard around twelve principles and eight performance domains, retaining the process groups as a widely used descriptive structure rather than a prescribed sequence.

WHY IT WAS DEVELOPED

Project failure was routinely attributed to poor execution, but post-project reviews located the causes earlier — in unclear authorisation, absent baselines and uncontrolled change. The lifecycle gives each of those a named home and an owner. Closing is included as a distinct group for the same reason: without a formal step, projects drift to a stop and the lessons are never recorded.

FIGURE 51 · FIVE PROCESS GROUPS ACROSS THE PROJECT

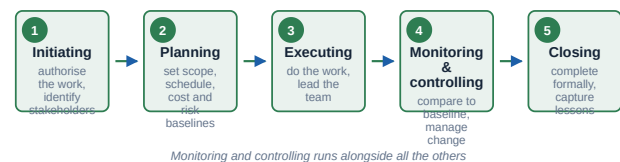


Figure 51. Five process groups. The fourth is not a phase but a continuous activity spanning the others (Project Management Institute, 2021).

HOW IT IS USED

Confirm which group the work is actually in before proposing an action: a scope question in execution is a change request, not a planning discussion.

WHEN IT IS USED

At project start-up, when a project is proceeding without a baseline, and when work is finishing without a formal close.

BY WHOM IT IS USED

Project managers structuring delivery, sponsors assessing project maturity, and team members seeking to understand where their work sits.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Before codification, project practice was transmitted by apprenticeship and varied between organisations, so failures could not be compared across projects. The process groups gave a common vocabulary and made professional certification possible. The criticism is real: the model presumes definable scope up front and suits stable, plan-driven work far better than genuinely emergent projects, which is what the seventh edition's principle-based reframing concedes.

RELATED FRAMEWORKS IN THIS COURSE

Iron Triangle · **Slibrary 11**Work Breakdown Structure · **Slibrary 11**Agile Manifesto Values · **Slibrary 12**Power–Interest Grid · **Slibrary 10**

SOURCES (HARVARD REFERENCING STYLE)

Project Management Institute (2021) *A Guide to the Project Management Body of Knowledge (PMBOK Guide)*. 7th edn. Newtown Square, PA: PMI.
 Project Management Institute (2017) *A Guide to the Project Management Body of Knowledge (PMBOK Guide)*. 6th edn. Newtown Square, PA: PMI.
 Office of Government Commerce (2017) *Managing Successful Projects with PRINCE2*. London: TSO.

Iron Triangle

SLIBRARY 11
SHEET 2 / 5

Scope · time · cost, with quality in the middle — change one and something else moves

PROFESSIONAL DEFINITION

The Iron Triangle states that a project's scope, time and cost are interdependent, so a change to any one requires a compensating change in at least one other if quality is to be held constant. Adding scope without adding time or budget does not produce more output; it produces lower quality or a missed date. The triangle is a negotiating instrument as much as a planning one (Atkinson, 1999).

WHO DEVELOPED IT

The three-constraint formulation is attributed to Martin Barnes, who taught it from about 1969 as the 'time, cost, quality' triangle in project-management courses in the United Kingdom. Roger Atkinson's 1999 paper in the *International Journal of Project Management* named it the iron triangle and argued it had become an inadequate definition of project success.

WHY IT WAS DEVELOPED

The formulation exists because sponsors routinely request additional scope mid-project while holding the date and budget fixed, and project managers accept it because the trade-off is not made visible. The triangle makes the arithmetic explicit at the moment of the request rather than at delivery, which converts an unwinnable conversation into a choice the sponsor has to make.

FIGURE 52 · THREE CONSTRAINTS, ONE DEPENDENT OUTCOME



Figure 52. Three constraints bounding quality. A change to one side must be met from another side or absorbed by the centre (Atkinson, 1999).

HOW IT IS USED

When new scope is requested, respond with the trade-off rather than a yes or no: name what moves in time or cost, and let the sponsor choose.

WHEN IT IS USED

At any change request, during initial estimation, and whenever a deadline is imposed after the scope has been agreed.

BY WHOM IT IS USED

Project managers negotiating with sponsors, teams under scope pressure, and clients deciding between competing requests.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The triangle replaced an implicit assumption that better management could deliver more for the same money in the same time, with a stated constraint relationship. Atkinson's own argument is the necessary caveat: delivering on the three constraints is not the same as delivering value, and projects meeting all three have failed commercially while projects breaching all three have succeeded.

RELATED FRAMEWORKS IN THIS COURSE

Project Lifecycle · **Slibrary 11**Critical Path Method · **Slibrary 11**Agile Manifesto Values · **Slibrary 12**Probability–Impact Risk Matrix · **Slibrary 12**

SOURCES (HARVARD REFERENCING STYLE)

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Work Breakdown Structure

SLIBRARY 11
SHEET 3 / 5

Deliverables decomposed until each package can be estimated, assigned and tracked

PROFESSIONAL DEFINITION

A Work Breakdown Structure is a hierarchical decomposition of the total scope into progressively smaller deliverables, terminating in work packages small enough to be estimated, assigned to one owner and tracked. It is governed by the hundred per cent rule: each level contains all of the work of its parent and no more, so anything absent from the structure is outside the project (Project Management Institute, 2019).

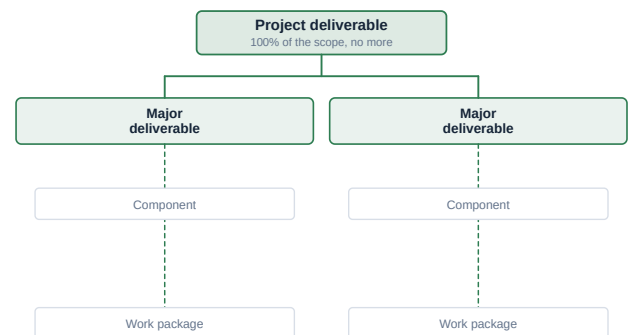
WHO DEVELOPED IT

The technique originated in US Department of Defense and NASA programme management in the 1960s, formalised in the Program Evaluation and Review Technique work and in DoD and NASA guidance of 1962. The Project Management Institute later published a dedicated practice standard and embedded it in the *PMBOK Guide* (Project Management Institute, 2019).

WHY IT WAS DEVELOPED

Large programmes were being estimated as single figures, which made both the estimate and any later variance impossible to interrogate. Decomposition to work-package level makes estimation additive and accountability specific. The hundred per cent rule addresses the other failure mode: scope that nobody explicitly excluded and nobody explicitly owns, which surfaces late as unplanned work.

FIGURE 53 · DECOMPOSITION TO WORK-PACKAGE LEVEL



Each level contains all of its parent's work and nothing beyond it

Figure 53. Deliverables decomposed until each terminal package has one owner and an estimate. Work absent from the structure is outside the project (PMI, 2019).

HOW IT IS USED

Decompose by deliverable rather than by activity, and stop when a package can be estimated within a reasonable margin and assigned to a single owner.

WHEN IT IS USED

During planning, before estimating cost or duration, and when a project is running over because of work that was never listed.

BY WHOM IT IS USED

Project managers building schedules and budgets, teams agreeing accountability, and sponsors verifying that scope has been fully captured.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Estimating a project as a whole conceals which assumptions the number rests on and makes variance untraceable. Decomposition makes estimation additive, assigns ownership at a usable granularity, and gives change control something concrete to act on. Its cost is significant up-front effort and a structure that resists change, which is why iterative approaches substitute a rolling backlog for a fixed breakdown.

RELATED FRAMEWORKS IN THIS COURSE

Project Lifecycle · **Slibrary 11**

Gantt Chart · **Slibrary 11**

Critical Path Method · **Slibrary 11**

MECE Issue Tree · **Slibrary 5**

SOURCES (HARVARD REFERENCING STYLE)

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Gantt Chart

SLIBRARY 11
SHEET 4 / 5

Tasks against a calendar — duration as length, dependency as position

PROFESSIONAL DEFINITION

A Gantt chart displays scheduled activities as horizontal bars on a time axis, where bar length represents duration and horizontal position represents start and finish. Dependencies are shown by relative position or by explicit links, milestones by markers with zero duration, and progress by shading within the bar. The chart's function is communication: it makes the schedule legible to people who did not build it (Gantt, 1919; Clark, 1922).

WHO DEVELOPED IT

Henry Laurence Gantt, an American mechanical engineer and associate of Frederick Winslow Taylor, developed the charts between 1910 and 1915 for production scheduling and wartime shipbuilding. Comparable bar charts had been produced by the Polish engineer Karol Adamiecki in 1896, though his work was published in Polish and remained unknown in the West (Clark, 1922).

WHY IT WAS DEVELOPED

Gantt's problem was industrial: work was scheduled in the foreman's head, so delay propagated invisibly and idle capacity could not be seen. Rendering time as horizontal distance makes both visible at a glance. Its endurance for over a century rests on legibility rather than analytical power — it is understood immediately by people with no scheduling training, which few later techniques manage.

FIGURE 54 · DURATION AS LENGTH, DEPENDENCY AS POSITION



Figure 54. Five activities on a weekly grid with a closing milestone. Overlap shows parallel work; the diamond marks a zero-duration decision point (Gantt, 1919).

HOW IT IS USED

Build the breakdown and the dependencies first, then draw the chart. A Gantt chart drawn directly from a deadline is a wish rather than a schedule.

WHEN IT IS USED

When communicating a plan to stakeholders, when tracking progress against dates, and when coordinating parallel workstreams.

BY WHOM IT IS USED

Project managers reporting to sponsors, teams coordinating dependencies, and anyone presenting a schedule to a non-specialist audience.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Before the bar chart, schedules existed as lists of dates in which overlap, sequence and slack were invisible. Mapping time to horizontal space made the whole plan readable at once. Its analytical limitations are well known: it does not by itself identify which delays matter, which is what critical path analysis was developed to supply, and it becomes unreadable on large projects.

RELATED FRAMEWORKS IN THIS COURSE

Work Breakdown Structure · [Slibrary 11](#)

Critical Path Method · [Slibrary 11](#)

Project Lifecycle · [Slibrary 11](#)

Earned Value S-Curve · [Slibrary 12](#)

SOURCES (HARVARD REFERENCING STYLE)

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Critical Path Method

SLIBRARY 11
SHEET 5 / 5

The longest path sets the duration — everything else has float

PROFESSIONAL DEFINITION

Critical Path Method calculates a project's minimum duration by identifying the longest sequence of dependent activities from start to finish. Activities on that path have zero float: any delay to one delays the project. Activities off it have float, the amount by which they can slip without affecting the end date. The method computes early and late start and finish dates for every activity through a forward and backward pass (Kelley and Walker, 1959).

WHO DEVELOPED IT

James E. Kelley of Remington Rand and Morgan R. Walker of DuPont developed the method in 1957 for scheduling chemical-plant maintenance shutdowns, publishing it in 1959. The US Navy's Program Evaluation and Review Technique was developed independently and near-simultaneously for the Polaris programme, adding probabilistic duration estimates (Malcolm et al., 1959).

WHY IT WAS DEVELOPED

Kelley and Walker's problem was economic: shutting a plant costs money per day, so the question was where to add resource to shorten the outage. Adding effort to activities with float changes nothing. The method identifies exactly where acceleration is worth paying for, which is why it was originally a cost-optimisation technique rather than a scheduling one.

FIGURE 55 · THE LONGEST PATH THROUGH THE NETWORK

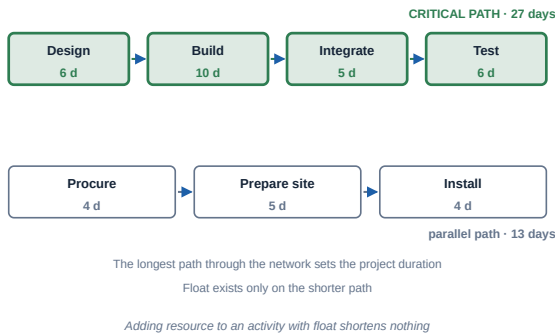


Figure 55. Two paths through the same network. The longer path fixes the project duration; float exists only on the shorter one (Kelley and Walker, 1959).

HOW IT IS USED

Identify the critical path before allocating extra resource, and re-identify it after every significant change — the critical path moves when durations change.

WHEN IT IS USED

When compressing a schedule, when deciding where to add resource, and when assessing the true consequence of a reported delay.

BY WHOM IT IS USED

Project managers under time pressure, planners performing schedule compression, and sponsors asking whether a delay matters.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Bar charts show when activities are scheduled but not which ones govern the end date, so management attention is distributed by visibility rather than consequence. CPM makes the governing sequence explicit and quantifies float everywhere else. Its assumptions are restrictive: durations are treated as deterministic and dependencies as fixed, which is why PERT and Monte Carlo methods extend it for uncertain work.

RELATED FRAMEWORKS IN THIS COURSE

Gantt Chart · [Slibrary 11](#)

Work Breakdown Structure · [Slibrary 11](#)

Iron Triangle · [Slibrary 11](#)

Earned Value S-Curve · [Slibrary 12](#)

SOURCES (HARVARD REFERENCING STYLE)

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Agile Manifesto Values

SLIBRARY 12
SHEET 1 / 5

Four preferences, each stated as 'over' rather than 'instead of'

PROFESSIONAL DEFINITION

The Agile Manifesto states four value preferences: individuals and interactions over processes and tools; working software over comprehensive documentation; customer collaboration over contract negotiation; and responding to change over following a plan. The manifesto is explicit that while there is value in the items on the right, the items on the left are valued more — a statement of priority under trade-off, not a rejection (Beck et al., 2001).

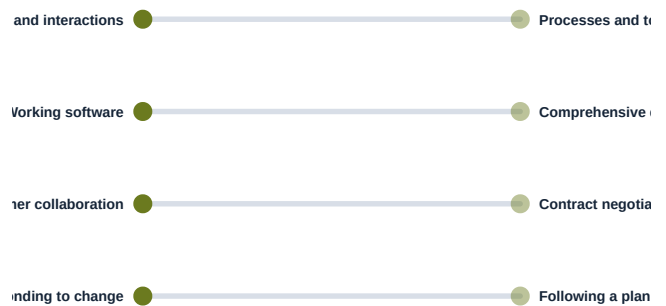
WHO DEVELOPED IT

Seventeen software practitioners, including Kent Beck, Martin Fowler, Ken Schwaber, Jeff Sutherland, Alistair Cockburn and Robert Martin, drafted the manifesto at a meeting in Snowbird, Utah, in February 2001. They represented previously separate lightweight methods — Extreme Programming, Scrum, DSDM, Crystal — and sought common ground rather than a merged method (Beck et al., 2001).

WHY IT WAS DEVELOPED

Heavyweight plan-driven methods assumed requirements could be specified in full before construction, an assumption that held poorly for software, where users discover what they need by using something. The manifesto's four preferences all follow from that: if requirements emerge, then feedback, working output and adaptation matter more than the artefacts produced to control a fixed plan.

FIGURE 56 · FOUR PREFERENCES UNDER TRADE-OFF



There is value in the items on the right; the items on the left are valued more

Figure 56. Four paired preferences. The right-hand items are not rejected — they are what is traded when the two conflict (Beck et al., 2001).

HOW IT IS USED

Use the values to resolve a specific trade-off, not to justify skipping the right-hand item: 'we value working software more' is not an argument for no documentation.

WHEN IT IS USED

When choosing between plan-driven and adaptive delivery, when a team is following process that no longer serves the outcome, and when interpreting what agile means in your context.

BY WHOM IT IS USED

Delivery teams choosing a method, managers adopting agile practice, and anyone in an organisation that has adopted agile ceremonies without the underlying preferences.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Plan-driven methods derived their control from documented conformance to a fixed specification. The manifesto substituted working output and direct collaboration as the control mechanism, which is a different theory of how quality is assured. It is a statement of values rather than a method, which is both its durability and its weakness: it is invoked to justify almost any practice, including the absence of discipline.

RELATED FRAMEWORKS IN THIS COURSE

Scrum Sprint Cycle · **Slibrary 12**Stacey Matrix · **Slibrary 12**Project Lifecycle · **Slibrary 11**Double Diamond · **Slibrary 5**

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Scrum Sprint Cycle

SLIBRARY 12
SHEET 2 / 5

Plan · work daily · review · retrospect — a fixed-length loop with inspection built in

PROFESSIONAL DEFINITION

Scrum organises work into sprints of fixed length, typically one to four weeks, each containing four events. Sprint planning selects work from the product backlog and defines a sprint goal; the daily scrum re-plans the next day's work against that goal; the sprint review inspects the increment with stakeholders; and the sprint retrospective inspects the team's own process. The sprint length does not change, so the scope adjusts to the time available (Schwaber and Sutherland, 2020).

WHO DEVELOPED IT

Ken Schwaber and Jeff Sutherland developed Scrum independently in the early 1990s and presented it jointly at OOPSLA in 1995. The name derives from Takeuchi and Nonaka's 1986 *Harvard Business Review* study of product development, which compared high-performing cross-functional teams to a rugby scrum moving the ball as a unit (Takeuchi and Nonaka, 1986).

WHY IT WAS DEVELOPED

The problem addressed was feedback latency: on a long delivery cycle, an error in understanding is discovered only at the end, when correction is most expensive. Fixed-length sprints bound that latency by construction. The two separate inspection events matter — the review inspects the product, the retrospective inspects the process, and teams that merge them typically stop improving the second.

FIGURE 57 · ONE SPRINT, FOUR EVENTS, ONE INCREMENT

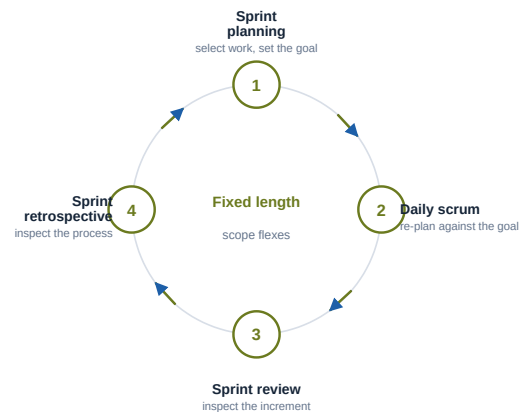


Figure 57. Four events in a fixed-length loop producing one increment. The sprint length is the constant; the scope is the variable (Schwaber and Sutherland, 2020).

HOW IT IS USED

Hold the sprint length constant and let scope vary. If sprints are routinely extended to finish the work, the feedback mechanism has been disabled.

WHEN IT IS USED

On work where requirements will change or are not fully knowable, and where a stakeholder is available to inspect output regularly.

BY WHOM IT IS USED

Delivery teams and product owners, managers introducing iterative delivery, and students running project simulations under changing requirements.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Stage-gate development places inspection at phase boundaries months apart, so correction cost rises with time. Scrum makes inspection periodic and short-cycle, and separates product inspection from process inspection. Its requirements are demanding and frequently unmet: a genuinely available stakeholder, a team empowered to change its own process, and stable membership — without which the ceremonies persist and the benefit does not.

RELATED FRAMEWORKS IN THIS COURSE

Agile Manifesto Values · **Slibrary 12**Stacey Matrix · **Slibrary 12**Kolb Experiential Learning Cycle · **Slibrary 1**Project Lifecycle · **Slibrary 11**

SOURCES (HARVARD REFERENCING STYLE)

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Stacey Matrix

SLIBRARY 12
SHEET 3 / 5

Agreement on what, certainty about how — where planning stops working

PROFESSIONAL DEFINITION

The Stacey Matrix positions a decision on two axes: the degree of agreement among stakeholders about what is wanted, and the degree of certainty about how to achieve it. Close to both agreement and certainty, conventional planning applies; as either falls, political and judgment-based approaches are required; in the zone of complexity neither prediction nor negotiation alone suffices; and far from both lies anarchy, where neither planning nor consensus is available (Stacey, 1996).

WHO DEVELOPED IT

Ralph D. Stacey, professor of management at the University of Hertfordshire, developed the diagram in *Complexity and Creativity in Organizations* (1996) as part of his work applying complexity science to strategic management. He later distanced himself from the popularised grid, particularly its use in agile training, arguing it had been reduced to a categorisation device he did not intend (Stacey, 2011).

WHY IT WAS DEVELOPED

Stacey's argument was that managers apply rational planning universally, including in conditions where its assumptions do not hold, and then attribute the resulting failure to execution. Locating a decision on the two axes makes the assumption explicit. The zone of complexity is the practically important region: it is where most consequential organisational decisions sit, and where planning produces false confidence.

FIGURE 58 · AGREEMENT AGAINST CERTAINTY



Most consequential decisions sit in the complex zone, not the simple one

Figure 58. Two axes bounding four regions. Plan-driven methods hold their assumptions only in the region closest to agreement and certainty (Stacey, 1996).

HOW IT IS USED

Position the decision before choosing a method. Away from agreement, invest in stakeholder alignment; away from certainty, invest in experiments rather than in a more detailed plan.

WHEN IT IS USED

When selecting a delivery approach, when a detailed plan keeps being invalidated, and when a technically sound proposal cannot get agreement.

BY WHOM IT IS USED

Project and programme managers choosing methods, leaders diagnosing why planning is failing, and teams deciding between predictive and adaptive delivery.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Contingency approaches asserted that method should fit context without specifying which features of context mattered. Stacey named two and showed they fail independently, which distinguishes a disagreement problem from an uncertainty problem — different interventions entirely. Stacey's own repudiation of the simplified grid is worth carrying: it was intended to provoke reflection about complexity, not to sort projects into boxes.

RELATED FRAMEWORKS IN THIS COURSE

Cynefin Framework · **Slibrary 5**

Agile Manifesto Values · **Slibrary 12**

Probability–Impact Risk Matrix · **Slibrary 12**

Double Diamond · **Slibrary 5**

SOURCES (HARVARD REFERENCING STYLE)

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Probability–Impact Risk Matrix

SLIBRARY 12
SHEET 4 / 5

How likely, how bad — and therefore which response

PROFESSIONAL DEFINITION

The probability–impact matrix scores each identified risk on its likelihood of occurrence and the severity of its consequence, and derives a response from the combination. High likelihood with high impact calls for avoidance or substantial mitigation; low likelihood with high impact for transfer, typically insurance or contract; high likelihood with low impact for active control; and low likelihood with low impact for acceptance under monitoring (ISO, 2018).

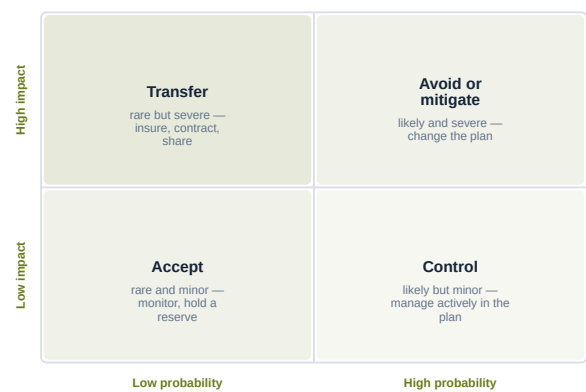
WHO DEVELOPED IT

The matrix is a standard instrument of formal risk management rather than the work of a single author, codified in ISO 31000 and its supporting risk-assessment techniques standard IEC 31010, and embedded in the *PMBOK Guide* and PRINCE2. Its intellectual roots lie in mid-twentieth century reliability and safety engineering (ISO, 2018; IEC, 2019).

WHY IT WAS DEVELOPED

Risk registers without scoring list every conceivable hazard with equal weight, which supplies no basis for allocating limited attention and is routinely ignored. The two-axis score orders the register and, more importantly, attaches a different class of response to each region — transfer for rare catastrophes, control for frequent nuisances — rather than a single generic instruction to mitigate.

FIGURE 59 · FOUR REGIONS, FOUR CLASSES OF RESPONSE



The response class differs by region; 'mitigate' is not a universal answer

Figure 59. Probability against impact, with a distinct response class in each region. The upper-left cell is where insurance and contractual transfer belong (ISO, 2018).

HOW IT IS USED

Score each risk, place it, and record the response class and its owner. Re-score at each phase change rather than treating the register as a start-up artefact.

WHEN IT IS USED

During planning, at every phase gate, and whenever a new dependency or external constraint enters the project.

BY WHOM IT IS USED

Project managers maintaining risk registers, sponsors reviewing exposure, and teams deciding where contingency should be held.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Unscored risk lists cannot prioritise and cannot differentiate response. The matrix supplies both from two judgments. Its documented weaknesses are serious and worth teaching: categorical bands compress very different magnitudes, the ordering can invert true expected loss, and the scores are subjective — so it is a communication and triage device, not a quantitative model (Cox, 2008).

RELATED FRAMEWORKS IN THIS COURSE

- Stacey Matrix · **Slibrary 12**
- Iron Triangle · **Slibrary 11**
- Cynefin Framework · **Slibrary 5**
- Earned Value S-Curve · **Slibrary 12**

SOURCES (HARVARD REFERENCING STYLE)

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Earned Value S-Curve

SLIBRARY 12
SHEET 5 / 5

Planned value, earned value, actual cost — progress and spend read against each other

PROFESSIONAL DEFINITION

Earned value management compares three cumulative measures over time. Planned value is the budgeted cost of work scheduled; earned value is the budgeted cost of work actually completed; and actual cost is what has been spent. Schedule variance is earned value minus planned value, cost variance is earned value minus actual cost. The method's central contribution is that progress is measured in the currency of the budget, not in percentage claimed (Fleming and Koppelman, 2010).

WHO DEVELOPED IT

Earned value originated in US Department of Defense cost and schedule control systems criteria issued in 1967 for major acquisition programmes. Quentin Fleming and Joel Koppelman brought it into general project-management practice, and it is codified in the PMI practice standard and in ANSI/EIA-748 (Fleming and Koppelman, 2010).

WHY IT WAS DEVELOPED

Reported progress and reported spend, taken separately, are compatible with any underlying situation: a project half spent and reported half complete may be on track or badly behind. Earned value resolves the ambiguity by valuing completed work at its budgeted cost, which makes schedule and cost performance separable and comparable — and detectable early rather than at delivery.

FIGURE 60 · THREE CUMULATIVE CURVES AGAINST TIME

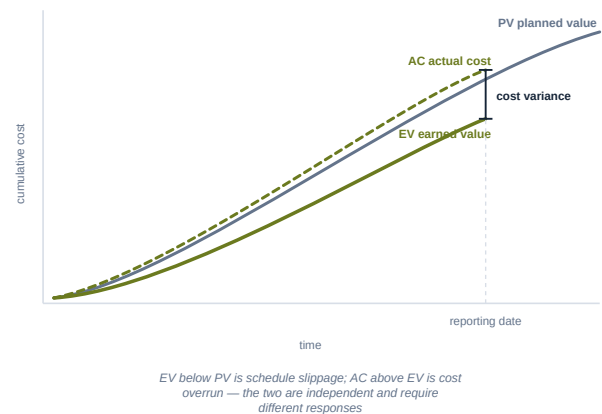


Figure 60. Three curves read at a common reporting date. The vertical gaps give cost and schedule variance directly (Fleming and Koppelman, 2010).

HOW IT IS USED

Establish the baseline before work starts, then read the two gaps at each report: earned against planned for schedule, earned against actual for cost.

WHEN IT IS USED

On projects with a defined scope baseline and measurable deliverables, particularly where external reporting or contractual performance is required.

BY WHOM IT IS USED

Project controllers and programme managers, sponsors assessing forecast outturn, and teams reporting on large capital or contracted work.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Traditional reporting compared spend against budget, which reveals nothing about how much has been accomplished for that spend. Earned value adds the third measure and thereby separates cost performance from schedule performance. It requires a stable, well-decomposed baseline and objective completion criteria; on projects with emergent scope the baseline is fiction and the method reports precise numbers about it.

RELATED FRAMEWORKS IN THIS COURSE

Critical Path Method · [Slibrary 11](#)Gantt Chart · [Slibrary 11](#)Iron Triangle · [Slibrary 11](#)Probability–Impact Risk Matrix · [Slibrary 12](#)

SOURCES (HARVARD REFERENCING STYLE)

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BATNA

SLIBRARY 13
SHEET 1 / 5

Best Alternative To a Negotiated Agreement — leverage is what you can walk to

PROFESSIONAL DEFINITION

A BATNA is the course of action a party will take if the present negotiation produces no agreement. It is not a wish or a target but the best of the alternatives actually available, and it determines the reservation value below which agreement is worse than walking away. Bargaining power derives from the quality of the alternative rather than from wealth, size or confidence, and it can be improved before the negotiation begins (Fisher and Ury, 1981).

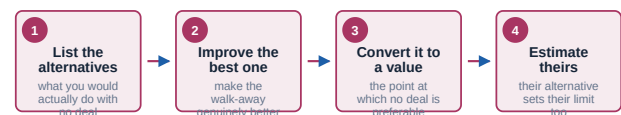
WHO DEVELOPED IT

Roger Fisher and William Ury of the Harvard Negotiation Project introduced the concept in *Getting to Yes* (1981), with Bruce Patton joining as co-author from the second edition. The underlying idea formalises the economic notion of a reservation price within a framework intended for practitioners rather than for game theorists (Fisher, Ury and Patton, 2011).

WHY IT WAS DEVELOPED

Fisher and Ury observed that negotiators protected themselves by setting a bottom line, which is arbitrary, rigid and set before the information that should inform it arrives. Anchoring protection in a real alternative fixes both problems: the reservation value is derived rather than invented, and the alternative can be improved, which changes the negotiation itself rather than merely the negotiator's resolve.

FIGURE 61 · FOUR STEPS TO A USABLE ALTERNATIVE



Leverage is prepared before the meeting, not summoned inside it

Figure 61. Four preparatory steps. The second is the one that changes the negotiation, because it alters the real cost of walking away (Fisher and Ury, 1981).

HOW IT IS USED

Develop the alternative before you negotiate, not during. Then state your reservation value to yourself in writing and refuse to renegotiate it under pressure in the room.

WHEN IT IS USED

Before any negotiation with material stakes — salary, contract, supplier terms — and particularly when the other party appears to hold all the leverage.

BY WHOM IT IS USED

Anyone negotiating terms, buyers and suppliers preparing positions, and candidates deciding how far an offer can be pushed.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The traditional bottom line is a number chosen in advance and defended regardless of what emerges, which both loses good deals and prevents learning during the negotiation. BATNA replaces it with a value derived from a real option, and thereby makes preparation productive rather than merely reassuring. Its limitation is that alternatives are often poorly known, so the calculated reservation value is only as sound as that assessment.

RELATED FRAMEWORKS IN THIS COURSE

Zone of Possible Agreement · **Slibrary 13**

Principled Negotiation · **Slibrary 13**

Anchoring & Adjustment · **Slibrary 6**

Distributive ↔ Integrative Spectrum · **Slibrary 13**

SOURCES (HARVARD REFERENCING STYLE)

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Zone of Possible Agreement

SLIBRARY 13
SHEET 2 / 5

The overlap between two reservation values — where a deal is possible at all

PROFESSIONAL DEFINITION

The Zone of Possible Agreement is the range of terms that both parties prefer to their respective alternatives. It is bounded by each side's reservation value, derived from their BATNA. Where the ranges overlap a deal is feasible and the negotiation concerns where within the zone it settles; where they do not overlap, no agreement at the current terms is possible and the constructive move is to change the terms or the alternatives (Raiffa, 1982).

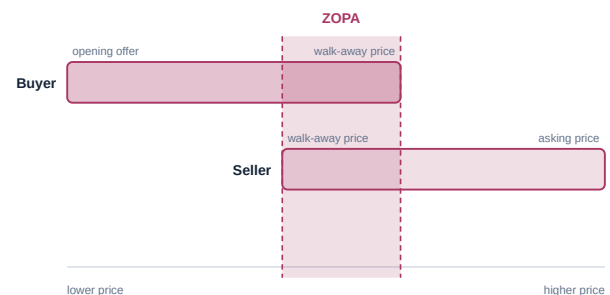
WHO DEVELOPED IT

Howard Raiffa, professor of managerial economics at Harvard and a founder of decision analysis, formalised the bargaining zone in *The Art and Science of Negotiation* (1982), which brought analytical method to negotiation without reducing it to game theory. The concept descends from Walton and McKersie's settlement range and from earlier bargaining models (Raiffa, Richardson and Metcalfe, 2002).

WHY IT WAS DEVELOPED

Negotiators routinely fail in one of two symmetrical ways: they walk away from deals that sat inside the zone because the opening position was far from theirs, or they persist for months where no zone existed. Making the zone explicit distinguishes a hard negotiation from an impossible one, which determines whether further effort is investment or waste.

FIGURE 62 · TWO RESERVATION RANGES AND THEIR OVERLAP



No overlap means no deal on these terms — change the terms, the scope, or your alternative

Figure 62. Each party's acceptable range runs to its reservation value. The shaded overlap is the set of terms both prefer to no deal (Raiffa, 1982).

HOW IT IS USED

Estimate both reservation values before opening. If you cannot construct a plausible overlap, change what is on the table rather than pressing on price.

WHEN IT IS USED

During negotiation preparation, when talks have stalled, and when deciding whether to continue investing time in a deal.

BY WHOM IT IS USED

Negotiators preparing positions, sales and procurement professionals qualifying deals, and mediators diagnosing whether settlement is available.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Positional bargaining treats the gap between opening positions as the object of negotiation, which conveys nothing about feasibility since openings are strategic. The ZOPA shifts attention to reservation values, which are real. The practical difficulty is that the other party's reservation value is private and must be inferred, so the zone as drawn is always an estimate — useful for diagnosis, unreliable as a precise boundary.

RELATED FRAMEWORKS IN THIS COURSE

BATNA · [Slibrary 13](#)Distributive ↔ Integrative Spectrum · [Slibrary 13](#)Anchoring & Adjustment · [Slibrary 6](#)Principled Negotiation · [Slibrary 13](#)

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Principled Negotiation

SLIBRARY 13
SHEET 3 / 5

People · interests · options · criteria — four rules that replace positional bargaining

PROFESSIONAL DEFINITION

Principled negotiation prescribes four rules. Separate the people from the problem, so that relationship damage is not traded against substance. Focus on interests rather than positions, since a stated position is one way of meeting an interest and rarely the only one. Invent options for mutual gain before deciding among them. And insist on objective criteria, so that the outcome rests on a standard rather than on whoever concedes last (Fisher and Ury, 1981).

WHO DEVELOPED IT

Roger Fisher, Director of the Harvard Negotiation Project, and William Ury, an anthropologist and its co-founder, set out the method in *Getting to Yes* (1981), with Bruce Patton co-authoring later editions. It emerged from the Project's work on international disputes and labour conflicts, and became the dominant framework in negotiation teaching worldwide (Fisher, Ury and Patton, 2011).

WHY IT WAS DEVELOPED

Positional bargaining produces three predictable costs: it is slow, since movement requires concessions from an arbitrary opening; it damages relationships, because holding a position requires resisting the person; and it produces poor agreements, because it never surfaces trades that would benefit both. Each of the four rules addresses one of these mechanisms directly.

FIGURE 63 · FOUR RULES REPLACING POSITIONAL BARGAINING

1 Separate the people from the problem
be hard on the substance, soft on the person

2 Focus on interests, not positions
ask why they want it, not what they demand

3 Invent options for mutual gain
generate before evaluating; separate the two

4 Insist on objective criteria
settle on a standard, not on who concedes last

The four are applied together; each addresses a different failure

Figure 63. Four rules, each targeting one cost of positional bargaining: relationship damage, hidden trades, premature evaluation and arbitrary settlement (Fisher and Ury, 1981).

HOW IT IS USED

Prepare interests rather than positions for both sides — for each demand, write down why they want it — and identify the objective standard before naming a number.

WHEN IT IS USED

In any negotiation intended to preserve an ongoing relationship, in multi-issue negotiations, and where a purely positional exchange has stalled.

BY WHOM IT IS USED

Managers negotiating internally, professionals dealing with clients and suppliers, and anyone who must reach agreement with someone they will work with afterwards.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Negotiation had been treated either as a zero-sum contest of will or as an appeal to goodwill; the first damages relationships, the second concedes substance. Principled negotiation separated the two dimensions and made it possible to be firm on substance while remaining workable with the person. Critics note it presumes reasonable good faith and offers less against a determinedly adversarial counterpart, which Ury addressed in later work.

RELATED FRAMEWORKS IN THIS COURSE

BATNA · [Slibrary 13](#)Dual Concerns Conflict Modes · [Slibrary 13](#)Tactical Empathy · [Slibrary 14](#)Distributive ↔ Integrative Spectrum · [Slibrary 13](#)

SOURCES (HARVARD REFERENCING STYLE)

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Dual Concerns Conflict Modes

SLIBRARY 13
SHEET 4 / 5

Assertiveness against cooperativeness — five modes, none of them always right

PROFESSIONAL DEFINITION

The dual concerns model locates conflict behaviour on two independent dimensions: assertiveness, the degree to which one pursues one's own concerns, and cooperativeness, the degree to which one attends to the other party's. The combinations yield five modes — competing, collaborating, avoiding, accommodating and, at the midpoint, compromising. Each is appropriate in some circumstances; the diagnostic question is whether a person can access all five (Thomas and Kilmann, 1974).

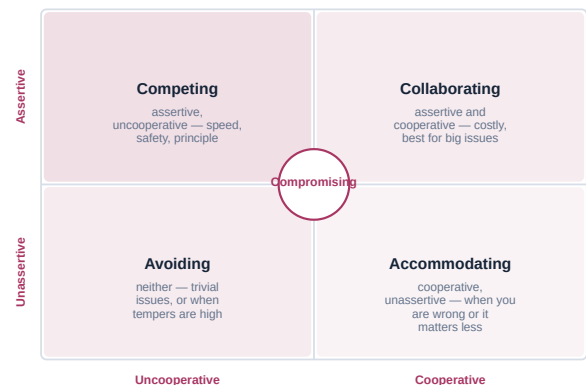
WHO DEVELOPED IT

Kenneth W. Thomas and Ralph H. Kilmann published the Thomas–Kilmann Conflict Mode Instrument in 1974, adapting a two-dimensional grid originally developed by Robert Blake and Jane Mouton for managerial style. The instrument remains among the most widely used in conflict training and management development (Blake and Mouton, 1964).

WHY IT WAS DEVELOPED

Conflict training before the model prescribed a single correct approach, usually cooperation, which fails in circumstances where speed, safety or a matter of principle requires assertion. The grid replaces prescription with diagnosis. Its practical finding is that most people have a habitual mode used regardless of circumstance, so the development task is range rather than correction.

FIGURE 64 · FIVE MODES ON TWO INDEPENDENT DIMENSIONS



The development question is range, not which mode is best

Figure 64. Five modes formed by two independent concerns, with compromise at the midpoint. Each is correct in some situations (Thomas and Kilmann, 1974).

HOW IT IS USED

Identify your default mode, then name a recent conflict where a different mode would have served better and rehearse that one specifically.

WHEN IT IS USED

Before a difficult conversation, when the same conflict pattern keeps recurring, and when building a team that handles disagreement badly.

BY WHOM IT IS USED

Managers handling team conflict, negotiators choosing an approach, and individuals whose conflict behaviour is unvarying across very different situations.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Single-dimension models treated conflict behaviour as a line from aggressive to yielding, which cannot represent collaboration — high on both concerns — or avoidance, low on both. Two independent dimensions accommodate all five. The instrument is self-report and measures preference rather than skill, and its forced-choice format has attracted psychometric criticism, so it is best used as a discussion device.

RELATED FRAMEWORKS IN THIS COURSE

Principled Negotiation · [Slibrary 13](#)

DISC Behavioural Quadrant · [Slibrary 4](#)

The Culture Map (Eight Scales) · [Slibrary 10](#)

Distributive ↔ Integrative Spectrum · [Slibrary 13](#)

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Distributive ↔ Integrative Spectrum

SLIBRARY 13
SHEET 5 / 5

Claiming value and creating it — the negotiator's dilemma in one line

PROFESSIONAL DEFINITION

Walton and McKersie distinguish distributive bargaining, in which a fixed quantity is divided so that one party's gain is the other's loss, from integrative bargaining, in which differences in priorities are traded to enlarge the total value available. Real negotiations contain both, and the tension between them is structural: the information-sharing that creates value also exposes a party to the claiming that divides it (Walton and McKersie, 1965).

WHO DEVELOPED IT

Richard E. Walton and Robert B. McKersie, both later at MIT's Sloan School, set out the distinction in *A Behavioral Theory of Labor Negotiations* (1965), based on study of collective bargaining. David Lax and James Sebenius later named the resulting tension the negotiator's dilemma (Lax and Sebenius, 1986).

WHY IT WAS DEVELOPED

Practitioner advice had polarised between hard bargaining and mutual-gains approaches, each treating the other as naive. Walton and McKersie showed both processes are present in the same negotiation and require opposite behaviours — one guarded, one open. That reframing explains why single-strategy advice fails: the skill is sequencing the two, typically creating before claiming.

FIGURE 65 · FOUR POSITIONS ALONG ONE CONTINUUM



Figure 65. A continuum from dividing fixed value to enlarging it. The dilemma is that the behaviour which creates value also makes claiming easier for the other side (Walton and McKersie, 1965).

HOW IT IS USED

Add issues before dividing them: identify where the two sides rank priorities differently, trade across those, and defer the single distributive issue until last.

WHEN IT IS USED

In any multi-issue negotiation, when talks are deadlocked on one number, and when preparing to negotiate with a counterpart you will deal with again.

BY WHOM IT IS USED

Negotiators preparing strategy, procurement and sales professionals structuring deals, and managers resolving resource disputes between teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Earlier bargaining theory modelled negotiation as purely distributive, which made every concession a loss and every negotiation adversarial by construction. Adding the integrative process made value creation a legitimate object of strategy. The dilemma has no general solution: the model describes the trade-off between openness and exposure rather than resolving it, and the resolution depends on trust and repetition.

RELATED FRAMEWORKS IN THIS COURSE

Zone of Possible Agreement · [Slibrary 13](#)

Principled Negotiation · [Slibrary 13](#)

Dual Concerns Conflict Modes · [Slibrary 13](#)

Influence Currencies · [Slibrary 14](#)

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Six Principles of Influence

SLIBRARY 14
SHEET 1 / 5

Reciprocity · commitment · social proof · authority · liking · scarcity

PROFESSIONAL DEFINITION

Cialdini identifies six psychological principles that reliably increase compliance. Reciprocity obliges return of a favour; commitment and consistency press people to act in line with prior public positions; social proof leads people to follow the behaviour of similar others; authority produces deference to credible expertise; liking increases compliance with those we find similar or agreeable; and scarcity raises the perceived value of what is limited (Cialdini, 1984).

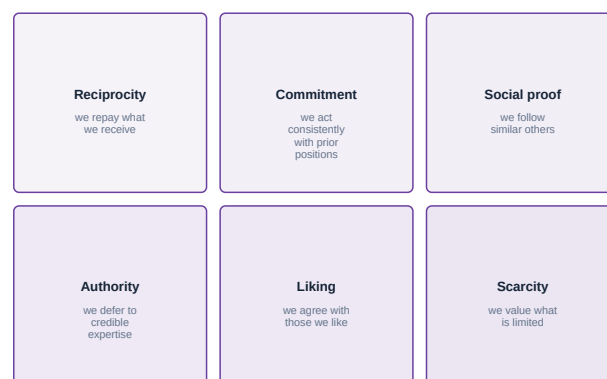
WHO DEVELOPED IT

Robert B. Cialdini, Regents' Professor Emeritus of psychology and marketing at Arizona State University, derived the principles from three years of participant observation inside sales, fundraising and advertising organisations, combined with laboratory experiments, and published them in *Influence* (1984). He added a seventh, unity, in 2016 (Cialdini, 2016).

WHY IT WAS DEVELOPED

Cialdini's method was deliberately unusual: rather than beginning from theory, he trained inside compliance professions to find which techniques survived commercial selection pressure. The six principles are what remained. His stated purpose was as much defensive as instructive — recognising a principle being deployed on you is the condition of resisting it.

FIGURE 66 · SIX PRINCIPLES OF COMPLIANCE



Recognising a principle in use is the condition of resisting it

Figure 66. Six principles established by field observation and experiment. A seventh, unity, was added in 2016 (Cialdini, 1984; Cialdini, 2016).

HOW IT IS USED

Use them to make a genuine case easier to accept, not to manufacture consent: give first, cite real evidence of others' behaviour, and state authority accurately.

WHEN IT IS USED

When seeking agreement without formal authority, when designing a request, and when you notice yourself agreeing to something faster than the merits warrant.

BY WHOM IT IS USED

Anyone influencing without authority, marketers and fundraisers, and individuals wanting to recognise techniques being used on them.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Persuasion research had been largely laboratory-based, examining attitude change under controlled conditions. Cialdini's field immersion identified which techniques practitioners had converged on commercially, giving external validity that laboratory work lacked. The caveat is that several classic supporting studies have had replication difficulties, and effect sizes in field settings vary considerably by context.

RELATED FRAMEWORKS IN THIS COURSE

French & Raven Power Bases · **Slibrary 14**

Influence Currencies · **Slibrary 14**

Persuasion ↔ Manipulation Line · **Slibrary 14**

Anchoring & Adjustment · **Slibrary 6**

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French & Raven Power Bases

SLIBRARY 14
SHEET 2 / 5

Legitimate · reward · coercive · expert · referent — five sources, only two of them formal

PROFESSIONAL DEFINITION

French and Raven distinguish five bases from which social power derives. Legitimate power rests on a recognised right to require compliance; reward power on the capacity to confer benefits; coercive power on the capacity to impose costs; expert power on recognised knowledge or skill; and referent power on identification with and respect for the person. Only the first three attach to a role; expert and referent power are held by individuals and are portable (French and Raven, 1959).

WHO DEVELOPED IT

John R. P. French Jr. and Bertram Raven published the typology in 1959 in *Studies in Social Power* at the University of Michigan's Institute for Social Research. Raven added a sixth base, informational power — deriving from control of information rather than from expertise — in 1965, and later refined the model further (Raven, 1965).

WHY IT WAS DEVELOPED

Organisational analysis at the time equated power with position, which could not explain why some junior people were consistently influential and some senior ones were not. Separating the bases explains it: expert and referent power accumulate through work and relationships rather than appointment, and both survive a change of role, which positional power does not.

FIGURE 67 · FIVE BASES OF SOCIAL POWER



Figure 67. Five bases, of which only the first three derive from position. Raven added informational power as a sixth in 1965 (French and Raven, 1959).

HOW IT IS USED

Audit which bases you actually hold in a given relationship, then build the portable ones — expert and referent — since those are the bases available without a title.

WHEN IT IS USED

When you must influence people who do not report to you, when moving into a new organisation, and when positional authority is producing compliance but not commitment.

BY WHOM IT IS USED

Professionals working across organisational boundaries, project managers without line authority, and new leaders establishing standing.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Treating power as a single quantity conferred by position gave no account of influence exercised without it, and no development path other than promotion. The typology makes power analysable and, in two of its bases, buildable. Later research finds the bases are not fully independent — coercive use, for example, tends to erode referent power — and that their relative effectiveness varies by culture.

RELATED FRAMEWORKS IN THIS COURSE

Six Principles of Influence · [Slibrary 14](#)

Influence Currencies · [Slibrary 14](#)

Power–Interest Grid · [Slibrary 10](#)

Hofstede Cultural Dimensions · [Slibrary 10](#)

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Influence Currencies

SLIBRARY 14
SHEET 3 / 5

Find what they value, then trade for it — influence as reciprocal exchange

PROFESSIONAL DEFINITION

Cohen and Bradford model influence without authority as an exchange in which each party trades what the other values. They group these currencies into five families: inspiration-related, such as vision and moral significance; task-related, such as resources, information and assistance; position-related, such as visibility, recognition and contacts; relationship-related, such as acceptance and personal support; and personal-related, such as gratitude and enhanced self-concept (Cohen and Bradford, 1989).

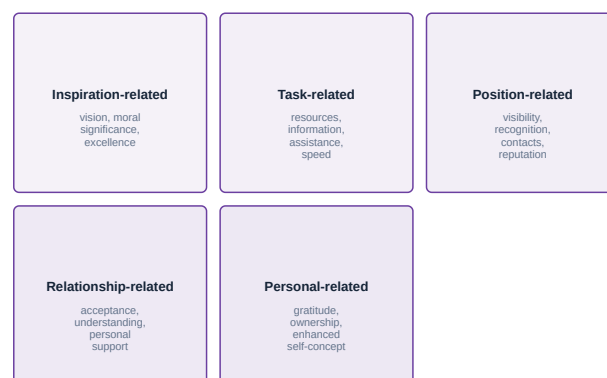
WHO DEVELOPED IT

Allan R. Cohen of Babson College and David L. Bradford of the Stanford Graduate School of Business developed the model in *Influence Without Authority* (1990), following a 1989 article in *Organizational Dynamics*. It applies social-exchange theory, particularly the norm of reciprocity, to lateral relationships inside organisations (Gouldner, 1960).

WHY IT WAS DEVELOPED

Matrix structures and cross-functional projects created a common situation with no established method: needing something from someone with no obligation to provide it. Cohen and Bradford's answer is that influence is a trade, and that most failures occur because the influencer offers what they themselves value rather than investigating what the other party is measured and rewarded on.

FIGURE 68 · FIVE FAMILIES OF EXCHANGE CURRENCY



Offer what they value, not what you value

Figure 68. Five currency families. The common error is paying in the currency the influencer values rather than the one the counterpart does (Cohen and Bradford, 1989).

HOW IT IS USED

Before asking, work out what the other person is measured on and what they are short of, then make the request in terms of that rather than of your own need.

WHEN IT IS USED

When you need cooperation from another function, when a request has been refused without explanation, and when building working relationships before you need them.

BY WHOM IT IS USED

Project managers working across silos, professionals in matrix organisations, and anyone whose delivery depends on people they cannot instruct.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Advice on lateral influence had consisted largely of relationship-building generalities. The currencies framework makes the exchange specific and therefore preparable, and reframes the diagnosis: a refused request usually indicates the wrong currency rather than an uncooperative colleague. The framework is practitioner-derived, and its transactional framing reads poorly where relationships are genuinely collegial rather than exchange-based.

RELATED FRAMEWORKS IN THIS COURSE

French & Raven Power Bases · [Slibrary 14](#)

Six Principles of Influence · [Slibrary 14](#)

Distributive ↔ Integrative Spectrum · [Slibrary 13](#)

Power–Interest Grid · [Slibrary 10](#)

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Tactical Empathy

SLIBRARY 14
SHEET 4 / 5

Demonstrate that you understand their position — before you ask them to move from it

PROFESSIONAL DEFINITION

Tactical empathy is the deliberate demonstration of understanding of the other party's perspective and emotional state, in order to lower resistance before any proposal is made. Its principal techniques are labelling, naming the emotion observed; mirroring, repeating the last few words to invite elaboration; the accusation audit, stating their likely objections before they do; calibrated questions beginning 'how' or 'what'; and working toward 'that's right' rather than 'you're right' (Voss and Raz, 2016).

WHO DEVELOPED IT

Chris Voss, formerly the FBI's lead international kidnapping negotiator, set out the approach in *Never Split the Difference* (2016) with Tahl Raz. It draws on crisis-negotiation practice developed at the FBI and Scotland Yard and on active-listening research from clinical psychology, notably Carl Rogers's work (Rogers and Farson, 1957).

WHY IT WAS DEVELOPED

Voss's argument, formed in hostage negotiation where no distributive trade is available, is that a party who does not feel understood will not move regardless of the quality of the offer. The techniques are practical rather than affective: labelling and mirroring produce evidence of understanding that the other party can hear, which is what reduces resistance, and require no actual agreement.

FIGURE 69 · FIVE TECHNIQUES, APPLIED BEFORE THE ASK

- 1 **Labelling**
"It sounds like the timing is the real problem"
- 2 **Mirroring**
repeat their last few words; let them elaborate
- 3 **Accusation audit**
state their objection before they have to
- 4 **Calibrated questions**
"how am I supposed to do that?"
- 5 **Aim for 'that's right'**
not 'you're right', which ends the conversation

Understanding is demonstrated, not claimed — and precedes the proposal

Figure 69. Five techniques used before a proposal is made. Each produces audible evidence of understanding without conceding anything (Voss and Raz, 2016).

HOW IT IS USED

Open by summarising their position and its difficulties in your own words, and continue until they confirm it. Only then introduce your proposal.

WHEN IT IS USED

When a counterpart is entrenched, in emotionally charged disagreement, and whenever your previous proposals have been rejected without engagement.

BY WHOM IT IS USED

Negotiators facing resistance, managers handling grievances, and anyone whose reasonable proposals are being refused for reasons not yet stated.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Principled negotiation supplied a rational structure but comparatively little on handling a counterpart who is not behaving rationally. Voss's contribution is a set of specific verbal techniques for reducing emotional resistance, developed under conditions where failure was catastrophic. The evidence base is professional experience rather than controlled study, and some of the techniques read as manipulative if used without genuine intent to understand.

RELATED FRAMEWORKS IN THIS COURSE

Principled Negotiation · [Slibrary 13](#)

Ladder of Inference · [Slibrary 6](#)

Persuasion ↔ Manipulation Line · [Slibrary 14](#)

High- and Low-Context Communication · [Slibrary 10](#)

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Persuasion ↔ Manipulation Line

SLIBRARY 14
SHEET 5 / 5

Does the technique work through their reasoning, or around it?

PROFESSIONAL DEFINITION

The distinction between persuasion and manipulation turns on whether an influence attempt engages the other party's capacity for reflective judgment or bypasses it. Sunstein's criterion is that an influence is manipulative to the extent that it does not sufficiently engage or appeal to a person's capacity for reflection and deliberation. Transparency is the practical test: an influence that would fail if the target knew what was being done to them sits on the manipulative side (Sunstein, 2016).

WHO DEVELOPED IT

Cass R. Sunstein, professor at Harvard Law School and former administrator of the US Office of Information and Regulatory Affairs, set out the criterion in 'Fifty Shades of Manipulation' (2016), extending the work on choice architecture he developed with Richard Thaler in *Nudge* (2008), where the ethical boundary became a central objection.

WHY IT WAS DEVELOPED

Behavioural insights made it possible to shape choices through framing, defaults and timing without argument and often without awareness, which put practitioners in possession of techniques whose legitimacy was unclear. Sunstein's contribution is a usable criterion rather than a list of banned techniques, since the same technique can be either — a default is helpful or exploitative depending on whose interest it serves.

FIGURE 70 · FOUR POSITIONS ON THE INFLUENCE CONTINUUM

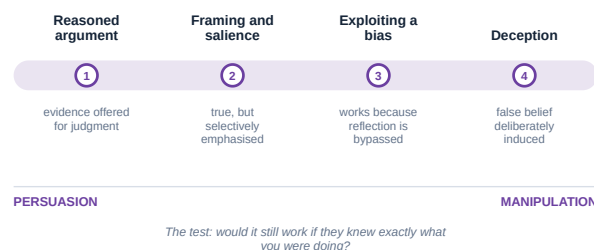


Figure 70. A continuum from engaging reflection to bypassing it. The transparency test locates a technique on the line more reliably than the technique's name (Sunstein, 2016).

HOW IT IS USED

Apply the transparency test before using a technique: if disclosing your method would defeat it or provoke justified resentment, choose a different one.

WHEN IT IS USED

When designing any influence attempt, when using behavioural techniques on customers or colleagues, and when deploying AI-generated persuasive content at scale.

BY WHOM IT IS USED

Marketers and communicators making ethical judgments, managers influencing teams, and professionals whose techniques outrun their stated intentions.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Business ethics generally addressed influence through rules prohibiting specific practices, which cannot keep pace with new techniques and misses context: the same default serves or exploits depending on whose interests it advances. Sunstein's criterion is generalisable and applies to methods not yet invented. It requires judgment rather than supplying an answer, and reasonable people disagree about cases in the middle of the line.

RELATED FRAMEWORKS IN THIS COURSE

Six Principles of Influence · [Slibrary 14](#)

Tactical Empathy · [Slibrary 14](#)

System 1 / System 2 · [Slibrary 6](#)

AI Attribution Ladder · [Slibrary 16](#)

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Eisenhower Matrix

SLIBRARY 15
SHEET 1 / 5

Urgent against important — and the quadrant that never shouts

PROFESSIONAL DEFINITION

The Eisenhower Matrix sorts tasks on two independent axes: importance, meaning contribution to a stated objective, and urgency, meaning proximity of a deadline. Important and urgent work is done now; important but not urgent work is scheduled, and is where planning, development and prevention live; urgent but unimportant work is delegated; and work that is neither is eliminated. The matrix exists because urgency is felt and importance is not (Covey, 1989).

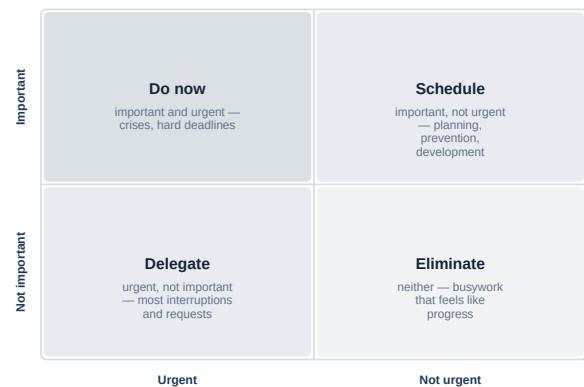
WHO DEVELOPED IT

The distinction is attributed to Dwight D. Eisenhower, who quoted a university president's remark that the urgent is seldom important and the important seldom urgent. Stephen R. Covey developed it into the four-quadrant time-management matrix in *The 7 Habits of Highly Effective People* (1989), where quadrant two is the organising idea (Covey, Merrill and Merrill, 1994).

WHY IT WAS DEVELOPED

Covey's observation was that people manage time against demands rather than against objectives, so attention is allocated by whatever signals urgency most loudly. Important work without a deadline generates no signal at all and is therefore permanently deferred — which is why the matrix's real subject is quadrant two, and why the practical instruction is to schedule it rather than intend it.

FIGURE 71 · FOUR QUADRANTS OF URGENCY AND IMPORTANCE



Quadrant two produces no signal, so it must be scheduled rather than intended

Figure 71. Two independent axes forming four responses. The upper-right quadrant is the one that never demands attention and therefore never receives it (Covey, 1989).

HOW IT IS USED

Place next week's tasks in the four quadrants, then put calendar time against the important and non-urgent ones before anything else claims the space.

WHEN IT IS USED

During weekly planning, when a week has been busy without progress, and when everything on a list appears equally pressing.

BY WHOM IT IS USED

Anyone allocating their own time, managers deciding what to delegate, and teams whose improvement work is permanently postponed by delivery work.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Conventional time management ranked a single list by deadline, which structurally guarantees that important work without a deadline is never reached. Separating the two axes makes that failure visible and gives it a name. The matrix supplies no method for judging importance, which is the harder question, and it works only where the person has genuine control over their own schedule.

RELATED FRAMEWORKS IN THIS COURSE

The Rule of 3 · **Slibrary 15**

Objectives & Key Results · **Slibrary 15**

Pareto 80/20 · **Slibrary 15**

Personal Energy Model · **Slibrary 3**

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Objectives & Key Results

SLIBRARY 15
SHEET 2 / 5

One qualitative objective, three to five measurable key results

PROFESSIONAL DEFINITION

An OKR pairs a single qualitative objective — a direction, stated memorably — with three to five key results that are quantitative, time-bound and verifiable. Key results measure outcomes rather than activity, so completing the listed work does not by itself constitute achievement. OKRs are typically set quarterly, made visible across the organisation, and scored rather than passed or failed, with ambitious targets expected to land short (Doerr, 2018).

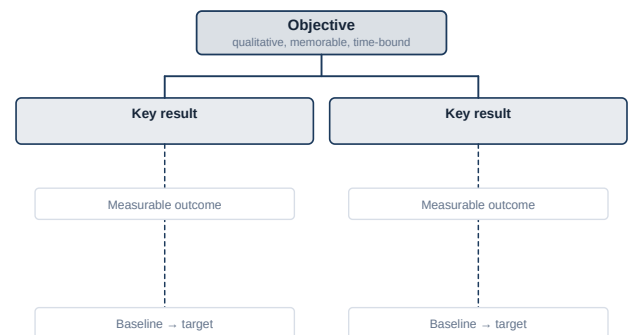
WHO DEVELOPED IT

Andrew Grove developed the method at Intel in the 1970s as an evolution of Drucker's Management by Objectives, describing it in *High Output Management* (1983). John Doerr, who learned it at Intel, introduced it to Google in 1999 and set it out for a general readership in *Measure What Matters* (2018) (Grove, 1983).

WHY IT WAS DEVELOPED

Management by Objectives had degenerated in many organisations into annual, private, compensation-linked targets that encouraged conservative goal-setting and disappeared from view between reviews. Grove's changes address each failure: quarterly cycles, public visibility, and separation from compensation, so that ambitious targets can be set without penalty for missing them.

FIGURE 72 · ONE OBJECTIVE, MEASURED BY KEY RESULTS



If it has no number and no date, it is not a key result

Figure 72. A directional objective supported by measurable outcomes. Key results measure results, not the activity undertaken to produce them (Doerr, 2018).

HOW IT IS USED

Write the objective so it would motivate someone without further explanation, then write key results that a sceptic could verify without asking you.

WHEN IT IS USED

At the start of a quarter or term, when a team is busy without agreed direction, and when goals are being confused with task lists.

BY WHOM IT IS USED

Teams and organisations setting quarterly direction, individuals structuring a term's study, and managers aligning work across functions.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

MBO established negotiated objectives but left cadence, visibility and the activity–outcome distinction unspecified, which is where it failed in practice. OKRs specify all three. The known risks are well documented: separating OKRs from compensation is essential and frequently abandoned, and heavily quantified goals can displace important work that resists measurement (Ordóñez et al., 2009).

RELATED FRAMEWORKS IN THIS COURSE

SMART Goal Frame · **Slibrary 1**

The Rule of 3 · **Slibrary 15**

Eisenhower Matrix · **Slibrary 15**

Firm Foundation: Vision, Mission, Values · **Slibrary 3**

SOURCES (HARVARD REFERENCING STYLE)

Doerr, J. (2018) *Measure What Matters*. New York: Portfolio/Penguin.

Grove, A.S. (1983) *High Output Management*. New York: Random House.

Drucker, P.F. (1954) *The Practice of Management*. New York: Harper & Row.

Ordóñez, L.D., Schweitzer, M.E., Galinsky, A.D. and Bazerman, M.H. (2009) 'Goals gone wild', *Academy of Management Perspectives*, 23(1), pp. 6–16.

The Rule of 3

SLIBRARY 15
SHEET 3 / 5

Three outcomes for the day, three for the week, three for the year

PROFESSIONAL DEFINITION

Meier's Rule of 3 limits commitment to three named outcomes at each time horizon: three for today, three for this week, three for this year, each nested within the level above. The constraint is the mechanism — three forces a choice where a list of ten does not — and outcomes are stated as results achieved rather than as tasks performed, so that completion is recognisable at the end of the period (Meier, 2010).

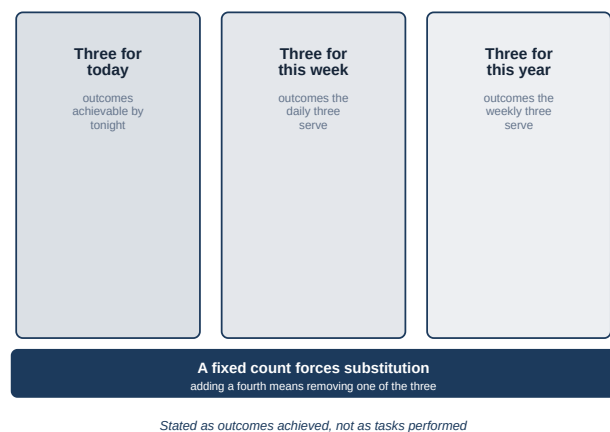
WHO DEVELOPED IT

J.D. Meier, a programme manager at Microsoft, set out the Rule of 3 as the core of *Getting Results the Agile Way* (2010) and revisits it as a thumbnail structure in *Thumbnail Thinking* (2025). It draws on the well-established finding that working memory handles a small number of items at once, and on agile practice's use of fixed capacity (Miller, 1956).

WHY IT WAS DEVELOPED

Meier's observation from managing large programmes was that to-do lists grow without bound because adding an item is free, so the list stops functioning as a decision and becomes a record of intentions. A fixed count makes every addition a substitution. Nesting the three horizons prevents the second failure mode: daily wins that are individually satisfying and cumulatively irrelevant.

FIGURE 73 · THREE NESTED HORIZONS



Stated as outcomes achieved, not as tasks performed

Figure 73. Three horizons, each nested in the one above. The count is fixed so that adding an item requires removing one (Meier, 2010).

HOW IT IS USED

Name three outcomes each morning and check them against the week's three. If a daily outcome serves none of the weekly three, it is either urgent or misplaced.

WHEN IT IS USED

Every morning and at the start of each week, and whenever a task list has grown long enough to be avoided rather than used.

BY WHOM IT IS USED

Individuals running a personal results system, managers setting team focus, and students balancing coursework across several modules.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Conventional task management optimises capture and retrieval, which improves memory without improving selection, so a well-organised list of forty items remains forty items. The Rule of 3 makes the constraint itself the method. It is practitioner-derived rather than empirically validated, and the fixed count fits poorly with work dominated by unpredictable reactive demand.

RELATED FRAMEWORKS IN THIS COURSE

Monday Vision, Daily Wins, Friday Reflection · **Slibrary 15**

Eisenhower Matrix · **Slibrary 15**

Firm Foundation: Vision, Mission, Values · **Slibrary 3**

Distil → Decompose → Prioritise · **Slibrary 9**

SOURCES (HARVARD REFERENCING STYLE)

Meier, J.D. (2010) *Getting Results the Agile Way: A Personal Results System for Work and Life*. Bellevue, WA: Innovation Playhouse.

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Miller, G.A. (1956) 'The magical number seven, plus or minus two', *Psychological Review*, 63(2), pp. 81–97.

Monday Vision, Daily Wins, Friday Reflection

SLIBRARY 15
SHEET 4 / 5

Set the week, work the day, close the loop — a weekly cycle with reflection built in

PROFESSIONAL DEFINITION

Meier's weekly pattern has three moves. Monday Vision sets three outcomes for the week ahead, chosen against the longer horizon rather than against the inbox. Daily Wins names three outcomes each morning that serve those. Friday Reflection asks what went well and what to improve, and feeds the answer into the next Monday. The cycle is deliberately short so that correction happens weekly rather than at an annual review (Meier, 2010).

WHO DEVELOPED IT

J.D. Meier developed the pattern as the operating rhythm of *Getting Results the Agile Way* (2010), applying the iteration-and-retrospective structure of agile delivery to individual work. The reflective step corresponds directly to the reflective observation stage of Kolb's experiential learning cycle (Kolb, 1984).

WHY IT WAS DEVELOPED

Meier's diagnosis was that productivity systems handle execution well and learning badly: people repeat the same week many times without improving it, because nothing in the system asks what happened. Friday Reflection supplies the missing stage, and the weekly cadence is chosen because it is short enough for the detail to be remembered and long enough for a pattern to appear.

FIGURE 74 · A THREE-MOVE WEEKLY CYCLE

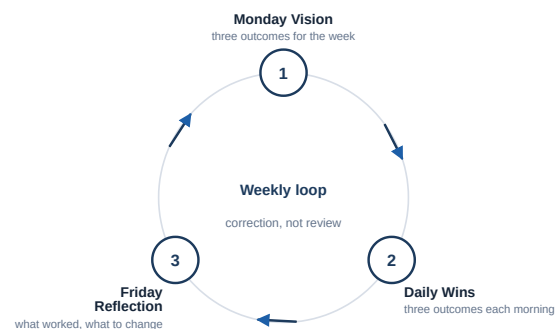


Figure 74. Three moves closing weekly. Without the reflective step the loop degrades into execution, which repeats rather than improves (Meier, 2010).

HOW IT IS USED

Book thirty minutes on Monday and twenty on Friday as recurring appointments. Write the reflection down — an unwritten reflection does not survive to Monday.

WHEN IT IS USED

Weekly, throughout a term or project, and particularly when workload is high enough that reflection feels unaffordable.

BY WHOM IT IS USED

Individuals building a sustainable working rhythm, students across a term, and teams running the same pattern collectively.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Most personal systems provide capture, planning and execution but no structured learning step, so improvement depends on the person noticing something. This pattern makes reflection a scheduled component with a defined output that feeds the next cycle, importing the agile retrospective into individual practice. It is practitioner-derived, and the reflective step is the one people abandon first under pressure.

RELATED FRAMEWORKS IN THIS COURSE

The Rule of 3 · **Slibrary 15**

Kolb Experiential Learning Cycle · **Slibrary 1**

Scrum Sprint Cycle · **Slibrary 12**

Personal Energy Model · **Slibrary 3**

SOURCES (HARVARD REFERENCING STYLE)

Meier, J.D. (2010) *Getting Results the Agile Way*. Bellevue, WA: Innovation Playhouse.

Kolb, D.A. (1984) *Experiential Learning*. Englewood Cliffs, NJ: Prentice Hall.

Schön, D.A. (1983) *The Reflective Practitioner: How Professionals Think in Action*. New York: Basic Books.

Pareto 80/20

SLIBRARY 15
SHEET 5 / 5

A minority of inputs accounts for a majority of outputs — find which minority

PROFESSIONAL DEFINITION

The Pareto principle observes that outcomes are typically distributed unevenly across causes, so that a small proportion of inputs accounts for a large proportion of results. The eighty-twenty split is an illustrative shorthand rather than a law: the actual ratio varies by domain and must be measured. The principle's value is diagnostic — it directs effort toward identifying which inputs are the productive minority (Juran, 1951; Pareto, 1896).

WHO DEVELOPED IT

Vilfredo Pareto, an Italian economist at Lausanne, observed in the 1890s that a small share of the population held a large share of the land. Joseph M. Juran, a quality-management pioneer, generalised the observation to quality defects in the 1950s and named it the Pareto principle, later regretting the attribution and preferring 'the vital few and the useful many' (Juran, 1975).

WHY IT WAS DEVELOPED

Juran's problem in manufacturing was that quality improvement effort was spread across all defect types equally, which produced slow progress. Sorting defects by frequency showed a small number accounted for most failures. The principle generalises to any domain where causes are unevenly weighted, which is most of them — but the specific minority has to be identified empirically, not assumed.

FIGURE 75 · AN UNEVEN DISTRIBUTION OF CAUSE AND EFFECT

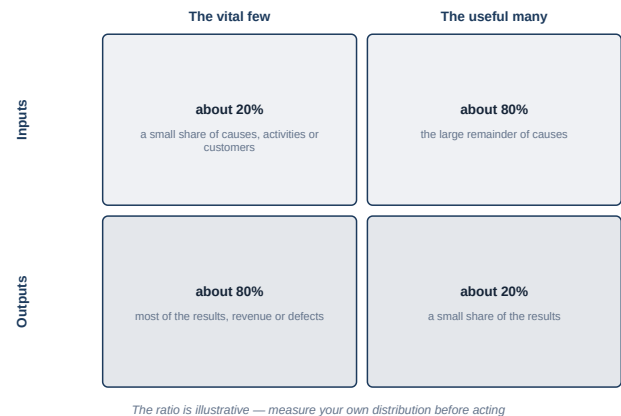


Figure 75. A small share of inputs associated with a large share of outputs. The figures are a shorthand; the real distribution must be measured (Juran, 1951).

HOW IT IS USED

Measure the actual distribution before reallocating effort — rank customers, defects or tasks by contribution — then concentrate on the measured minority.

WHEN IT IS USED

When effort is spread evenly across a long list, when improving quality or revenue, and when deciding what to stop doing.

BY WHOM IT IS USED

Managers allocating limited resource, analysts prioritising improvement work, and individuals deciding which activities to drop.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Uniform allocation of improvement effort treats all causes as equally weighted, which is almost never true and produces slow, diffuse progress. Sorting by contribution concentrates effort where return is highest. The principle is routinely misapplied: it is an empirical observation about particular distributions, not a rule, and the eighty-twenty figures are quoted as findings far more often than they are measured.

RELATED FRAMEWORKS IN THIS COURSE

[Eisenhower Matrix · Slibrary 15](#)
[The Rule of 3 · Slibrary 15](#)
[MECE Issue Tree · Slibrary 5](#)
[Objectives & Key Results · Slibrary 15](#)

SOURCES (HARVARD REFERENCING STYLE)

Juran, J.M. (1951) *Quality Control Handbook*. New York: McGraw-Hill.

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Pareto, V. (1896) *Cours d'économie politique*. Lausanne: F. Rouge.

Koch, R. (1997) *The 80/20 Principle*. London: Nicholas Brealey.

Prompt Anatomy

Role · task · context · format · constraints · examples — six components of a specified request

PROFESSIONAL DEFINITION

A well-formed prompt specifies six components: the role the model should adopt, the task to be performed stated as a single instruction, the context the model would otherwise supply from assumption, the format the output must take, the constraints it must respect, and worked examples where the task is hard to describe but easy to demonstrate. Unspecified components are not omitted by the model; they are filled with the statistical default (Mollick, 2024).

WHO DEVELOPED IT

Prompt structure is a practitioner convention that converged across independent sources rather than the work of one author. Its empirical basis includes the few-shot learning results reported with GPT-3 and the chain-of-thought prompting work of Wei and colleagues, both of which demonstrated large performance differences from prompt structure alone (Brown et al., 2020; Wei et al., 2022).

WHY IT WAS DEVELOPED

Language models produce output conditioned on the request, so an underspecified request returns the most probable generic response — which is why unspecified prompts produce writing that reads as characterless. Naming the components makes the specification auditable: when output is wrong, the failure can be traced to a missing component rather than attributed to the model.

FIGURE 76 · SIX COMPONENTS OF A SPECIFIED REQUEST

Role	who the model should be
Task	one instruction, stated plainly
Context	what it cannot infer
Format	the shape of the output
Constraints	length, tone, what to avoid
Examples	demonstrate what you cannot describe

Anything unspecified is filled with the default, not left out

Figure 76. Six components. A disappointing output usually indicates a missing component rather than a limit of the model (Mollick, 2024).

HOW IT IS USED

When output disappoints, identify which of the six was unspecified and add it, rather than rewriting the request from scratch or trying a different model.

WHEN IT IS USED

Whenever delegating substantive work to a model, when building a reusable prompt, and when output is fluent but generic.

BY WHOM IT IS USED

Anyone working with generative AI, teams standardising prompts, and students producing work that must be defensible.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Early practice treated prompting as an undocumented knack, which made results irreproducible and unteachable. Decomposing the request into named components makes prompts diagnosable, transferable and improvable by inspection. The caveat is that specific prompting techniques are tied to model generations and age quickly; the components endure, the particular formulations do not.

RELATED FRAMEWORKS IN THIS COURSE

- Co-Intelligence Working Rules · [Slibrary 2](#)
- Delegate–Verify–Own Loop · [Slibrary 16](#)
- The Jagged Frontier · [Slibrary 16](#)
- Technology Acceptance Model · [Slibrary 2](#)

SOURCES (HARVARD REFERENCING STYLE)

Mollick, E. (2024) *Co-Intelligence: Living and Working with AI*. New York: Portfolio/Penguin.

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Delegate–Verify–Own Loop

SLIBRARY 16
SHEET 2 / 5

Delegate the task, verify the output, own the result — the accountability never moves

PROFESSIONAL DEFINITION

The loop states the working relationship with an AI system in three moves. Delegate: assign a task whose output you are competent to evaluate. Verify: check the output against a source, a calculation or your own expertise, at a depth proportional to the consequence. Own: accept full accountability for what is submitted, since responsibility does not transfer to the tool. The second move is the one under pressure, and the first constrains it (Mollick, 2024).

WHO DEVELOPED IT

The loop synthesises the human-in-the-loop principle from Mollick's working rules with the verification requirements established in human-factors research on automation bias by Parasuraman and Riley, and with the accountability provisions of the EU AI Act, which requires meaningful human oversight in high-risk applications (Parasuraman and Riley, 1997; European Parliament, 2024).

WHY IT WAS DEVELOPED

Two distinct failures recur. The first is delegating work the delegator cannot evaluate, which makes verification impossible in principle rather than merely burdensome. The second is verification decaying as the tool proves reliable, which is the automation-bias finding applied to generative systems. The loop addresses both: competence gates delegation, and ownership is stated in advance.

FIGURE 77 · THREE MOVES THAT DO NOT TRANSFER RESPONSIBILITY

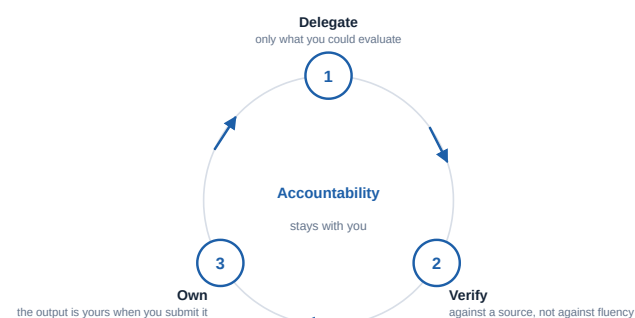


Figure 77. Three moves in a repeating loop. Competence to evaluate is the precondition for delegation, not a consideration afterwards (Mollick, 2024).

HOW IT IS USED

Before delegating, ask whether you could tell if the output were wrong. If not, either build the competence first or do not delegate that task.

WHEN IT IS USED

Every time AI output enters work you will submit, sign, publish or act on, and particularly once the tool has been reliable for a long stretch.

BY WHOM IT IS USED

Professionals using AI in regulated or high-consequence settings, students under academic-integrity rules, and managers setting team standards.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Organisational responses have tended toward blanket policy — prohibition or permission — neither of which distinguishes tasks by consequence or by the user's ability to check. The loop makes the decision task-by-task and locates it with the person accountable. It is a practitioner synthesis rather than a validated model, and it depends on honest self-assessment of competence, which Dunning and Kruger's work suggests is unreliable exactly where it matters.

RELATED FRAMEWORKS IN THIS COURSE

Automation Bias Curve · [Slibrary 2](#)Co-Intelligence Working Rules · [Slibrary 2](#)AI Attribution Ladder · [Slibrary 16](#)Dunning–Kruger Curve · [Slibrary 6](#)

SOURCES (HARVARD REFERENCING STYLE)

Mollick, E. (2024) *Co-Intelligence: Living and Working with AI*. New York: Portfolio/Penguin.

Parasuraman, R. and Riley, V. (1997) 'Humans and automation: use, misuse, disuse, abuse', *Human Factors*, 39(2), pp. 230–253.

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Automation–Augmentation Spectrum

SLIBRARY 16
SHEET 3 / 5

Replace the task or amplify the person — a design choice, not a technical inevitability

PROFESSIONAL DEFINITION

The spectrum distinguishes automation, in which a system performs a task previously done by a person, from augmentation, in which a system extends what a person can do while the person retains the judgment. Most deployments sit between the two, and the position is chosen rather than determined by the technology. Automation optimises cost on defined tasks; augmentation optimises capability on undefined ones (Raisch and Krakowski, 2021).

WHO DEVELOPED IT

The distinction descends from Douglas Engelbart's 1962 framework for augmenting human intellect, articulated in opposition to the automation-oriented AI research of the period. Sebastian Raisch and Sebastian Krakowski restated it as a management paradox in the *Academy of Management Review* (2021), arguing the two are interdependent rather than alternatives (Engelbart, 1962).

WHY IT WAS DEVELOPED

Organisations adopting AI frequently pose the question as whether a role will be automated, which is not a question the technology answers: the same capability can be deployed to remove a task or to extend a practitioner. Raisch and Krakowski's argument is that treating the choice as inevitable removes it from management, and that pure automation erodes the human expertise the system will later require.

FIGURE 78 · FOUR POSITIONS BETWEEN REPLACEMENT AND AMPLIFICATION

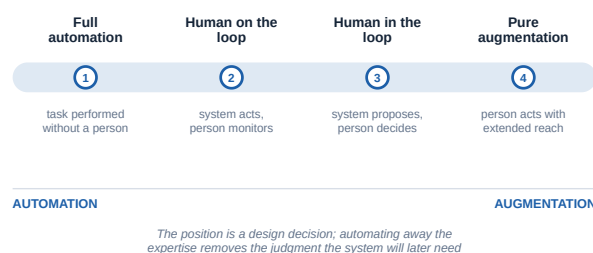


Figure 78. Four positions on one continuum. The technology does not select the position; the deployment decision does (Raisch and Krakowski, 2021).

HOW IT IS USED

For each task, decide the position deliberately and write down the reason. Where judgment matters and the volume is low, augmentation is usually the better trade.

WHEN IT IS USED

When designing an AI-supported workflow, when a role is being redefined around new tools, and when deciding what to keep doing manually in order to retain the skill.

BY WHOM IT IS USED

Managers designing workflows, professionals deciding where to apply tools in their own work, and students deciding which capabilities to develop rather than delegate.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Public and organisational debate frames AI adoption as a binary of replacement or resistance, which obscures the range of intermediate designs and hides the choice. Naming the spectrum makes the position explicit and reversible. The framework identifies the trade-off clearly but offers little guidance on where a given task should sit, which remains a matter of judgment about consequence and volume.

RELATED FRAMEWORKS IN THIS COURSE

The Jagged Frontier · [Slibrary 16](#)

Dreyfus Skill Acquisition Model · [Slibrary 1](#)

Delegate–Verify–Own Loop · [Slibrary 16](#)

Automation Bias Curve · [Slibrary 2](#)

SOURCES (HARVARD REFERENCING STYLE)

- Raisch, S. and Krakowski, S. (2021) 'Artificial intelligence and management: the automation–augmentation paradox', *Academy of Management Review*, 46(1), pp. 192–210.
- Engelbart, D.C. (1962) *Augmenting Human Intellect: A Conceptual Framework*. Menlo Park, CA: Stanford Research Institute.
- Brynjolfsson, E. (2022) 'The Turing trap: the promise and peril of human-like artificial intelligence', *Daedalus*, 151(2), pp. 272–287.

The Jagged Frontier

SLIBRARY 16
SHEET 4 / 5

AI capability has an uneven boundary — and it is invisible from outside

PROFESSIONAL DEFINITION

The jagged frontier describes the finding that AI capability is highly uneven across tasks of apparently similar difficulty: some tasks fall inside the frontier, where the system performs at or above a capable professional, and adjacent tasks fall outside it, where performance is poor. The boundary does not follow human intuitions about difficulty and is not visible from the output, because failures are expressed as fluently as successes (Dell'Acqua et al., 2023).

WHO DEVELOPED IT

The term comes from a field experiment run by Fabrizio Dell'Acqua and colleagues at Harvard Business School with the Boston Consulting Group, involving over seven hundred consultants working with and without GPT-4. Ethan Mollick, a co-author, made it central to *Co-Intelligence* (2024) (Dell'Acqua et al., 2023; Mollick, 2024).

WHY IT WAS DEVELOPED

The experiment produced two results that only make sense together. On tasks inside the frontier, consultants using AI completed more work of higher quality, with the largest gains among lower performers. On a task designed to sit outside it, consultants using AI were substantially more likely to reach the wrong answer than those working without it — because the output was persuasive and the boundary invisible.

FIGURE 79 · AN UNEVEN AND INVISIBLE CAPABILITY BOUNDARY

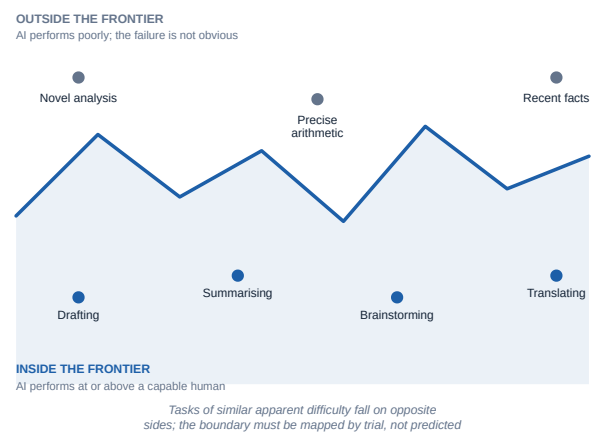


Figure 79. An uneven boundary between strong and weak performance. Adjacent tasks of similar apparent difficulty fall on opposite sides (Dell'Acqua et al., 2023).

HOW IT IS USED

Map the frontier empirically for your own work: test tasks you can already evaluate, record which fall inside, and re-test after each model change.

WHEN IT IS USED

When adopting a tool for a new class of task, when deciding what to check most carefully, and after any model update, since the boundary moves.

BY WHOM IT IS USED

Professionals adopting AI in expert work, managers setting team norms, and anyone whose output would be materially damaged by a confident error.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Capability had been discussed through benchmark scores, which report averages and conceal exactly the unevenness that matters in practice. The field experiment measured performance on real professional tasks and identified both the gain and the specific mechanism of harm. The finding is tied to particular models and tasks and requires re-establishment as systems change, which is itself the practical instruction.

RELATED FRAMEWORKS IN THIS COURSE

Co-Intelligence Working Rules · [Slibrary 2](#)Automation–Augmentation Spectrum · [Slibrary 16](#)Delegate–Verify–Own Loop · [Slibrary 16](#)Automation Bias Curve · [Slibrary 2](#)

SOURCES (HARVARD REFERENCING STYLE)

Dell'Acqua, F. et al. (2023) 'Navigating the jagged technological frontier: field experimental evidence of the effects of AI on knowledge worker productivity and quality', *Harvard Business School Working Paper* 24-013.

Mollick, E. (2024) *Co-Intelligence: Living and Working with AI*. New York: Portfolio/Penguin.

Noy, S. and Zhang, W. (2023) 'Experimental evidence on the productive effects of generative artificial intelligence', *Science*, 381(6654), pp. 187–192.

AI Attribution Ladder

SLIBRARY 16
SHEET 5 / 5

From no use to generated content — declare the rung, and be able to show it

PROFESSIONAL DEFINITION

The attribution ladder sets out ascending levels of AI involvement in a piece of work, from no use at all, through assistance with mechanics such as spelling and grammar, assistance with structure and ideas, generation of drafted material subsequently edited, to substantially generated content. The requirement is that the rung be declared accurately and that the underlying process be evidenceable, since the level of involvement is not detectable from the text (Perkins et al., 2024).

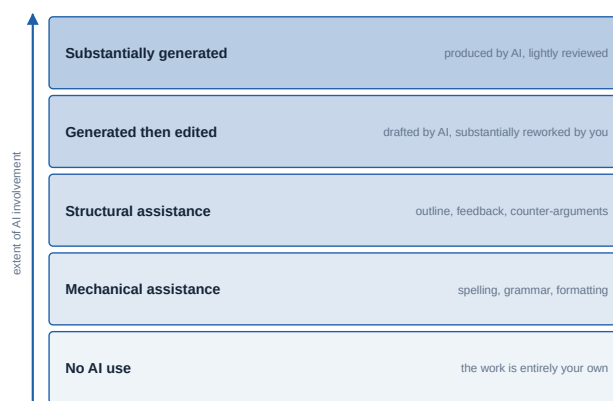
WHO DEVELOPED IT

The ladder is a synthesis of institutional academic-integrity frameworks rather than the work of one author, drawing on the AI Assessment Scale developed by Mike Perkins and colleagues, on Russell Group principles on generative AI in education (2023), and on QAA guidance to the UK higher-education sector (QAA, 2023; Perkins et al., 2024).

WHY IT WAS DEVELOPED

Institutions initially attempted to police AI use through detection, which failed on the evidence: detection tools have unreliable accuracy and disproportionately misclassify writing by non-native English speakers. The response has shifted to declaration and process evidence, which asks a different question — not whether AI was used, but whether the use was declared and the judgment retained.

FIGURE 80 · FIVE RUNGS OF DECLARED INVOLVEMENT



The rung is declared, not detected — keep the evidence of process

Figure 80. Five ascending rungs of involvement. The level cannot be read from the text, which is why declaration and process evidence replace detection (Perkins et al., 2024).

HOW IT IS USED

Declare the highest rung reached, keep drafts and prompt records as evidence of process, and check the specific rule for each assessment rather than assuming a general policy.

WHEN IT IS USED

On every submitted assessment, in professional work where provenance matters, and whenever you are unsure whether a use requires declaration.

BY WHOM IT IS USED

Students under academic-integrity regulations, educators designing assessment, and professionals in fields where the origin of work carries legal or ethical weight.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Binary permitted-or-prohibited policies could not describe the range of real use, so they produced either blanket bans that were widely ignored or permissions with no accountability. A graded scale allows an institution to specify the permitted rung per assessment. Its limitation is that it rests on honest declaration: it establishes a norm and an audit trail, not an enforcement mechanism.

RELATED FRAMEWORKS IN THIS COURSE

Delegate–Verify–Own Loop · [Slibrary 16](#)

Co-Intelligence Working Rules · [Slibrary 2](#)

CRAAP Source Test · [Slibrary 2](#)

Persuasion ↔ Manipulation Line · [Slibrary 14](#)

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